Contents

Exact Generating-Function Analysis of a Two-Class Non-Preemptive Priority $M/M/1$ Queue with Jockeying and Abandonment

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Abstract

We study the stationary queue-length distribution of a two-class, non-preemptive priority $M/M/1$ queue augmented with two waiting-room mechanisms: *jockeying*, in which a waiting customer switches to the other queue, and *abandonment*, in which a waiting customer leaves the system. Jockeying conserves the total number of customers, whereas abandonment does not, and combining both yields a model (Model-X) that is analytically intractable. We therefore present a collection of simpler models: Model-A (baseline), Model-B (bidirectional jockeying), Model-B_2 (jockeying from the priority class to the non-priority class) and Model-C_2 (abandonments in the priority class). We also study two head-of-line models: Model-B_2^{H} (only the first customer in the priority queue can jockey) and Model-C_2^{H} (only the first customer in the priority queue can abandon).

The central object is the bivariate Probability Generating Function (PGF) $P(x,y)$ of the two queue lengths, obtained by analytical and, where the structure allows, probabilistic arguments. The purpose of this work is to present a unified framework for deriving the functional equation of each model and to expose the mathematical challenges intrinsic to each. Model-A is classical and collapses to an algebraic expression by finding a kernel root inside the unit disc; Model-B_2 is solved by imposing analyticity at an interior point and Model-C_2 by imposing boundedness at the unit-disc boundary $x = 1$. The head-of-line models, like the baseline Model-A, also yield an algebraic expression for the PGF. Two models are left open: Model-B reduces to an integral expression for the PGF involving a Volterra-type integral equation, and Model-X compounds this with nonlinear characteristic curves.

Every closed form is validated against a truncated continuous-time Markov Chain (CTMC) that recovers the baseline model as the mechanism rates vanish. The comparison reveals a key difference between jockeying and abandonment: the former merely redistributes delay between classes, whereas the latter substantially shortens the queue lengths.

1 Introduction

Motivation

Priority queues arise naturally wherever a common server must satisfy heterogeneous demand. The primary motivation for this thesis is healthcare operations, specifically the management of waiting lists for elderly-care facilities, where medical severity dictates a user's —hereafter *customer*— priority class. Here prolonged waiting may itself alter the system's dynamics: a lack of appropriate care can deteriorate a waiting customer's health, leading them to change priority status (jockeying) or leave the system (abandonment), whether to seek alternative care or through death. Despite this motivation, the work is fundamentally modelling-driven: we characterise the long-run behaviour of queueing systems in which jockeying and/or abandonment play a role.

This thesis analyses a 2-class $M/M/1$ queue with non-preemptive priority: each class arrives according to a Poisson process (rates λ_1 and λ_2), service times are exponential at a common rate μ, and the single server completes the current service even when a higher-priority customer arrives. To this we add the mechanisms of jockeying and abandonment. The model and its mechanisms are described in more detail in §4.

How these two mechanisms interact within a priority discipline, and how that interaction affects the difficulty of obtaining a closed-form Probability Generating Function (PGF), is the main focus of this thesis.

The analytical challenge

In full generality, the bivariate PGF for Model-X satisfies a functional equation with terms of three kinds: algebraic coefficients, partial derivatives in x —from jockeying and abandonment in class-1— and partial derivatives in y —from the same mechanisms in class-2. When both kinds of derivative term are present, the equation belongs to a family of integro-differential equations whose solution would require more advanced tools. The problem is moreover asymmetric: derivative terms from the non-priority class complicate it more than those from the priority class alone.

We therefore solve less complex models obtained by setting some of the jockeying and abandonment parameters to zero. Table 1 lists the models studied here and indicates whether a PGF expression is obtained for each:

Model	Active mechanism	Equation type	Status
X	jockeying and abandonment	PDE, transcendental characteristics	open
A	none	algebraic PGF equation	solved
B	bidirectional jockeying	PDE, Volterra family	open
B_2	one-way jockeying ($1 \to 2$)	ODE in x, Kummer	solved
C_2	class-1 abandonment	ODE in x, Kummer	solved
B_2^{H}	one-way jockeying, head-of-line	algebraic PGF equation	solved
C_2^{H}	class-1 abandonment, head-of-line	algebraic PGF equation	solved

Table 1: Models studied in this thesis, with the active mechanism, the resulting functional-equation type, and the solution status. The head-of-line variants B_2^{H} and C_2^{H} restrict the mechanism to the customer at the head of Queue 1, which collapses the derivative term to an algebraic one and renders the model solvable in closed form.

Some models —notably Model-X and Model-B— are left incomplete because they involve derivative terms in y; others, such as a potential Model-B_1, Model-C or Model-C_1, are not considered for the same reason, as they all reduce to a Volterra integral equation lying beyond the scope of this thesis. Model-B serves as our representative example of how such problems reduce to that integral equation; the remaining models are solvable. The last two entries, Models B_2^{H} and C_2^{H} (§§9 and 10), are *head-of-line* simplifications in which only the customer at the head of Queue 1 may jockey or abandon. The mechanism then acts at the constant rate $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$ rather than the length-proportional $\gamma_1 n_1$ or $\theta_1 n_1$, so the fundamental equation stays algebraic and is solved by the Kernel Method exactly as in Model-A.

Methods

Three complementary proof strategies are used throughout.

Kernel method / ODE with integrating factor. For Model-A, which has no derivative terms, the algebraic coefficient of $P(x, y)$ vanishes on a curve (the *kernel*); evaluating the equation there determines the remaining unknowns. Model-B_2 and Model-C_2, which do carry derivative terms in x, reduce to a first-order linear ODE in x with an explicit integrating factor, and their unknowns follow from a regularity argument at a singularity of that factor. We present both strategies together, as they share the same underlying idea: the latter extends the former to the case where the unknowns are independent of the derivative terms and can be taken outside the integral.

Probabilistic argument via Pollaczek–Khinchine and the effective service time. For Model-A and Model-C_2, class-2 arrivals are Poisson, so the PASTA property holds and we can apply the Pollaczek–Khinchine formula together with the *effective service time* seen by a class-2 customer. This is an $M/G/1$ queue whose service times equal the busy period of the class-1 subsystem —an $M/M/1$ queue for Model-A and an $M/M/1+M$ system for Model-C_2. In both cases the argument independently confirms the analytic results. Since jockeying destroys the Poisson structure, it does not apply to the other models.

Partial PGF (PPGF) in the alternative state space $\widetilde{S}$. For the baseline Model-A we introduce an alternative state space indexed by the number of class-2 customers —whose distribution is hard— and the total number of customers —which is easy, since the total is conserved and behaves like a regular $M/M/1$ queue. This yields a PPGF with a non-homogeneous term. Cohen's recursion argument [Coh56] was developed for the regular state space; we explore, with limited success, the challenges of carrying it over to this alternative space and assess whether Cohen's trick can deliver a feasible rough approximation.

Structure of the thesis

Section 2 reviews the relevant literature and Section 3 collects the probabilistic and analytic background used throughout this work. Section 4 defines the full model, Model-X, and derives the general balance equations and the fundamental PGF equations, both for the regular and the alternative state space. The analysis then proceeds from baseline to synthesis: Section 5 solves the baseline Model-A in closed form; Section 6 shows that bidirectional jockeying (Model-B) reduces the problem to a Volterra-type integral equation and is left open. Sections 7 and 8 then recover tractability in the one-way-jockeying (Model-B_2) and class-1-abandonment (Model-C_2) specialisations, while Sections 9 and 10 show how the head-of-line variants, Models B_2^{H} and C_2^{H}, collapse the functional equation to an algebraic one solved in closed form. Finally, Section 11 validates every closed form against an exact CTMC and compares the solved models (§11.2), and Section 12 concludes by abstracting the obstruction taxonomy and setting out the future work.

2 Literature

The study of two-class priority queues with departure mechanisms sits at the intersection of three well-developed streams: the exact generating-function analysis of non-preemptive priority queues, the theory of queues with customer impatience (abandonment), and the theory of jockeying and dynamic priority changes. The applied motivation throughout is the management of access to care —elderly-care waiting lists, emergency departments, and transplant registries— in which the two mechanisms studied here carry direct clinical readings: a waiting patient whose condition deteriorates changes priority class (jockeying), while one who leaves the list by death or transfer reneges (abandonment). This thesis contributes an exact, single-server treatment that admits jockeying and abandonment within a unified bivariate-PGF framework, filling a gap that the multi-server and approximate literature has so far left open.

Non-preemptive priority queues: exact queue-length analysis

The direct methodological ancestor of this work is Cohen [Coh56], who studied a single server loaded by two independent Poisson streams and introduced the partial-generating-function recursion —the "Cohen trick"— that resolves the lower-priority queue length by transforming only one coordinate of the joint distribution while keeping the other discrete. That device is exactly the one reused, in modern notation, in the partial-PGF analysis of Model-A on the total-count state space (§5.2). The stationary queue-length law of the two-class non-preemptive $M/M/1$ priority queue —the content of Theorem 2 (Model-A)— is in this sense classical, recoverable by several routes; the contribution of the present work is not that result in isolation but the *unified framework* built around it, into which jockeying and abandonment are introduced one mechanism at a time and solved by a common functional-equation method.

The most recent exact closed-form treatment in this line is that of Zuk & Kirszenblat [ZK23], who obtain the joint and marginal queue-length distributions of a non-preemptive $M/M/c$ queue with two priority classes and a common service rate by generating-function methods and a quadratic recurrence; their single-server specialisation ($c = 1$) coincides with Model-A. A complementary strand studies dynamic rather than static priority: Stanford et al. [STZ14] analyse the *accumulating-priority* $M/G/1$ queue, in which each class earns priority linearly in its waiting time so that older customers eventually overtake newer ones, and derive exact waiting-time distributions through a maximum-priority process and Laplace–Stieltjes transforms. Accumulating priority is a continuous-time analogue of the discrete, rate-γ_i jockeying studied here; like the present work it is exact and single-server, but its object is the sojourn time rather than the bivariate queue-length PGF.

Priority queues with abandonment

Abandonment enters queueing theory through the Erlang-A ($M/M/c + M$) model, in which each waiting customer holds an independent exponential patience clock; the modern treatment and its heavy-traffic scaling are due to Garnett et al. [GMR02], and the single-server case $M/M/1 + M$ supplies the busy-period structure on which Model-C_2 rests (§3.3). The analytic signature of impatience is well documented: per-customer patience clocks make the transition rates state-inhomogeneous, so the generating-function solutions, where they close at all, close in terms of hypergeometric series.

Kapodistria [Kap11] exhibits this pattern at its sharpest in a single-server queue with *synchronised* abandonments —all waiting customers abandon simultaneously, each with probability p, at Poisson opportunity epochs— solving the stationary law exactly through basic (q-)hypergeometric series and recovering, in the no-synchronisation limit $p \to 0^+$, the $M/M/1+M$ queue-length PGF as a ratio of Kummer functions: precisely the special-function structure that carries the boundary function of Model-C_2.

Exact stationary results for priority queues *with* reneging are scarce and almost always multi-server. Jouini & Roubos [JR14] treat an $M/M/s+M$ queue with two non-preemptive classes and per-class exponential patience, deriving Laplace–Stieltjes transforms for conditional and unconditional waiting times together with the probabilities of service and of abandonment —the multi-server analogue of Model-C_2, but with the waiting time, not the queue-length PGF, as its object. Where exact transforms are unavailable the literature turns to fluid and diffusion approximations or to many-server asymptotics, in which the Erlang-A stationary law and its descendants are expressed through confluent hypergeometric and incomplete-gamma functions [ZM05]; the same special-function structure appears exactly, and at finite scale, in the closed form for Model-C_2.

The exact multi-server priority results that do exist carry no class-asymmetric reneging —those of Wang et al. [WBSW15] and Zuk & Kirszenblat [ZK23] are abandonment-free— so Model-C_2 appears to be the first exact, single-server, queue-length treatment of *class-asymmetric* abandonment, with impatience acting on the priority class alone. The clinical reading that licenses this asymmetry is long-standing: Zenios [Zen99] models the organ-transplant waiting list as a queue whose reneging is patient mortality, precisely the regime in which losing high-priority demand is the outcome of interest.

Jockeying and priority changes

The closest predecessor on the jockeying side is de Waal [dW92], who studies an $M/G/1$ priority queue in which impatient customers *change priority class* while waiting. Its entry in Table 2 is marked approximate not by choice of method but by necessity: with general service times the governing equations do not close, and the analysis delivers Laplace–Stieltjes-based *approximations* of the waiting-time distributions rather than an exact stationary law. The healthcare incarnation of the same idea is the model of Hu, Chan & Dong [HCD22], who let customers transition between a moderate and an urgent class as their condition deteriorates or improves and characterise a near-optimal $c\mu/\theta$-type scheduling index in heavy traffic; the system is multi-server and its results are asymptotic, so it too is approximate in the sense of Table 2. Against this backdrop Model-B_2 is the exact, single-server counterpart of the approximate multi-server "priority-change" literature: it retains the deterioration mechanism —a waiting class-1 customer relocating to the class-2 queue at rate γ_1— but pays for exactness with the single-server, Markovian, one-directional restriction, returning a closed-form boundary function in place of an approximation.

Multi-server extensions and performance design

The last reference bounds the present work from the side of scale. Wang et al. [WBSW15] give the exact queue-length PGF of an $M/M/c$ queue with two *preemptive* priority classes and class-dependent service rates, a useful contrast to the non-preemptive, common-rate discipline used throughout this thesis. That the result carries no departure mechanism is symptomatic of the multi-server literature as a whole: once $c > 1$ is combined with jockeying or abandonment, the analysis moves to approximations and asymptotics, confirming that exact, mechanism-rich analysis is essentially a single-server phenomenon.

Methodological lineage

The functional-equation technique used in Sections 7 and 8 belongs to the kernel-method and boundary-value tradition for two-dimensional Markovian queues. The systematic theory —reducing a bivariate balance equation to the curve on which its kernel vanishes, and from there to a boundary-value problem for the unknown boundary functions— is developed by Fayolle, Iasnogorodski & Malyshev [FIM99] for random walks in the quarter-plane. That programme still delivers explicit two-class results: Devos, Walraevens, Fiems & Bruneel [DWFB20] obtain the closed-form joint queue-length distribution of a discrete-time two-class queue with randomly alternating service by analysing the kernel of just such a functional equation and pinning its two unknown boundary functions through analytic continuation.

The pinning arguments of this thesis are the elementary, one-dimensional residue of that programme. In Model-B_2 the boundary function is fixed by demanding *analyticity* of the PGF at the interior point where the kernel root meets the diagonal (§7), and in Model-C_2 by demanding mere *boundedness* at the unit-disc boundary (§8); both select the analytic solution of the reduced equation, exactly the role the boundary conditions play in the general theory.

Two narrower tools recur. The confluent hypergeometric (Kummer) function that carries the Model-C_2 boundary function is the same special function through which the Erlang-A and reneging-queue stationary laws are expressed [ZM05, Kap11, DLM]; and the selection of the decaying solution of the three-term recurrence behind the partial-PGF analysis is governed by Pincherle's theorem, for which we cite the recurrence-relation treatment of Gautschi [Gau67] alongside the DLMF [DLM].

Position of this thesis

Table 2: Literature overview: models and mechanisms. ✓ = present, – = absent, ≈ = approximate only.

Reference	Servers	Non-preem.	Jockeying	Aband.	Queue-len.	Exact
Cohen (1956)	1	✓	–	–	✓	✓
de Waal (1992)	1	✓	✓	–	–	≈
Jouini & Roubos (2014)	c	✓	–	✓	–	✓
Wang et al. (2015)	c	–	–	–	✓	✓
Stanford et al. (2014)	1	✓	✓	–	–	✓
Zuk & Kirszenblat (2023)	c	✓	–	–	✓	✓
Hu, Chan & Dong (2022)	c	✓	✓	✓	–	≈
This thesis (Model-A)	**1**	✓	–	–	✓	✓
This thesis (Model-B_2)	**1**	✓	✓	–	✓	✓
This thesis (Model-C_2)	**1**	✓	–	✓	✓	✓

Table 2 places this work against its closest neighbours, and reading it by row makes the gap precise. The Cohen (1956) row is the single-server, exact, non-preemptive queue-length baseline with neither extra mechanism —the ancestor of Model-A. de Waal (1992) adds jockeying (priority change) at a single server, but only approximately and for waiting times rather than queue lengths. Jouini & Roubos (2014) add abandonment exactly, but at c servers and again for waiting times. Wang et al. (2015) is exact and queue-length-focused, yet multi-server, preemptive, and mechanism-free. Stanford et al. (2014) is single-server and exact with a dynamic-priority (jockeying-like) mechanism, but its object is the sojourn time. Zuk & Kirszenblat (2023) is the exact multi-server queue-length result whose $c = 1$ case is Model-A, with neither jockeying nor abandonment. Hu, Chan & Dong (2022) carries both mechanisms —jockeying-as-deterioration and abandonment— but multi-server and only approximately.

No row above combines, at a single server, the exact bivariate queue-length PGF with either jockeying or abandonment —which is precisely the column pattern of the three "This thesis" rows. The gap can therefore be stated in one sentence: prior to this work there is no exact, single-server, queue-length analysis that places non-preemptive priority together with jockeying or abandonment inside a single PGF framework.

The non-claims must be stated with equal precision. The framework is unified but not complete: the full model (Model-X, Remark 3), in which jockeying and abandonment act together, and the bidirectional-jockeying model (Model-B, §6) are left open, each reducing to a Volterra-type integral equation that this thesis characterises but does not solve in closed form. The exact contributions are confined to the three "This thesis" rows of Table 2 —the baseline Model-A, the one-way jockeying Model-B_2, and the class-1 abandonment Model-C_2, together with their head-of-line simplifications. The claim is exactness and unification at a single server with one mechanism active at a time; it is explicitly not a solution of the combined or bidirectional problems, which remain, as in the multi-server literature, beyond exact reach.

3 Preliminaries

This section collects the background material used throughout the derivations, so that the later arguments can stay focused on their primary objectives.

The central object throughout is the (P)PGF of the model in question, together with the system's idle probability for some corollaries.

3.1 $M/M/1$ queue

The canonical $M/M/1$ queue is the foundation of queueing theory and is treated in many textbooks; we follow [AR02]. Customers arrive according to a Poisson process —i.e. the inter-arrival times are exponentially distributed with parameter λ—, service times are exponentially distributed with parameter μ, and a single server operates in first-come-first-served (FCFS) —also known as first-in-first-out (FIFO)— order. The system is stable if and only if the load $\rho = \lambda/\mu < 1$.

Let π_n be the limiting probability of finding n customers in the system. Under stability, the detailed balance equations $\lambda\,\pi_n = \mu\,\pi_{n+1}$ $(n \geq 0)$ together with the normalisation $\sum_{n=0}^{\infty} \pi_n = 1$ give us the geometric steady-state distribution of the number of customers in the system:

$$\pi_n = (1-\rho)\,\rho^n, \quad n \geq 0.$$

By the PASTA property (§3.2), this also corresponds to the distribution seen by arriving customers. Let L be the number of customers in the system, including any in service. Its expectation is

$$\mathbb{E}[L] = \frac{\rho}{1-\rho}.$$

From this, the expected sojourn time W—the time between a customer's arrival and their departure after service—follows by Little's Law ($\mathbb{E}[L] = \lambda\mathbb{E}[W]$):

$$\mathbb{E}[W] = \frac{\mathbb{E}[L]}{\lambda} = \frac{1/\mu}{1-\rho}.$$

A less common but highly relevant concept is the Busy Period (BP): the time between a customer arriving to an idle system and the departure that next leaves it empty. Its Laplace–Stieltjes Transform (LST) and expectation are known:

$$\widetilde{BP}(s) = \frac{(\mu+\lambda+s) - \sqrt{(\mu+\lambda+s)^2 - 4\mu\lambda}}{2\lambda}, \tag{3.1}$$

$$\mathbb{E}[BP] = \frac{1}{\mu(1-\rho)}. \tag{3.2}$$

3.2 Non-preemptive priority and the PASTA property

Throughout this thesis the two customer classes are served under a *non-preemptive* priority discipline: whenever the server becomes free it admits a waiting class-1 customer if any are present, and a class-2 customer only otherwise, but a customer already in service is *never* interrupted—a class-1 arrival that finds a class-2 customer in service waits until that service completes. Priority therefore governs only the order in which the queues are drained, not the service in progress. Because the service rate μ is common to both classes, the *memorylessness* of the exponential service time makes the residual service of the customer in progress $\mathrm{Exp}(\mu)$ regardless of its class; hence the class of the in-service customer does not influence the future evolution of the queue lengths, which is why a single in-service customer of either class suffices to mark the server busy.

We also invoke repeatedly the *PASTA* property (*Poisson Arrivals See Time Averages*) [AR02]: if a system is fed by a Poisson arrival process that satisfies the *lack-of-anticipation assumption* (at each instant the future increments of the arrival process are independent of the current system state), then the long-run fraction of arrivals that find the system in a given state equals the stationary probability of that state, so the distribution seen by an arriving customer coincides with the time-stationary one. Both hypotheses hold here: the exogenous class-i arrivals are independent Poisson processes at rates λ_i, and they are independent of the system's internal exponential service, jockeying and abandonment clocks, so they cannot anticipate the future. By Poisson *superposition* the merged arrival stream is itself Poisson at rate $\lambda = \lambda_1 + \lambda_2$, so PASTA applies to the aggregate state; and since each class stream is an independent Poisson process—equivalently, an i.i.d. λ_i/λ *thinning* of the merged stream—PASTA applies to each class separately.

3.3 $M/M/1{+}M$ queue (Erlang-A)

Another model relevant to this thesis is the $M/M/1{+}M$ queue, also known as the *Erlang-A* model. It extends the $M/M/1$ queue —Poisson arrivals at rate λ, exponential service at rate μ— with customer impatience: each *waiting* customer (the one in service is not impatient) independently abandons after an exponentially distributed patience time with rate θ. Crucially, unlike the $M/M/1$ it does not require $\rho = \lambda/\mu < 1$ for stability: it is positive recurrent for all $\lambda, \mu, \theta > 0$.

The steady-state distribution follows from its balance equations, where n denotes the number of customers in the *system*:

$$\lambda\,\pi_n = (\mu + n\theta)\,\pi_{n+1}, \qquad n \geq 0.$$

Iterating yields

$$\pi_n = \pi_0\,\frac{(\lambda/\theta)^n}{(\mu/\theta)^{(n)}} \quad n \geq 0; \quad \text{where} \quad a^{(n)} = a(a+1)\cdots(a+n-1)$$

Normalisation gives the non-trivial idle probability $\pi_0 = [{}_1F_1(1;\,\mu/\theta;\,\lambda/\theta)]^{-1}$ (see §3.5 for the Kummer function ${}_1F_1$), and summing the geometric-like terms shows that the queue-length PGF is correspondingly a *ratio* of Kummer functions, $\Pi(z) = {}_1F_1(1;\,\mu/\theta;\,(\lambda/\theta)z)\,/\,{}_1F_1(1;\,\mu/\theta;\,\lambda/\theta)$. The same closed form is derived by Kapodistria [Kap11, Thm. 3] as the no-synchronisation limit of a queue with synchronised abandonments, there in the sojourn-time-impatience convention (the customer in service also abandons), which amounts to replacing μ by $\mu+\theta$.

Unlike the textbook facts collected above, the busy-period identity that follows is the one preliminary we derive from scratch, as Model-C_2 rests on it directly. The *Busy Period* of this $M/M/1+M$ is far more complex than its $M/M/1$ counterpart. We record it for the class-1 subsystem used in Model-C_2, with arrival rate λ_1, service rate μ and abandonment rate θ_1, so that the relevant idle probability is $\pi_0 = [{}_1F_1(1;\,\mu/\theta_1;\,\lambda_1/\theta_1)]^{-1}$. Writing B_C for this busy period, its LST and expectation are

$$\widetilde{B}_C(s) = \cfrac{\mu}{\lambda_1 + \mu + s - \cfrac{\lambda_1(\mu+\theta_1)}{\lambda_1 + \mu + \theta_1 + s - \cfrac{\lambda_1(\mu+2\theta_1)}{\lambda_1+\mu+2\theta_1+s-\cdots}}}, \tag{3.3}$$

$$\mathbb{E}[B_C] = \frac{1}{\lambda_1}\Big({}_1F_1\big(1;\,\mu/\theta_1;\,\lambda_1/\theta_1\big) - 1\Big) \tag{3.4}$$

The continued fraction (3.3) is generated by a first-passage recursion. Let $\phi_n(s)$ denote the LST of the first-passage (sub-busy-period) time from n customers in the system down to $n-1$. Conditioning on the first transition out of state n—an arrival (rate λ_1) to state $n+1$, or a departure by service completion or abandonment (combined rate $\mu + (n-1)\theta_1$: one customer in service plus $n-1$ waiting, each impatient at rate θ_1) to state $n-1$—the strong Markov property gives

$$\phi_n(s) \;=\; \frac{\mu + (n-1)\theta_1}{\lambda_1 + \mu + (n-1)\theta_1 + s - \lambda_1\,\phi_{n+1}(s)}, \qquad n \geq 1, \qquad \widetilde{B}_C(s) = \phi_1(s). \tag{3.5}$$

Unrolling Equation (3.5) from $n = 1$ reproduces the continued fraction (3.3) term by term. The continued fraction also sums in closed form, and the argument is short enough to give in full. Fix $s > 0$ and scale the rates by θ_1, setting

$$a = \frac{\mu}{\theta_1}, \qquad b = \frac{s}{\theta_1}, \qquad c = \frac{\lambda_1}{\theta_1},$$

so that Equation (3.5) reads $\phi_n(s) = (a+n-1)/\big((a+n-1) + b + c - c\,\phi_{n+1}(s)\big)$. Consider the integrals

$$I(p) \;=\; \int_0^1 t^{\,p}\,(1-t)^{\,b-1}\,e^{-ct}\,dt, \qquad p > -1,$$

which by the Euler representation (3.10) are Beta-function multiples of Kummer values, $I(p) = B(p+1,\,b)\,{}_1F_1(p+1;\,p+1+b;\,-c)$. Integrating the exact derivative $\frac{d}{dt}\big[t^{\,p}(1-t)^{\,b}e^{-ct}\big]$ over $[0,1]$ —the boundary terms vanish for $p, b > 0$— and splitting $(1-t)^b = (1-t)^{b-1} - t\,(1-t)^{b-1}$ in the two terms where it appears yields the three-term recurrence

$$(p+b+c)\,I(p) \;=\; p\,I(p-1) \;+\; c\,I(p+1), \tag{3.6}$$

a relation of contiguous type for ${}_1F_1$ (obtainable equivalently by combining the contiguous relations of [DLM, §13.3]). Dividing Equation (3.6) at $p = a+n-1$ by $I(a+n-2)$ shows that the ratios $R_n := I(a+n-1)/I(a+n-2)$ satisfy exactly the recursion of ϕ_n. Two further facts pin the identification. First, by Laplace's method at the endpoint $t=1$, $I(p) \sim e^{-c}\,\Gamma(b)\,p^{-b}$ as $p \to \infty$: the solution $\{I(p)\}$ decays only polynomially, whereas the dominant solution of Equation (3.6) grows superexponentially (its consecutive ratio grows like p/c), so $\{I(p)\}$ is the *minimal (recessive)* solution. Second, by *Pincherle's theorem* the convergent continued fraction equals precisely the ratio built from the minimal solution. Hence

$$\widetilde{B}_C(s) \;=\; \phi_1(s) \;=\; \frac{I(a)}{I(a-1)} \;=\; \frac{B(a+1,\,b)\;{}_1F_1\big(a+1;\,a+1+b;\,-c\big)}{B(a,\,b)\;{}_1F_1\big(a;\,a+b;\,-c\big)}, \tag{3.7}$$

a contiguous ratio of the form (3.11). This identity is used, with $s = \lambda_2(1-y)$, in the boundedness route for Model-C_2 (§8, Equation (8.11)).

3.4 Pollaczek–Khinchine formula

The $M/G/1$ queue [AR02] is a single-server queue with Poisson arrivals at rate λ and generally distributed i.i.d. service times B with LST $\widetilde{B}(s) = \mathbb{E}[e^{-sB}]$. For stability purposes, the system load $\rho_B = \lambda\,\mathbb{E}[B] < 1$ is required (written ρ_B to keep the symbol ρ for the two-class load of this thesis). Under these two conditions —Poisson arrivals and stability— the Pollaczek–Khinchine (PK) formula gives the PGF of the stationary number of customers in the system:

$$L(z) \;=\; \frac{(1-\rho_B)(1-z)\,\widetilde{B}(\lambda(1-z))}{\widetilde{B}(\lambda(1-z)) - z}. \tag{3.8}$$

3.5 Confluent hypergeometric function

The *Kummer confluent hypergeometric function of the first kind* appears in several derivations in this thesis; it reads as follows:

$${}_1F_1(a;\,b;\,z) \;=\; \sum_{n=0}^{\infty} \frac{a^{(n)}}{b^{(n)}\,n!}\,z^n, \qquad a^{(n)} = \prod_{k=0}^{n-1}(a+k), \quad a^{(0)} = 1, \tag{3.9}$$

where b is not 0 or a negative integer —i.e. $b \notin \{0, -1, -2, \dots\}$— and the series converges for all $z \in \mathbb{C}$. The function ${}_1F_1(a;\,b;\,z)$ is the unique solution of *Kummer's equation* [DLM]:

$$z\,u'' + (b-z)\,u' - a\,u = 0,$$

normalised by ${}_1F_1(a;\,b;\,0) = 1$.

For this thesis, two identities related to it are used. First, the *Euler integral representation*, for $a, b > 0$ we have:

$${}_1F_1(a;\,a+b;\,z) \;=\; \frac{1}{B(a,b)} \int_0^1 t^{a-1}(1-t)^{b-1} e^{zt}\,dt, \tag{3.10}$$

where B is the *Beta function* $B(a,b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$, and Γ is the *Gamma function*. Secondly, *contiguous ratios*: ratios of two ${}_1F_1$ values whose parameters differ by integer shifts. Two patterns arise in this thesis. The *simultaneous shift*

$$\frac{{}_1F_1(a+1;\,a+1+b;\,z)}{{}_1F_1(a;\,a+b;\,z)}, \tag{3.11}$$

in which both parameters move by one so that their difference b is preserved, is by the Euler representation (3.10) a ratio of two weighted Beta integrals differing only in the exponent of t; it appears as the closed-form value of the Erlang-A busy-period continued fraction (§3.3, Equation (3.7)) and hence in the boundary function of Model-C_2. The *first-parameter shift* ${}_1F_1(a+1;\,b;\,z)\big/{}_1F_1(a;\,b;\,z)$, with the second parameter fixed, appears instead in the boundary function of Model-B_2 (§7).

3.6 First-order linear ODEs and PDEs

Since jockeying and abandonment add state-dependent rates to the balance equations, derivative terms appear in the derivations, requiring us to solve first-order linear ODEs and PDEs for the PGF. The three methods below are invoked at specific steps of the analysis.

3.6.1 Integrating factor

This method is standard to solve —for each fixed y— a *first-order linear ODE in x*. Such an equation has standard form

$$\frac{dP}{dx} + p(x;\, y)\, P \;=\; q(x;\, y). \tag{3.12}$$

The *integrating factor* $\mu_I(x; y) = \exp\!\big(\int_0^x p(t; y)\, dt\big)$ converts Equation (3.12) to $\frac{d}{dx}[P \cdot \mu_I] = q \cdot \mu_I$. Integrating from 0 to x gives the general solution

$$P(x, y) \;=\; \frac{1}{\mu_I(x;\, y)} \left[C_0(y) + \int_0^x q(t;\, y)\, \mu_I(t;\, y)\, dt \right], \tag{3.13}$$

where $C_0(y)$ is an arbitrary function of y, determined by a boundary condition.

3.6.2 Method of Characteristics

This method is standard for solving a *first-order linear PDE in both x and y* simultaneously. It reduces the PDE to a family of ODEs along the so-called *characteristic curves* in the (x, y)-plane; along each curve, the PDE collapses to a single-variable ODE.

A PDE of the form

$$a\frac{\partial P}{\partial x} + b\frac{\partial P}{\partial y} = cP + d \tag{3.14}$$

is solved by tracing the characteristic curve $(x(t), y(t))$ satisfying

$$\frac{dx}{dt} = a, \qquad \frac{dy}{dt} = b. \tag{3.15}$$

By the chain rule, $\frac{d}{dt}P(x(t), y(t)) = a\, P_x + b\, P_y$, so Equation (3.14) reduces to

$$\frac{dP}{dt} = cP + d, \tag{3.16}$$

a first-order linear ODE in t. Its homogeneous part $\frac{dP}{dt} = cP$ is solved at once by separation of variables; the inhomogeneous term d is then absorbed either by the integrating factor (§3.6.1) or, once that homogeneous solution is in hand, by variation of parameters (§3.6.3). When a and b are constants, the characteristics are straight lines and the quantity

$$\xi \;:=\; b\, x - a\, y$$

is preserved along each curve —it is the *first integral* that labels each characteristic. The integration constant of Equation (3.16) therefore varies between characteristics: it is an arbitrary function $F(\xi)$. Re-expressing the solution in the original variables (x, y) via Equation (3.15) gives the general solution of Equation (3.14). In the PGF setting, F is identified with a boundary generating function, and analyticity of $P(x, y)$ for $|x|, |y| \leq 1$ uniquely determines it.

3.6.3 Variation of parameters

The integrating factor of §3.6.1 builds a solution from scratch by constructing μ_I. Frequently, however, we *already* hold a nonzero solution of the homogeneous equation —most importantly, the one the Method of Characteristics (§3.6.2) produces along each characteristic curve. *Variation of parameters* is the technique that turns any such known homogeneous solution directly into the general solution of the inhomogeneous equation, with no further integrating factor required. We describe it in the form in which it is used later.

Consider a first-order linear ODE written as

$$\frac{dP}{dx} = f(x)\, P + h(x), \tag{3.17}$$

and suppose a nonzero homogeneous solution P_h is known, that is, $\frac{dP_h}{dx} = f(x)\, P_h$. The homogeneous equation then has general solution $C\, P_h(x)$ for an arbitrary constant C. To solve the full equation (3.17), we *vary the constant*: we promote C to a function $C(x)$ and look for a solution of the form

$$P(x) = C(x)\, P_h(x).$$

Differentiating by the product rule and using $P_h' = f\,P_h$,

$$\begin{aligned}\frac{dP}{dx} &= C'(x)\,P_h(x) + C(x)\,P_h'(x)\\ &= C'(x)\,P_h(x) + f(x)\,\big[C(x)\,P_h(x)\big] = C'(x)\,P_h(x) + f(x)\,P(x).\end{aligned}$$

Substituting this back into Equation (3.17), the common term $f(x)\,P(x)$ cancels from both sides and only the derivative of the varied constant survives:

$$C'(x)\,P_h(x) = h(x) \qquad \Longrightarrow \qquad C'(x) = \frac{h(x)}{P_h(x)}.$$

Integrating from a base point x_0 gives $C(x) = F + \int_{x_0}^{x} h(t)/P_h(t)\,dt$ with F an arbitrary constant, and hence the general solution

$$P(x) = P_h(x)\left[F + \int_{x_0}^{x} \frac{h(t)}{P_h(t)}\,dt\right]. \tag{3.18}$$

The known homogeneous solution P_h thus does double duty: it is the overall prefactor and, as $1/P_h$, the weight inside the integral. The constant F is fixed by a boundary condition. Taking in particular $P_h = 1/\mu_I$ reproduces the integrating-factor formula (3.13) of §3.6.1.

This is exactly the step that completes the Method of Characteristics for an *inhomogeneous* PDE. Once §3.6.2 has reduced the PDE to an ODE of the form (3.17) along a characteristic and supplied its homogeneous solution P_h, formula (3.18) delivers the full solution along that characteristic, the constant F becoming an arbitrary function $F(\xi)$ of the characteristic label. Model-B (§6) is solved by precisely this pairing.

4 The general model (Model-X) and its fundamental equation

This section defines the general two-class system with both jockeying (γ_i) and abandonment (θ_i); we call this full model *Model-X*. The simpler variants are recovered as special cases by zeroing the appropriate parameters, as summarised in Table 3:

Model	Jockeying	Abandonment	Parameter restriction
X	both $(1 \leftrightarrow 2)$	both $(1,2)$	$\gamma_i \geq 0,\ \theta_i \geq 0$ (general)
A	none	none	$\gamma_1 = \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
B	both $(1 \leftrightarrow 2)$	none	$\gamma_i \geq 0,\ \theta_1 = \theta_2 = 0$
B_2	$1 \to 2$ only	none	$\gamma_1 \geq 0,\ \gamma_2 = 0,\ \theta_1 = \theta_2 = 0$
C_2	none	class 1 only	$\theta_1 \geq 0,\ \theta_2 = 0,\ \gamma_1 = \gamma_2 = 0$
B_2^{H}	$1 \to 2$, head-of-line	none	$\gamma_1\,\mathbf{1}_{\{n_1 \geq 1\}} > 0,\ \gamma_2 = \theta_1 = \theta_2 = 0$
C_2^{H}	none	class 1, head-of-line	$\theta_1\,\mathbf{1}_{\{n_1 \geq 1\}} > 0,\ \gamma_1 = \gamma_2 = \theta_2 = 0$

Table 3: Nested family of models. Model-X is the parent; Models A, B, B_2 and C_2 are obtained by restricting the jockeying directions and the abandoning classes. The two head-of-line variants B_2^{H} and C_2^{H} are listed separately, as their zeroing mechanism involves $\mathbf{1}_{\{n_1 \geq 1\}}$ rather than being proportional to the queue length.†

Figure 1 gives a visual overview of Model-X and each variant we consider. It omits the two head-of-line variants, which are less conventional and for which such a diagram would add little:

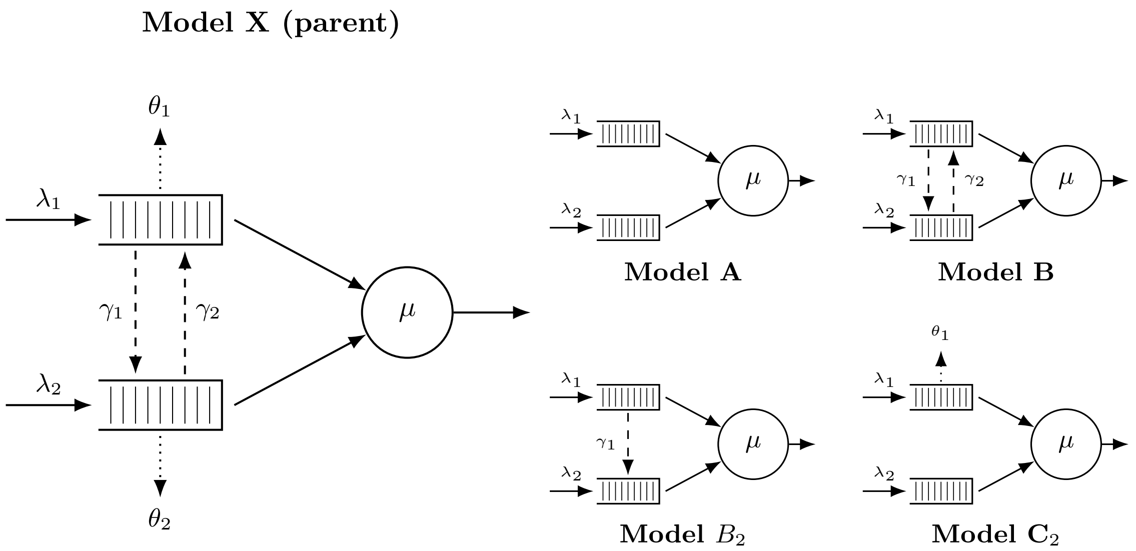

Figure 1: Physical queue layout for Model-X (left) and its four specialisations (right, 2×2 grid). Queue 1 (top buffer) has non-preemptive priority over Queue 2. Solid arrows are for Poisson arrivals (λ_i) and exponential service (μ); dashed arrows are for jockeying(γ_i); and dotted arrows are for abandonments (θ_i)

The model in this section is an $M/M/1$ queue with two priority classes. Customers are served First-Come, First-Served (FCFS) within their own class, and class-1 (the priority class) has non-preemptive priority over class-2: a service in progress is never interrupted, so a class-2 customer in service is not displaced by an arriving class-1 customer. Beyond these basic dynamics, Model-X adds two queue-leaving mechanisms. First, *jockeying*: a waiting customer leaves its own queue to *join the other queue*, at per-customer rate γ_1 for direction $1 \to 2$ and γ_2 for $2 \to 1$, so the total number of customers is preserved. Second, *abandonment* (reneging): a waiting customer leaves the *system* at per-customer rate θ_i, decreasing the total by one. Concretely, each waiting class-i customer (not the one in service) holds an independent $\mathrm{Exp}(\theta_i)$ patience clock and abandons when it expires, so the aggregate class-i abandonment rate in a state with n_i waiting customers is the linear term $\theta_i n_i$ in the balance equations. This is precisely the linear-reneging mechanism of the $M/M/1+M$ queue from §3.3.

In Kendall's notation this is an $M/M/1$-type model: Poisson arrivals at rate λ_i for class-i, exponentially distributed service times at rate μ, and a single server. To define the stochastic processes that drive this system, we introduce the following random variables:

- $\mathcal{B}$: binary indicator for the server being idle or busy, $\mathcal{B} \in \{0, 1\}$;
- N_1, N_2 : number of customers of class 1 and 2, respectively, in their respective queues;
- N : total number of customers in the queues.

Note that N_1, N_2 and N count customers in the *queues*, not the *system*. Since the service rate is common to both classes, we write the idle state as (0) and all other states as (n_1, n_2), so $(0, 0)$ means someone is in service and the queues are empty. Alternatively, an auxiliary state space $\widetilde{S}$ tracks only n_2 and the total $n = n_1 + n_2$, subject to $n \geq n_2 \geq 0$. Formally:

$$\begin{aligned} S &:= \{(0)\} \cup \{(n_1, n_2) \mid n_1, n_2 \geq 0\}, \\ \widetilde{S} &:= \{(0)\} \cup \{(n_2, n) \mid n \geq n_2 \geq 0\}. \end{aligned} \tag{4.1}$$

We define the long-run probabilities of finding the system in each state:

- π_0 for the idle state (0), it belongs to both spaces S and $\widetilde{S}$;
- $\pi(n_1, n_2)$ for states $(n_1, n_2) \in S$, where $n_1, n_2 \geq 0$;
- $\widetilde{\pi}(n_2, n)$ for states $(n_2, n) \in \widetilde{S}$, where $n \geq n_2 \geq 0$.

Arrivals are Poisson for each class and service times exponential (common across classes), with the following notation:

- λ_1, λ_2: arrival rates of classes 1 and 2, respectively;
- μ: service rate of a customer, regardless of class;
- $\rho_i = \frac{\lambda_i}{\mu}$: system load of class $i \in \{1, 2\}$;
- γ_1, γ_2: per-customer jockeying rates from Class $1 \to 2$ and $2 \to 1$, respectively; jockeying transfers a customer between queues, preserving the total number of customers;
- θ_1, θ_2: per-customer abandonment (reneging) rates from classes 1 and 2, respectively; abandonment removes a customer from the system, reducing the total n by one.

For the aggregate system parameters, we define:

- $\lambda = \lambda_1 + \lambda_2$: total arrival rate;
- $\rho = \frac{\lambda}{\mu}$: total system load.

Stability. Jockeying does not affect stability (customers are merely relocated). Model-X has abandonments from both classes, so the departure rate $\mu + \theta_1 n_1 + \theta_2 n_2$ grows unbounded, ensuring positive recurrence for all loads. In contrast, models without abandonments require $\rho < 1$ (they behave like M/M/1). Mixed cases, such as Model-C_2, are discussed separately (§8).

Computation of π_0 and $\pi(0,0)$. The two empty-state probabilities are pinned, for the full Model-X, by two exact identities that hold irrespective of the jockeying and abandonment mechanisms: a cut between the idle and busy states, and a renewal–reward identity over busy cycles. Together they reduce the determination of π_0 and $\pi(0,0)$ to a single scalar, the mean busy period $\mathbb{E}[B]$.

Theorem 1 (Empty-state probabilities of the general model)**.** *Assume the CTMC of Model-X is positive recurrent, so that the stationary distribution exists. Let $\lambda = \lambda_1 + \lambda_2$ and let $\mathbb{E}[B]$ denote the mean length of a server busy period —the time from an arrival that finds the system empty until the system is next empty. Then*

$$\pi(0,0) \;=\; \rho\, \pi_0, \tag{4.2}$$

and

$$\pi_0 \;=\; \frac{1}{1 + \lambda\, \mathbb{E}[B]}, \qquad \pi(0,0) \;=\; \frac{\rho}{1 + \lambda\, \mathbb{E}[B]}. \tag{4.3}$$

Proof. Cut identity. Consider the cut separating the idle state (0) from the set of busy states. The only transitions out of (0) are the class-1 and class-2 arrivals, at total rate λ; the only transition into (0) is a service completion from $(0,0)$, at rate μ —from any state with a waiting customer a departure leaves the server busy, and neither jockeying nor abandonment can act when both queues are empty. Equating the probability flux across the cut gives $\lambda\, \pi_0 = \mu\, \pi(0,0)$, which is Equation (4.2).

Renewal–reward identity. The idle state is a regeneration point: by the memorylessness of the exponential inter-arrival and service times (the strong Markov property), the evolution after each visit to (0) is an independent probabilistic replica. Successive cycles therefore alternate an idle period of mean $1/\lambda$ (the wait for the next arrival, exponential at rate λ) and a busy period of mean $\mathbb{E}[B]$. By the renewal–reward theorem, the stationary probability of the idle state equals the long-run fraction of time spent idle,

$$\pi_0 \;=\; \frac{1/\lambda}{1/\lambda + \mathbb{E}[B]} \;=\; \frac{1}{1 + \lambda\, \mathbb{E}[B]},$$

and substituting into Equation (4.2) yields the stated $\pi(0,0)$. □

Both identities are exact for the full model. They reduce the problem to the mean busy period $\mathbb{E}[B]$, which is elementary when there is no abandonment and is computed model by model once abandonment is present; the resulting π_0 and $\pi(0,0)$ for each tractable model then follow as corollaries (Corollary 1 for the no-abandonment family; §8 and §10 for the abandonment variants). For the full asymmetric Model-X, $\mathbb{E}[B]$ itself has no closed form (Remark 3).

4.1 Balance equations

This subsection presents the state-transition diagrams of Model-X on both state spaces S and $\widetilde{S}$, and derives the corresponding sets of balance equations.

4.1.1 Balance equations for state space S

The state-transition diagram of Model-X on S is given in Figure 2.

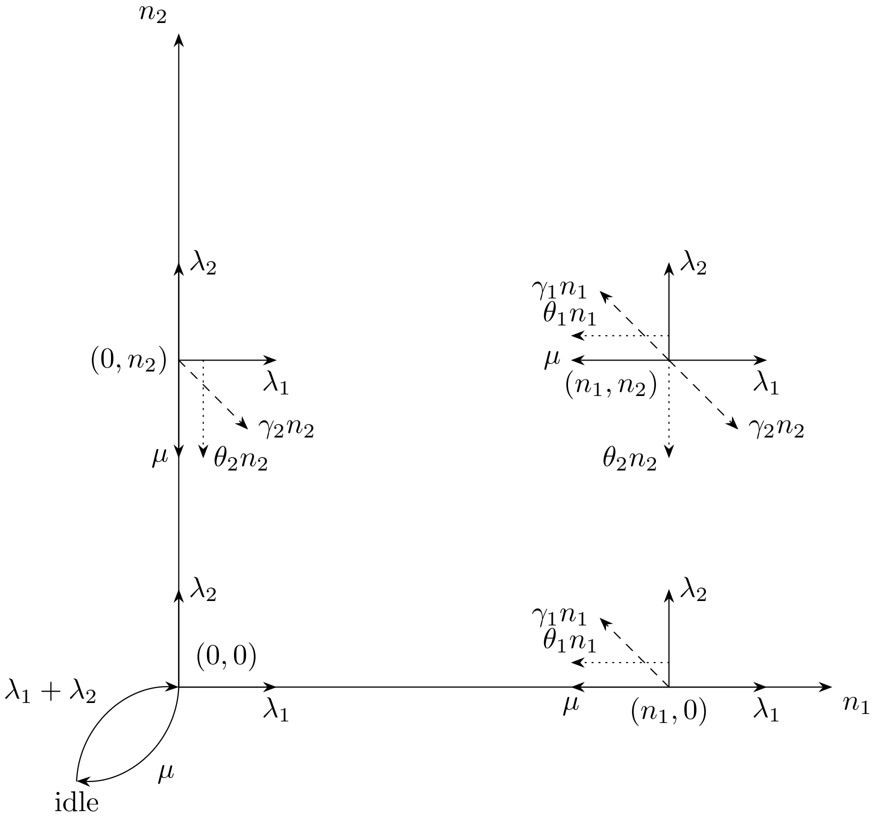

Figure 2: State transition diagram on state space S for the model with jockeying ($\gamma_i n_i$, dashed) and abandonments ($\theta_i n_i$, dotted).

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1) n_1 + (\gamma_2 + \theta_2) n_2]\, \pi(n_1, n_2) \\
&\quad = \quad \mu \pi(n_1 + 1, n_2) \\
&\qquad + \lambda_1 \pi(n_1 - 1, n_2) \\
&\qquad + \lambda_2 \pi(n_1, n_2 - 1) \\
&\qquad + \gamma_1 (n_1 + 1) \pi(n_1 + 1, n_2 - 1) \\
&\qquad + \gamma_2 (n_2 + 1) \pi(n_1 - 1, n_2 + 1) \\
&\qquad + \theta_1 (n_1 + 1) \pi(n_1 + 1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(n_1, n_2 + 1)
\end{aligned}
\qquad \text{for} \quad n_1 \geq 1, n_2 \geq 1 \tag{4.4}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2) n_2]\, \pi(0, n_2) \\
&\quad = \quad \lambda_2 \pi(0, n_2 - 1) \\
&\qquad + \mu \pi(1, n_2) \\
&\qquad + \mu \pi(0, n_2 + 1) \\
&\qquad + \gamma_1 \pi(1, n_2 - 1) \\
&\qquad + \theta_1 \pi(1, n_2) \\
&\qquad + \theta_2 (n_2 + 1) \pi(0, n_2 + 1)
\end{aligned}
\qquad \text{for} \quad n_1 = 0, n_2 \geq 1 \tag{4.5}
$$

$$
\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1) n_1]\, \pi(n_1, 0) \\
&\quad = \quad \lambda_1 \pi(n_1 - 1, 0) \\
&\qquad + \mu \pi(n_1 + 1, 0) \\
&\qquad + \gamma_2 \pi(n_1 - 1, 1) \\
&\qquad + \theta_1 (n_1 + 1) \pi(n_1 + 1, 0) \\
&\qquad + \theta_2 \pi(n_1, 1)
\end{aligned}
\qquad \text{for} \quad n_1 \geq 1, n_2 = 0 \tag{4.6}
$$

$$[\lambda_1 + \lambda_2]\,\pi(0,0) = (\mu + \theta_1)\pi(1,0) + (\mu + \theta_2)\pi(0,1) \tag{4.7}$$

$$(\lambda_1 + \lambda_2)\pi_0 = \mu\pi(0,0) \tag{4.8}$$

4.1.2 Balance equations for state space $\widetilde{S}$

Recall the state space $\widetilde{S}$ defined in Equation (4.1).

For readability we denote its limiting probabilities by $\widetilde{\pi}(n_2, n)$, for $(n_2, n) \in \widetilde{S}$. Figure 3 shows the corresponding state transition diagram.

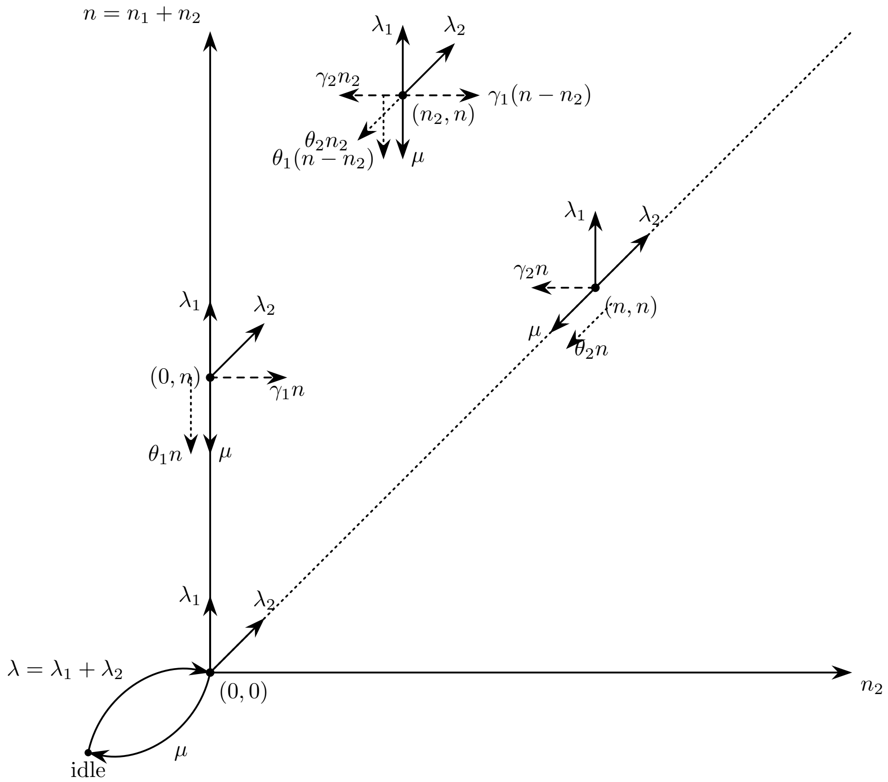

Figure 3: State transition diagram on state space $\widetilde{S}$.

The balance equations, after combining the equations where $\widetilde{\pi}_0$ shows up, are:

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)(n - n_2) + (\gamma_2 + \theta_2)n_2]\,\widetilde{\pi}(n_2, n) \\
&\quad = \quad \lambda_1 \widetilde{\pi}(n_2, n-1) \\
&\qquad + \lambda_2 \widetilde{\pi}(n_2 - 1, n-1) \\
&\qquad + \mu \widetilde{\pi}(n_2, n+1) \\
&\qquad + \gamma_1(n - n_2 + 1)\widetilde{\pi}(n_2 - 1, n) \\
&\qquad + \gamma_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n) \\
&\qquad + \theta_1(n + 1 - n_2)\widetilde{\pi}(n_2, n+1) \\
&\qquad + \theta_2(n_2 + 1)\widetilde{\pi}(n_2 + 1, n+1)
\end{aligned} \qquad \text{for} \quad 1 \le n_2 \le n-1,\ n \ge 2 \tag{4.9}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_2 + \theta_2)n]\,\widetilde{\pi}(n, n) \\
&\quad = \quad \lambda_2 \widetilde{\pi}(n-1, n-1) \\
&\qquad + \mu \widetilde{\pi}(n+1, n+1) \\
&\qquad + \mu \widetilde{\pi}(n, n+1) \\
&\qquad + \gamma_1 \widetilde{\pi}(n-1, n) \\
&\qquad + \theta_1 \widetilde{\pi}(n, n+1) \\
&\qquad + \theta_2(n+1)\widetilde{\pi}(n+1, n+1)
\end{aligned} \qquad \text{for} \quad n_2 = n,\ n \ge 1 \tag{4.10}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + (\gamma_1 + \theta_1)n]\,\widetilde{\pi}(0, n) \\
&\quad = \quad \lambda_1 \widetilde{\pi}(0, n-1) \\
&\qquad + \mu \widetilde{\pi}(0, n+1) \\
&\qquad + \gamma_2 \widetilde{\pi}(1, n) \\
&\qquad + \theta_1 (n+1) \widetilde{\pi}(0, n+1) \\
&\qquad + \theta_2 \widetilde{\pi}(1, n+1)
\end{aligned} \qquad \text{for} \quad n_2 = 0,\ n \geq 1 \tag{4.11}$$

$$\begin{aligned}
&[\lambda_1 + \lambda_2]\,\widetilde{\pi}(0, 0) \\
&\quad = \quad (\mu + \theta_1)\widetilde{\pi}(0, 1) + (\mu + \theta_2)\widetilde{\pi}(1, 1)
\end{aligned} \tag{4.12}$$

$$(\lambda_1 + \lambda_2)\widetilde{\pi}_0 = \mu \widetilde{\pi}(0, 0) \tag{4.13}$$

4.2 Derivation of equations for (Partial) Probability Generating functions

To study the limiting behaviour of this two-class system, we define the bivariate Probability Generating Function (PGF) of the joint queue lengths N_1 and N_2:

$$P(x, y) = \mathbb{E}[x^{N_1} y^{N_2}] = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2) x^{n_1} y^{n_2}.$$

The derivation also uses the boundary terms, where at least one queue is empty:

$$P_x(x) = P(x, 0) = \sum_{n_1=0}^{\infty} \pi(n_1, 0) x^{n_1}$$

$$P_y(y) = P(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2}.$$

Note that both terms include the busy-but-empty state $\pi(0, 0)$, whereas the fully idle state is treated separately.

A further term arises in the balance equations, describing the boundary dynamics when the class-1 queue holds exactly one waiting customer:

$$P_1(y) = \sum_{n_2=1}^{\infty} \pi(1, n_2) y^{n_2}.$$

This is an intermediate term; as the derivation proceeds, it will be expressed in terms of the primary boundary functions $P_x(x)$ and $P_y(y)$.

In the same fashion, we introduce the **Partial Probability Generating Function (PPGF)**. This technique, notably used by Cohen [Coh56], transforms only a subset of the random variables while keeping the discrete indices of the rest. For a priority queue it is particularly advantageous: it lets us exploit the well-known properties of the priority class (often a standard $M/M/1$ system) and concentrate the analytical effort on the non-trivial PGF of the non-priority class.

Cohen developed this framework for multi-server systems; we apply it here to the single-server case. To avoid confusion from mid-20th-century notation, we give a modern formulation consistent with the variables defined above.

We define the PPGF with respect to N_2, conditioned on the first queue being in state n_1:

$$\widetilde{P}(n_1, y) = \mathbb{E}\left[y^{N_2} \nVdash_{\{N_1 = n_1\}}\right] = \sum_{n_2=0}^{\infty} \pi(n_1, n_2) y^{n_2}. \tag{4.14}$$

Similarly, the transform with respect to N_1, keeping the second queue in discrete state n_2, is:

$$\widetilde{P}(x, n_2) = \mathbb{E}\left[x^{N_1} \nVdash_{\{N_2 = n_2\}}\right] = \sum_{n_1=0}^{\infty} \pi(n_1, n_2) x^{n_1}.$$

These partial transforms relate to the joint PGF by:

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y) x^{n_1} = \sum_{n_2=0}^{\infty} \widetilde{P}(x, n_2) y^{n_2}.$$

4.2.1 Equation for PGF in state space S

Lemma 1 (Fundamental equation for the PGF on S)**.** *Consider the two-class non-preemptive priority queue on the state space $S = \{(n_1, n_2) : n_1, n_2 \geq 0\}$, with Poisson arrival rates λ_1, λ_2, exponential service rate μ, jockeying rates γ_1, γ_2 and abandonment rates θ_1, θ_2, where n_1 and n_2 count the class-*1 *and class-*2 *customers* waiting *in the queues. Assume the associated CTMC is positive recurrent, so that the stationary distribution $\{\pi(n_1, n_2)\}$ exists and the joint PGF*

$$P(x, y) = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \pi(n_1, n_2)\, x^{n_1} y^{n_2}$$

is analytic on the closed unit polydisc $|x| \leq 1,\ |y| \leq 1$. Then $P(x, y)$, together with the boundary function $P_y(y) = P(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2) y^{n_2}$, satisfies the first-order linear partial differential equation

$$\begin{aligned}
&\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) \\
&+ xy \left[\gamma_1(x - y) + \theta_1(x - 1)\right] \frac{\partial}{\partial x} P(x, y) \\
&+ xy \left[\gamma_2(y - x) + \theta_2(y - 1)\right] \frac{\partial}{\partial y} P(x, y) \\
&\quad = \quad \mu(x - y)P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{4.15}$$

Remark 1 (Reading the fundamental equation). Equation (4.15) repays a moment's mechanical reading, since its structure dictates the entire analysis that follows. The bracketed algebraic coefficient multiplying $P(x, y)$ is exactly the kernel of the baseline Model-A: the flux balance between the arrival contributions $\lambda_1 x^2 y$ and $\lambda_2 x y^2$, the service contribution μy, and the total outflow $(\lambda_1 + \lambda_2 + \mu)xy$.

The two convection coefficients carry the added mechanisms. In the x-convection coefficient $xy\,[\gamma_1(x - y) + \theta_1(x - 1)]$, the jockeying factor $(x - y)$ vanishes on the diagonal $x = y$ —the analytic signature that jockeying merely relocates a customer and so conserves the total count— while the abandonment factor $(x - 1)$ vanishes at $x = 1$, the signature that abandonment respects normalisation ($P(1, 1) = 1$) yet removes a customer and thereby destroys conservation. The y-convection coefficient $xy\,[\gamma_2(y-x)+\theta_2(y-1)]$ states the symmetric facts for the class-2 mechanisms, with $(y - x)$ vanishing on the diagonal and $(y - 1)$ at $y = 1$.

The right-hand side carries the two boundary unknowns that any solution must pin down: the boundary function $P_y(y) = P(0, y)$, the generating function of the states with an empty priority queue, and the scalar $\pi(0, 0)$, the probability that the server is busy with both queues empty.

The position of these vanishing factors fixes, model by model, where the operative singularity sits and hence which technique applies. With both convection terms absent (Model-A, §5) the equation is algebraic and $P_y(y)$ is recovered from the kernel root $x^*(y)$ inside the unit disc. With one-way jockeying alone (Model-B_2, §7) the factor $(x - y)$ places the singularity at the interior point $x = y$ with a negative exponent, and $P_y(y)$ is pinned by analyticity there. With class-1 abandonment alone (Model-C_2, §8) the factor $(x - 1)$ places it on the boundary $x = 1$ with a positive exponent, and $P_y(y)$ is pinned by mere boundedness. Restricting a mechanism to the head of the queue removes the derivative outright, collapsing the equation to the algebraic Model-A form (Models B_2^{H} and C_2^{H}, §9). These four regimes are gathered side by side in Table 5 of §11.2, and are synthesised, together with the two irreducible obstructions, as the obstruction taxonomy of §12.

Proof. We multiply each balance equation by $x^{n_1} y^{n_2}$ and sum over its domain.

From the interior equation (4.4):

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&+\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&\quad = \quad \mu\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&\quad +\lambda_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)x^{n_1}y^{n_2} \\
&\quad +\lambda_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)x^{n_1}y^{n_2} \\
&\quad +\gamma_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2} \\
&\quad +\gamma_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2} \\
&\quad +\theta_1\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&\quad +\theta_2\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1,n_2+1)x^{n_1}y^{n_2} \\
&LHS_1+LHS_2+LHS_3+LHS_4+LHS_5 \\
&\quad = \quad RHS_1+RHS_2+RHS_3+RHS_4+RHS_5+RHS_6+RHS_7
\end{aligned}
$$

Evaluating each summand:

$$
\begin{aligned}
&LHS_1 \\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\pi(n_1,0)x^{n_1}y^{0}\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty}\pi(n_1,n_2)x^{n_1}y^{n_2}-\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right] \\
&- (\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[P(x,y)-P_y(y)-P_x(x)+\pi(0,0)\right]
\end{aligned}
$$

LHS_2

$$\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \gamma_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
&= \gamma_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \gamma_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \gamma_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]
\end{aligned}$$

LHS_3

$$\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty}\left[\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} - 0\cdot\pi(n_1,0)x^{n_1}\right] \\
&= \gamma_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right] \\
&= \gamma_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]
\end{aligned}$$

LHS_4

$$\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \theta_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} \\
&= \theta_1 x \left[\sum_{n_1=1}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=1}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \theta_1 x \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_1\pi(n_1,n_2)x^{n_1-1}y^{n_2} - \sum_{n_1=0}^{\infty} n_1\pi(n_1,0)x^{n_1-1}\right] \\
&= \theta_1 x \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x)\right]
\end{aligned}$$

LHS_5

$$\begin{aligned}
&= \theta_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2} \\
&= \theta_2 y \left[\sum_{n_1=0}^{\infty}\sum_{n_2=0}^{\infty} n_2\pi(n_1,n_2)x^{n_1}y^{n_2-1} - \sum_{n_2=0}^{\infty} n_2\pi(0,n_2)y^{n_2-1}\right] \\
&= \theta_2 y \left[\frac{\partial}{\partial y}P(x,y) - \frac{d}{dy}P_y(y)\right]
\end{aligned}$$

RHS_1

$$\begin{aligned}
&= \mu \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&= \mu x^{-1} \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1+1,n_2)x^{n_1+1}y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty}\sum_{n_2=1}^{\infty} \pi(m,n_2)x^{m}y^{n_2} \\
&= \mu x^{-1} \sum_{m=2}^{\infty}\left[\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \pi(m,0)x^m\right] \\
&= \mu x^{-1} \left[\sum_{m=2}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \sum_{m=2}^{\infty}\pi(m,0)x^m\right] \\
&= \mu x^{-1} \left[\sum_{m=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - x\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2} - \sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}\right] \\
&- \mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m - x\pi(1,0) - \pi(0,0)\right] \\
&= \mu x^{-1}\left[P(x,y) - xP_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0)\right]
\end{aligned}$$

RHS_2

$$\begin{aligned}
&= \lambda_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1-1,n_2)x^{n_1}y^{n_2} \\
&= \lambda_1 x \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1-1,n_2)x^{n_1-1}y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty}\sum_{n_2=1}^{\infty} \pi(m,n_2)x^{m}y^{n_2} \\
&= \lambda_1 x \sum_{m=0}^{\infty}\left[\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \pi(m,0)x^m\right] \\
&= \lambda_1 x \left[\sum_{m=0}^{\infty}\sum_{n_2=0}^{\infty} \pi(m,n_2)x^{m}y^{n_2} - \sum_{m=0}^{\infty}\pi(m,0)x^m\right] \\
&= \lambda_1 x\left[P(x,y) - P_x(x)\right]
\end{aligned}$$

RHS_3

$$\begin{aligned}
&= \lambda_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1,n_2-1)x^{n_1}y^{n_2} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty} \pi(n_1,n_2-1)x^{n_1}y^{n_2-1} \\
&= \lambda_2 y \sum_{n_1=1}^{\infty}\sum_{m=0}^{\infty} \pi(n_1,m)x^{n_1}y^{m} \\
&= \lambda_2 y \left[\sum_{n_1=0}^{\infty}\sum_{m=0}^{\infty} \pi(n_1,m)x^{n_1}y^{m} - \sum_{m=0}^{\infty}\pi(0,m)y^m\right] \\
&= \lambda_2 y\left[P(x,y) - P_y(y)\right]
\end{aligned}$$

RHS_4

$$
\begin{aligned}
&= \gamma_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2-1)x^{n_1}y^{n_2} \\
&= \gamma_1 y \sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty}(n_1+1)\pi(n_1+1,m_2)x^{n_1}y^{m_2} \\
&= \gamma_1 y \sum_{m_1=2}^{\infty}\sum_{m_2=0}^{\infty} m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2} \\
&= \gamma_1 y \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} m_1\pi(m_1,m_2)x^{m_1-1}y^{m_2} - \sum_{m_2=0}^{\infty}\pi(1,m_2)y^{m_2}\right] \\
&= \gamma_1 y \left[\frac{\partial}{\partial x}P(x,y) - P_y(y|n_1=1)\right]
\end{aligned}
$$

RHS_5

$$
\begin{aligned}
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_2+1)\pi(n_1-1,n_2+1)x^{n_1}y^{n_2} \\
&= \gamma_2 \sum_{n_1=1}^{\infty}\sum_{m_2=2}^{\infty} m_2\pi(n_1-1,m_2)x^{n_1}y^{m_2-1} \\
&= \gamma_2 x \left[\sum_{n_1=1}^{\infty}\sum_{m_2=0}^{\infty} m_2\pi(n_1-1,m_2)x^{n_1-1}y^{m_2-1} - \sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1-1}\right] \\
&= \gamma_2 x \left[\sum_{m_1=0}^{\infty}\sum_{m_2=0}^{\infty} m_2\pi(m_1,m_2)x^{m_1}y^{m_2-1} - \sum_{m_1=0}^{\infty}\pi(m_1,1)x^{m_1}\right] \\
&= \gamma_2 x \left[\frac{\partial}{\partial y}P(x,y) - P_x(x|n_2=1)\right]
\end{aligned}
$$

RHS_6

$$
\begin{aligned}
&= \theta_1 \sum_{n_1=1}^{\infty}\sum_{n_2=1}^{\infty}(n_1+1)\pi(n_1+1,n_2)x^{n_1}y^{n_2} \\
&= \theta_1 \sum_{m_1=2}^{\infty}\sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=1}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\right] \\
&= \theta_1 \left[\sum_{m_1=0}^{\infty}\sum_{n_2=0}^{\infty} m_1\pi(m_1,n_2)x^{m_1-1}y^{n_2} - \sum_{m_1=0}^{\infty} m_1\pi(m_1,0)x^{m_1-1}\right] \\
&\quad - \theta_1 \left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2} - \pi(1,0)\right] \\
&= \theta_1 \left[\frac{\partial}{\partial x}P(x,y) - \frac{d}{dx}P_x(x) - P_y(y|n_1=1) + \pi(1,0)\right]
\end{aligned}
$$

$$
\begin{aligned}
&RHS_7\\
&= \quad \theta_2 \sum_{n_1=1}^{\infty} \sum_{n_2=1}^{\infty} (n_2+1)\pi(n_1, n_2+1) x^{n_1} y^{n_2}\\
&= \quad \theta_2 \sum_{n_1=1}^{\infty} \sum_{m_2=2}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1}\\
&= \quad \theta_2 \left[\sum_{n_1=1}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{n_1=1}^{\infty} \pi(n_1, 1) x^{n_1} \right]\\
&= \quad \theta_2 \left[\sum_{n_1=0}^{\infty} \sum_{m_2=0}^{\infty} m_2 \pi(n_1, m_2) x^{n_1} y^{m_2-1} - \sum_{m_2=0}^{\infty} m_2 \pi(0, m_2) y^{m_2-1} \right]\\
&\quad - \theta_2 \left[\sum_{n_1=0}^{\infty} \pi(n_1, 1) x^{n_1} - \pi(0,1) \right]\\
&= \quad \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

Collecting and multiplying through by xy to clear the x^{-1} factor:

$$
\begin{aligned}
&(\lambda_1 + \lambda_2 + \mu) \left[P(x,y) - P_y(y) - P_x(x) + \pi(0,0) \right]\\
&+ \gamma_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right]\\
&+ \gamma_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right]\\
&+ \theta_1 x \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) \right]\\
&+ \theta_2 y \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) \right]\\
&\quad = \quad \mu x^{-1} \left[P(x,y) - x P_y(y|n_1=1) - P_y(y) - P_x(x) + x\pi(1,0) + \pi(0,0) \right]\\
&\quad\quad + \lambda_1 x \left[P(x,y) - P_x(x) \right]\\
&\quad\quad + \lambda_2 y \left[P(x,y) - P_y(y) \right]\\
&\quad\quad + \gamma_1 y \left[\frac{\partial}{\partial x} P(x,y) - P_y(y|n_1=1) \right]\\
&\quad\quad + \gamma_2 x \left[\frac{\partial}{\partial y} P(x,y) - P_x(x|n_2=1) \right]\\
&\quad\quad + \theta_1 \left[\frac{\partial}{\partial x} P(x,y) - \frac{d}{dx} P_x(x) - P_y(y|n_1=1) + \pi(1,0) \right]\\
&\quad\quad + \theta_2 \left[\frac{\partial}{\partial y} P(x,y) - \frac{d}{dy} P_y(y) - P_x(x|n_2=1) + \pi(0,1) \right]
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x-\lambda_2 y\right]P(x,y)\\
&+\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_2 y\right]P_y(y)\\
&\qquad+\left[-(\lambda_1+\lambda_2+\mu)+\mu x^{-1}\right]\pi(0,0)\\
&\qquad+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+\mu\pi(1,0)\\
&\qquad+\theta_1\pi(1,0)\\
&\qquad+\theta_2\pi(0,1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 x y^2\right]P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 x y^2\right]P_y(y)\\
&\qquad-\left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\qquad+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad-xy\left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\qquad+(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad+\theta_2 xy\,\pi(0,1)
\end{aligned}
\tag{4.16}
$$

The x-boundary equation (4.6), treated analogously, gives:

$$
\begin{aligned}
&\sum_{n_1=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n_1\right]\pi(n_1,0)x^{n_1}\\
&\quad=\quad \lambda_1\sum_{n_1=1}^{\infty}\pi(n_1-1,0)x^{n_1}+\mu\sum_{n_1=1}^{\infty}\pi(n_1+1,0)x^{n_1}\\
&\qquad+\gamma_2\sum_{n_1=1}^{\infty}\pi(n_1-1,1)x^{n_1}+\theta_1\sum_{n_1=1}^{\infty}(n_1+1)\pi(n_1+1,0)x^{n_1}+\theta_2\sum_{n_1=1}^{\infty}\pi(n_1,1)x^{n_1}\\
&(\lambda_1+\lambda_2+\mu)\sum_{n_1=1}^{\infty}\pi(n_1,0)x^{n_1}+(\gamma_1+\theta_1)x\sum_{n_1=1}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^{m}+\mu x^{-1}\sum_{m=2}^{\infty}\pi(m,0)x^{m}\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^{m}+\theta_1\sum_{m=2}^{\infty}m\pi(m,0)x^{m-1}+\theta_2\sum_{m=1}^{\infty}\pi(m,1)x^{m}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_1=0}^{\infty}\pi(n_1,0)x^{n_1}-\pi(0,0)\right]+(\gamma_1+\theta_1)x\sum_{n_1=0}^{\infty}n_1\pi(n_1,0)x^{n_1-1}\\
&\quad=\quad \lambda_1 x\sum_{m=0}^{\infty}\pi(m,0)x^m\\
&\qquad+\mu x^{-1}\left[\sum_{m=0}^{\infty}\pi(m,0)x^m-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 x\sum_{m=0}^{\infty}\pi(m,1)x^m\\
&\qquad+\theta_1\left[\sum_{m=0}^{\infty}m\pi(m,0)x^{m-1}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}\pi(m,1)x^m-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_x(x)-\pi(0,0)\right]+(\gamma_1+\theta_1)x\frac{d}{dx}P_x(x)\\
&\quad=\quad \lambda_1 xP_x(x)+\mu x^{-1}\left[P_x(x)-x\pi(1,0)-\pi(0,0)\right]\\
&\qquad+\gamma_2 xP_x(x|n_2=1)+\theta_1\left[\frac{d}{dx}P_x(x)-\pi(1,0)\right]+\theta_2\left[P_x(x|n_2=1)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}-\lambda_1 x\right]P_x(x)+\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)-\mu x^{-1}\right]\pi(0,0)-(\mu+\theta_1)\pi(1,0)-\theta_2\pi(0,1)\\
&\qquad+\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y\right]P_x(x)+xy\left[\gamma_1 x+\theta_1(x-1)\right]\frac{d}{dx}P_x(x)\\
&\quad=\quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y\right]\pi(0,0)\\
&\qquad-(\mu+\theta_1)xy\,\pi(1,0)\\
&\qquad-\theta_2 xy\,\pi(0,1)\\
&\qquad+xy\left[\gamma_2 x+\theta_2\right]P_x(x|n_2=1)
\end{aligned}
\tag{4.17}
$$

The y-boundary equation (4.5) gives:

$$
\begin{aligned}
&\sum_{n_2=1}^{\infty}\left[\lambda_1+\lambda_2+\mu+(\gamma_2+\theta_2)n_2\right]\pi(0,n_2)y^{n_2}\\
&\quad=\quad \lambda_2\sum_{n_2=1}^{\infty}\pi(0,n_2-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(0,n_2+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{\infty}\pi(1,n_2-1)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{\infty}(n_2+1)\pi(0,n_2+1)y^{n_2}
\end{aligned}
$$

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(0,n_2)y^{n_2}\\
&+(\gamma_2+\theta_2)y\sum_{n_2=1}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\mu y^{-1}\sum_{m=2}^{\infty}\pi(0,m)y^m\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\sum_{n_2=1}^{\infty}\pi(1,n_2)y^{n_2}\\
&\qquad+\theta_2\sum_{m=2}^{\infty}m\pi(0,m)y^{m-1}\\
&(\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(0,n_2)y^{n_2}-\pi(0,0)\right]\\
&+(\gamma_2+\theta_2)y\sum_{n_2=0}^{\infty}n_2\pi(0,n_2)y^{n_2-1}\\
&\quad=\quad \lambda_2 y\sum_{m=0}^{\infty}\pi(0,m)y^m\\
&\qquad+\mu\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[\sum_{m=0}^{\infty}\pi(0,m)y^m-y\pi(0,1)-\pi(0,0)\right]\\
&\qquad+\gamma_1 y\sum_{m=0}^{\infty}\pi(1,m)y^m\\
&\qquad+\theta_1\left[\sum_{n_2=0}^{\infty}\pi(1,n_2)y^{n_2}-\pi(1,0)\right]\\
&\qquad+\theta_2\left[\sum_{m=0}^{\infty}m\pi(0,m)y^{m-1}-\pi(0,1)\right]\\
&(\lambda_1+\lambda_2+\mu)\left[P_y(y)-\pi(0,0)\right]+(\gamma_2+\theta_2)y\frac{d}{dy}P_y(y)\\
&\quad=\quad \lambda_2 yP_y(y)+\mu\left[P_y(y|n_1=1)-\pi(1,0)\right]\\
&\qquad+\mu y^{-1}\left[P_y(y)-y\pi(0,1)-\pi(0,0)\right]+\gamma_1 yP_y(y|n_1=1)\\
&\qquad+\theta_1\left[P_y(y|n_1=1)-\pi(1,0)\right]+\theta_2\left[\frac{d}{dy}P_y(y)-\pi(0,1)\right]\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right]P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad=\quad \left[\mu+\gamma_1 y+\theta_1\right]P_y(y|n_1=1)\\
&\qquad-(\mu+\theta_1)\pi(1,0)-(\mu+\theta_2)\pi(0,1)\\
&\qquad+\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}\right]\pi(0,0)
\end{aligned}
$$

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right] P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad= \quad \left[\mu+\gamma_1 y+\theta_1\right] P_y(y|n_1=1)\\
&\qquad +\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-(\lambda_1+\lambda_2)\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)-\mu y^{-1}-\lambda_2 y\right] P_y(y)+\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad= \quad \left[\mu+\gamma_1 y+\theta_1\right] P_y(y|n_1=1)+\left[\mu-\mu y^{-1}\right]\pi(0,0)\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right] P_y(y)+xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\quad= \quad \left[\mu+\gamma_1 y+\theta_1\right] xy\, P_y(y|n_1=1)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
\tag{4.18}
$$

Substituting Equation (4.17) into Equation (4.16) eliminates all P_x terms:

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right] P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right] P_y(y)\\
&\qquad +xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad -xy\left[\mu+\gamma_1 y+\theta_1\right] P_y(y|n_1=1)
\end{aligned}
\tag{4.19}
$$

Isolating $[\mu+\gamma_1 y+\theta_1]\, xy\, P_y(y|n_1=1)$ from Equation (4.18) and substituting into Equation (4.19):

$$
\begin{aligned}
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right] P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad= \quad \left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_2 xy^2\right] P_y(y)\\
&\qquad +xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)\\
&\qquad -\Big\{\left[(\lambda_1+\lambda_2+\mu)xy-\mu x-\lambda_2 xy^2\right] P_y(y)\\
&\qquad\quad +xy\left[\gamma_2 y+\theta_2(y-1)\right]\frac{d}{dy}P_y(y)+\mu x(1-y)\,\pi(0,0)\Big\}\\
&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 xy^2\right] P(x,y)\\
&+xy\left[\gamma_1(x-y)+\theta_1(x-1)\right]\frac{\partial}{\partial x}P(x,y)\\
&+xy\left[\gamma_2(y-x)+\theta_2(y-1)\right]\frac{\partial}{\partial y}P(x,y)\\
&\quad= \quad \mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0)
\end{aligned}
$$

□

The fundamental equation already delivers the first specialisation of Theorem 1: the case of no abandonment, which covers Models A, B, B_2 and B_2^{H} at once.

Corollary 1 (Empty states without abandonment)**.** *Suppose $\theta_1=\theta_2=0$ (Models A, B, B_2 and B_2^{H}) and $\rho<1$. Then the total number in the system is a birth–death process with birth rate λ and death rate μ —an $M/M/1$ queue— for* any *jockeying rates γ_1,γ_2, since jockeying merely relocates a waiting customer and so conserves the total count. Its mean busy period is $\mathbb{E}[B]=(\mu-\lambda)^{-1}$, and Theorem 1 gives*

$$
\pi_0=1-\rho, \qquad \pi(0,0)=\rho(1-\rho). \tag{4.20}
$$

Moreover, setting $x=y=z$ in the fundamental equation (4.15) annihilates both convection terms (each carries the factor $x-y$), leaving the algebraic relation $-(1-z)\big(\mu-\lambda z\big)P(z,z)=-\mu(1-$

$z)\,\pi(0,0)$, *whence the diagonal of the PGF and the normalisation read*

$$P(z,z) = \frac{\pi(0,0)}{1-\rho z}, \qquad P(1,1) = 1 - \pi_0 = \rho. \tag{4.21}$$

4.2.2 Equation for PPGF in state space $\widetilde{S}$

Lemma 2 (Fundamental equation for the PPGF on $\widetilde{S}$)**.** *Consider the same system on the state space* $\widetilde{S} = \{(n_2, n) : 0 \le n_2 \le n\}$*, where* n *is the total number of customers* waiting *in the queues and* n_2 *the number of waiting class-2 customers. Assume positive recurrence, so that the stationary distribution* $\{\widetilde{\pi}(n_2,n)\}$ *exists, and define the partial PGF (transforming only* n_2*, keeping* n *discrete)*

$$\widetilde{P}(y,n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)\, y^{n_2},$$

a polynomial of degree n *in* y*. Then, for every* $n \ge 1$*, the family* $\{\widetilde{P}(y,n)\}_{n\ge0}$ *satisfies the differential–difference equation*

$$\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y) + (\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n) + [\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n)\\
&\quad= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + [\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)\\
&\qquad + (\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1) + \mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1).
\end{aligned}$$

Proof. Multiply the interior equation (4.9) by y^{n_2} and sum over $1 \le n_2 \le n-1$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n-1}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n-1}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad= \quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n-1}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n-1}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}
\end{aligned}$$

Adding the diagonal equation (4.10) multiplied by y^n extends most summation limits to n; the λ_1- and γ_2-terms on the RHS remain restricted to $n_2 \le n-1$, and a spare $\mu\widetilde{\pi}(n+1,n+1)y^n$ survives:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\gamma_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_1\sum_{n_2=1}^{n}(n-n_2)\widetilde{\pi}(n_2,n)y^{n_2}\\
&+\theta_2\sum_{n_2=1}^{n}n_2\widetilde{\pi}(n_2,n)y^{n_2}\\
&\quad=\quad \lambda_1\sum_{n_2=1}^{n-1}\widetilde{\pi}(n_2,n-1)y^{n_2}\\
&\qquad+\lambda_2\sum_{n_2=1}^{n}\widetilde{\pi}(n_2-1,n-1)y^{n_2}\\
&\qquad+\mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\mu\widetilde{\pi}(n+1,n+1)y^{n}\\
&\qquad+\gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2}\\
&\qquad+\gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2}\\
&\qquad+\theta_1\sum_{n_2=1}^{n}(n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2}\\
&\qquad+\theta_2\sum_{n_2=1}^{n}(n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2}.
\end{aligned}
$$

Evaluating each summand:

LHS_1

$$
\begin{aligned}
&= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n)y^{n_2}\\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{n}\widetilde{\pi}(n_2,n)y^{n_2}-\widetilde{\pi}(0,n)y^{0}\right]\\
&= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n)-\widetilde{\pi}(0,n)\right]
\end{aligned}
$$

LHS_2

$$
\begin{aligned}
&= \gamma_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \gamma_1 \left[n \sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \right] \\
&= \gamma_1 \left[n \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)y^{n_2} - \widetilde{\pi}(0,n)y^0 \right) - y \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1} \right] \\
&= \gamma_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned}
$$

LHS_3

$$
\begin{aligned}
&= \gamma_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \gamma_2 \left[y \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2-1} \right] \\
&= \gamma_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned}
$$

LHS_4

$$
\begin{aligned}
&= \theta_1 \sum_{n_2=1}^{n} (n-n_2)\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \theta_1 \left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n) \right]
\end{aligned}
$$

LHS_5

$$
\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n)y^{n_2} \\
&= \theta_2 y\frac{d}{dy}\widetilde{P}(y,n)
\end{aligned}
$$

RHS_1

$$
\begin{aligned}
&= \lambda_1 \sum_{n_2=1}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2} \\
&= \lambda_1 \left[\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^n \right] \\
&= \lambda_1 \left[\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^n - \widetilde{\pi}(0,n-1)y^0 \right] \\
&= \lambda_1 \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^n - \widetilde{\pi}(0,n-1) \right]
\end{aligned}
$$

RHS_2

$$
\begin{aligned}
&= \lambda_2 \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2} \\
&= \lambda_2 \left[y \sum_{n_2=1}^{n} \widetilde{\pi}(n_2-1,n-1)y^{n_2-1} \right] \\
&= \lambda_2 \left[y \sum_{n_2=0}^{n-1} \widetilde{\pi}(n_2,n-1)y^{n_2} \right] \\
&= \lambda_2 \left[y \left(\sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n-1)y^{n_2} - \widetilde{\pi}(n,n-1)y^n \right) \right] \\
&= \lambda_2 y \left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^n \right]
\end{aligned}
$$

RHS_3

$$
\begin{aligned}
&= \mu\widetilde{\pi}(n+1,n+1)y^n + \mu\sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&= \mu\left[\widetilde{\pi}(n+1,n+1)y^n + \sum_{n_2=1}^{n}\widetilde{\pi}(n_2,n+1)y^{n_2}\right] \\
&= \mu\left[\widetilde{\pi}(n+1,n+1)y^n + \sum_{n_2=0}^{n+1}\widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1)y^0 - \widetilde{\pi}(n+1,n+1)y^{n+1}\right] \\
&= \mu\left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1)y^n(1-y)\right]
\end{aligned}
$$

RHS_4

$$
\begin{aligned}
&= \gamma_1\sum_{n_2=1}^{n}(n-n_2+1)\widetilde{\pi}(n_2-1,n)y^{n_2} \\
&= \gamma_1\left[y\sum_{n_2=1}^{n}(n-(n_2-1))\widetilde{\pi}(n_2-1,n)y^{n_2-1}\right] \\
&= \gamma_1\left[y\sum_{k=0}^{n-1}(n-k)\widetilde{\pi}(k,n)y^k\right] \\
&= \gamma_1\left[y\sum_{k=0}^{n}(n-k)\widetilde{\pi}(k,n)y^k\right] \quad \text{(since the } k=n \text{ term is 0)} \\
&= \gamma_1 y\left[n\sum_{k=0}^{n}\widetilde{\pi}(k,n)y^k - y\sum_{k=1}^{n}k\widetilde{\pi}(k,n)y^{k-1}\right] \\
&= \gamma_1 y\left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n)\right]
\end{aligned}
$$

RHS_5

$$
\begin{aligned}
&= \gamma_2\sum_{n_2=1}^{n-1}(n_2+1)\widetilde{\pi}(n_2+1,n)y^{n_2} \\
&= \gamma_2\sum_{k=2}^{n}k\widetilde{\pi}(k,n)y^{k-1} \quad \text{(letting } k=n_2+1\text{)} \\
&= \gamma_2\left[\sum_{k=1}^{n}k\widetilde{\pi}(k,n)y^{k-1} - 1\cdot\widetilde{\pi}(1,n)y^0\right] \\
&= \gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n)\right]
\end{aligned}
$$

RHS_6

$$
\begin{aligned}
&= \theta_1 \sum_{n_2=1}^{n} (n+1-n_2)\widetilde{\pi}(n_2,n+1)y^{n_2} \\
&= \theta_1 \left[(n+1)\sum_{n_2=1}^{n} \widetilde{\pi}(n_2,n+1)y^{n_2} - \sum_{n_2=1}^{n} n_2\widetilde{\pi}(n_2,n+1)y^{n_2}\right] \\
&= \theta_1 \left[(n+1)\left(\sum_{n_2=0}^{n+1} \widetilde{\pi}(n_2,n+1)y^{n_2} - \widetilde{\pi}(0,n+1) - \widetilde{\pi}(n+1,n+1)y^{n+1}\right)\right] \\
&\quad - \theta_1 \left[y\left(\sum_{n_2=0}^{n+1} n_2\widetilde{\pi}(n_2,n+1)y^{n_2-1} - (n+1)\widetilde{\pi}(n+1,n+1)y^{n}\right)\right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1}\right] \\
&\quad - \theta_1 \left[y\frac{d}{dy}\widetilde{P}(y,n+1) - (n+1)\widetilde{\pi}(n+1,n+1)y^{n+1}\right] \\
&= \theta_1 \left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1)\right]
\end{aligned}
$$

RHS_7

$$
\begin{aligned}
&= \theta_2 \sum_{n_2=1}^{n} (n_2+1)\widetilde{\pi}(n_2+1,n+1)y^{n_2} \\
&= \theta_2 \sum_{k=2}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} \quad \text{(letting } k = n_2+1\text{)} \\
&= \theta_2 \left[\sum_{k=1}^{n+1} k\widetilde{\pi}(k,n+1)y^{k-1} - 1\cdot\widetilde{\pi}(1,n+1)y^{0}\right] \\
&= \theta_2 \left[\frac{d}{dy}\widetilde{P}(y,n+1) - \widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Collecting:

$$
\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\right] \\
&+ \gamma_1\left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+ \gamma_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&+ \theta_1\left[n\big(\widetilde{P}(y,n) - \widetilde{\pi}(0,n)\big) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&+ \theta_2 y\frac{d}{dy}\widetilde{P}(y,n) \\
&\quad = \lambda_1\left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^{n} - \widetilde{\pi}(0,n-1)\right] \\
&\qquad + \lambda_2 y\left[\widetilde{P}(y,n-1) - \widetilde{\pi}(n,n-1)y^{n}\right] \\
&\qquad + \mu\left[\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1) + \widetilde{\pi}(n+1,n+1)y^{n}(1-y)\right] \\
&\qquad + \gamma_1 y\left[n\widetilde{P}(y,n) - y\frac{d}{dy}\widetilde{P}(y,n)\right] \\
&\qquad + \gamma_2\left[\frac{d}{dy}\widetilde{P}(y,n) - \widetilde{\pi}(1,n)\right] \\
&\qquad + \theta_1\left[(n+1)\big(\widetilde{P}(y,n+1) - \widetilde{\pi}(0,n+1)\big) - y\frac{d}{dy}\widetilde{P}(y,n+1)\right] \\
&\qquad + \theta_2\left[\frac{d}{dy}\widetilde{P}(y,n+1) - \widetilde{\pi}(1,n+1)\right]
\end{aligned}
$$

Regrouping:

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n)\\
&= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)\\
&\quad + \{[\lambda_1+\lambda_2+\mu+(\gamma_1+\theta_1)n]\,\widetilde{\pi}(0,n)-\lambda_1\widetilde{\pi}(0,n-1)-[\mu+\theta_1(n+1)]\,\widetilde{\pi}(0,n+1)\\
&\qquad -\gamma_2\widetilde{\pi}(1,n)-\theta_2\widetilde{\pi}(1,n+1)\}\\
&\quad - y^n(\lambda_1+\lambda_2 y)\widetilde{\pi}(n,n-1)+\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

The y-boundary equation (4.11) makes the bracketed terms vanish; since $\widetilde{\pi}(n,n-1)=0$ (outside $\widetilde{S}$):

$$
\begin{aligned}
&[-(\gamma_2+\gamma_1 y)(1-y)+(\theta_2-\theta_1)y]\,\frac{d}{dy}\widetilde{P}(y,n)+[\lambda_1+\lambda_2+\mu+\gamma_1 n(1-y)+\theta_1 n]\,\widetilde{P}(y,n)\\
&= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1)+[\mu+\theta_1(n+1)]\,\widetilde{P}(y,n+1)+(\theta_2-\theta_1 y)\frac{d}{dy}\widetilde{P}(y,n+1)\\
&\quad + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1)
\end{aligned}
$$

□

Remark 2 (The $\widetilde{S}$ recursion is a failed simplification). The partial-PGF recurrence of Lemma 2 was introduced in the expectation that generating only over the class-2 index, while carrying the total count n as a discrete parameter, would *simplify* the analysis — collapsing the two-dimensional balance on S to a one-dimensional recurrence in n. It does the opposite. The forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ ties each level n to the unknown diagonal sequence $\{\pi(0,n)\}$, so the recurrence does not close on itself: solving it would require precisely the boundary probabilities it was meant to deliver. Only for the baseline Model-A does that forcing decay fast enough to be dropped, and even there the result is an *approximation*, accurate only in a restricted parameter region (Approximation 1). For every model with an active mechanism the $\widetilde{S}$ route is strictly harder than the S analysis it was meant to replace. We therefore record it as a deliberate negative result: the $\widetilde{S}$ coordinatisation is retained only for Model-A, and the S formulation is used throughout the remainder of this thesis.

5 Analysis of Model-A ($\gamma_1=\gamma_2=0,\ \theta_1=\theta_2=0$)

Model-A is the baseline of this collection: all jockeying and abandonment parameters are zero, leaving an ordinary two-class $M/M/1$ queue with non-preemptive priority. We analyse it in both state spaces S and $\widetilde{S}$: on the former we give two complete proofs, on the latter a conditionally accurate approximation, evaluated in Approximation 1.

The balance equations are obtained by plugging $\gamma_1=\gamma_2=\theta_1=\theta_2=0$ into Equations (4.4), (4.6), (4.5), (4.7), and (4.8) for state space S, and Equations (4.9), (4.10), (4.11), (4.12), and (4.13) for state space $\widetilde{S}$. The corresponding starting points are Lemma 1 for S and Lemma 2 for $\widetilde{S}$.

Stability. Since there is no abandonment, the system is positive recurrent if and only if the total load satisfies

$$\rho=\frac{\lambda_1+\lambda_2}{\mu}<1. \tag{5.1}$$

Figure 4 shows the queueing diagram of Model-A.

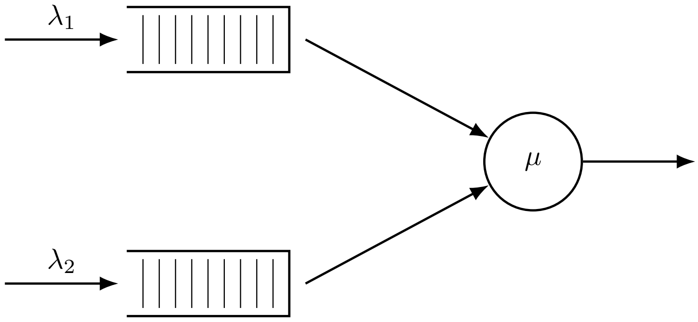

Figure 4: Queue layout for Model-A. Only Poisson arrivals and exponential service are present; all active mechanisms rates are zero.

5.1 Analysis of Model-A: state space S

We begin from Equation (4.15) with $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, in which the boundary function $P_y(y) = P(0, y)$ is still to be determined:

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0). \tag{5.2}$$

The following theorem summarises the main result of this subsection; two standalone proofs are given, one analytical and one probabilistic.

Theorem 2. *Consider a queueing system with two priority classes with Poisson arrival rates λ_1 and λ_2, where class-1 customers have non-preemptive priority over those of class 2, and exponentially distributed service times with rate μ independently of the class. Under the stability condition $\rho < 1$ (5.1), the joint PGF $P(x, y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is*

$$P(x, y) = \left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1} \cdot y(1 - y) \left[\frac{x - x^*(y)}{x^*(y) - y}\right] \cdot \rho(1 - \rho), \tag{5.3}$$

where

$$x^*(y) = \frac{[\mu + \lambda_1 + \lambda_2(1 - y)] - \sqrt{[\mu + \lambda_1 + \lambda_2(1 - y)]^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{5.4}$$

Corollary 2. *Under the stability condition $\rho < 1$ (5.1), the empty-state probabilities for Model-A are*

$$\pi_0 = 1 - \rho, \qquad \pi(0, 0) = \rho(1 - \rho). \tag{5.5}$$

In particular, $P(1, 1) = 1 - \pi_0 = \rho$ and the diagonal generating function satisfies $P(z, z) = \pi(0, 0)/(1 - \rho z)$.

Proof of Corollary 2. Model-A is the no-mechanism case $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$, so the statement is the specialisation of Corollary 1 —equivalently Theorem 1 with the $M/M/1$ mean busy period $\mathbb{E}[B] = (\mu - \lambda)^{-1}$, giving $\pi_0 = 1 - \rho$ and $\pi(0, 0) = \rho(1 - \rho)$. The diagonal value $P(z, z) = \pi(0, 0)/(1 - \rho z)$ and the normalisation $P(1, 1) = 1 - \pi_0 = \rho$ are the reduction (4.21) of the Model-A kernel (5.2) on $x = y = z$. □

5.1.1 Analytical approach for Model-A in state space S

Proof of Theorem 2, analytical approach. We begin from the fundamental equation for Model-A (5.2):

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x, y) = \mu(x - y)P_y(y) - \mu x(1 - y)\pi(0, 0).$$

We apply the *Kernel Method*, used in [Coh56]: fix $y \in (0, 1]$ and find $x^*(y)$ such that the LHS of the expression above is zero. Since $P(x, y)$ is a PGF, it is bounded. At any such root the LHS vanishes, so the RHS must also vanish:

$$\mu\left[(x^*(y) - y)\, P_y(y) - x^*(y)(1 - y)\, \pi(0, 0)\right] = 0.$$

To find the roots of the coefficient of $P(x, y)$, divide by $y \neq 0$ and set

$$0 = (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \lambda_2 x y$$

$$0 = \lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]\, x + \mu,$$

whose roots are

$$x^{\pm}(y) = \frac{[\mu + \lambda_1 + \lambda_2(1-y)] \pm \sqrt{[\mu + \lambda_1 + \lambda_2(1-y)]^2 - 4\mu\lambda_1}}{2\lambda_1}.$$

We label the root with the negative sign $x^*(y)$ and the root with the positive sign $x^+(y)$. At $y = 1$:

$$x^*(1) = 1 \quad \text{and} \quad x^+(1) = \mu/\lambda_1 > 1.$$

We require our root to lie inside the closed unit disc for $y \in [0, 1)$. Computing the derivative of $x^*(y)$:

$$\frac{d}{dy} x^*(y) = \frac{\lambda_2}{2\lambda_1}\left[-1 + \frac{\mu + \lambda_1 + \lambda_2(1-y)}{\sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}\right] > 0, \tag{5.6}$$

since $(\mu+\lambda_1+\lambda_2(1-y))^2 > (\mu+\lambda_1+\lambda_2(1-y))^2 - 4\lambda_1\mu$ whenever $\lambda_1\mu > 0$, by squaring numerator and denominator in the inner fraction. So $x^*(y)$ is strictly increasing in $[0, 1]$, and since $x^*(1) = 1$ we have $x^*(y) < 1$ for $y \in [0, 1)$. We make use of Vieta's formulas to discard the other root for the kernel polynomial $\lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]x + \mu = 0$: the product of the two roots satisfies $x^*(y)\, x^+(y) = \mu/\lambda_1 > 1$, so $x^+(y) = \mu/(\lambda_1 x^*(y)) > 1$ throughout $[0, 1)$. So the only root strictly inside the unit disc for $y \in [0, 1)$ is $x^*(y)$:

$$x^*(y) = \frac{(\mu + \lambda_1 + \lambda_2(1-y)) - \sqrt{(\mu + \lambda_1 + \lambda_2(1-y))^2 - 4\lambda_1\mu}}{2\lambda_1}. \tag{5.7}$$

Returning to the fundamental equation (5.2) at $x = x^*(y)$, the RHS also equals zero, giving

$$\mu \cdot [(x^*(y) - y)P_y(y) - x^*(y)(1-y)\pi(0,0)] = 0 \quad\Longrightarrow\quad P_y(y) = \frac{x^*(y)(1-y)}{x^*(y) - y}\pi(0,0). \tag{5.8}$$

By Corollary 2, $\pi(0,0) = \rho(1-\rho)$. Substituting this and Equation (5.8) into the fundamental equation (5.2):

$$\begin{aligned}
\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho) \\
\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho) \\
\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{y(x - x^*(y))}{x^*(y)-y}\right]\rho(1-\rho) \\
\left[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) &= y(1-y)\left[\frac{x - x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ recovers $P(x, y)$ in Theorem 2 and concludes the proof. □

5.1.2 Probabilistic approach for Model-A in state space S

Proof of Theorem 2, probabilistic approach. This proof mirrors the analytical one, but derives the boundary function $P_y(y)$ by means of probabilistic arguments. We again begin from the fundamental equation for Model-A (Equation (5.2)):

$$\left[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) = \mu\big[(x-y)P_y(y) - x(1-y)\pi(0,0)\big]. \tag{5.9}$$

Class-2 conditional distribution via $M/G/1$. Since class 1 has non-preemptive priority, class-2 customers are not taken into service whenever the priority queue is not empty ($N_1 > 0$); their *effective service time* B is therefore the duration of a complete busy period of the single-class ordinary $M/M/1$ sub-queue with arrival rate λ_1 and service rate μ. From class-2's viewpoint the system is thus an $M/G/1$ queue with arrival rate λ_2 and generally distributed service times B and mean $\mathbb{E}[B]$. The LST and mean of B are presented in §3.1 (Equations (3.1)–(3.2)), now with λ_1 instead of the generic λ:

$$\widetilde{B}(s) = \frac{(\mu + \lambda_1 + s) - \sqrt{(\mu + \lambda_1 + s)^2 - 4\mu\lambda_1}}{2\lambda_1},$$

$$\mathbb{E}[B] = \frac{1}{\mu(1-\rho_1)}.$$

Applying the Pollaczek–Khinchine (PK) formula (§3.4, Equation (3.8)) to this $M/G/1$ system, with $\widetilde{B}(\lambda_2(1-y)) = x^*(y)$ (cf. (5.7)) and $\lambda_2\mathbb{E}[B] = \rho_2/(1-\rho_1)$:

$$L_2(y) = \frac{(1-\lambda_2\mathbb{E}[B])(1-y)\widetilde{B}(\lambda_2(1-y))}{\widetilde{B}(\lambda_2(1-y)) - y} = \frac{1-\rho}{1-\rho_1} \cdot \frac{x^*(y)(1-y)}{x^*(y)-y}.$$

By the PASTA property, the time-average distribution of N_2 conditional on (busy, $N_1 = 0$) coincides with the $M/G/1$ steady-state distribution, so $\mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = L_2(y)$.

By the law of total expectation, $P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0]$. So

$$P_y(y) = \mathbb{P}(\text{busy}, N_1 = 0) \cdot \mathbb{E}[y^{N_2} \mid \text{busy}, N_1 = 0] = \mathbb{P}(\text{busy}, N_1 = 0) \cdot L_2(y).$$

Under non-preemptive priority, the priority queue operates autonomously (memorylessness of residual service time), so $\mathbb{P}(N_1 = 0 \mid \text{busy}) = 1-\rho_1$ by the geometric law of the ordinary $M/M/1(\lambda_1, \mu)$. Combined with $\mathbb{P}(\text{busy}) = \rho$, we have

$$\mathbb{P}(\text{busy}, N_1 = 0) = \rho(1-\rho_1);$$

and therefore

$$P_y(y) = \rho(1-\rho_1) \cdot L_2(y) = \rho(1-\rho_1) \cdot \frac{1-\rho}{1-\rho_1} \cdot \frac{x^*(y)(1-y)}{x^*(y)-y} = \frac{x^*(y)(1-y)}{x^*(y)-y} \cdot (1-\rho)\rho \tag{5.10}$$

Substituting Equation (5.10) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 2) into Equation (5.9):

$$\begin{aligned}
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(x-y)\frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho) - \mu x(1-y)\rho(1-\rho) \\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{x^*(y)(x-y)}{x^*(y)-y} - x\right]\rho(1-\rho) \\
\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y) &= \mu(1-y)\left[\frac{y(x-x^*(y))}{x^*(y)-y}\right]\rho(1-\rho) \\
\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right] P(x,y) &= y(1-y)\left[\frac{x-x^*(y)}{x^*(y)-y}\right]\rho(1-\rho).
\end{aligned}$$

Multiplying both sides by $\left[(1+\rho_1+\rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\right]^{-1}$ to isolate $P(x,y)$ recovers Theorem 2 and concludes the proof. □

5.1.3 Partial-PGF recurrence approach for Model-A in state space S

This proof follows Cohen [Coh56]'s method. We keep the class-1 index n_1 as a discrete parameter and generate only with respect to N_2, via the PPGF $\widetilde{P}(n_1, y) = \sum_{n_2 \geq 0} \pi(n_1, n_2)\, y^{n_2}$ from Equation (4.14). Multiplying the balance equations by y^{n_2} and summing over n_2 turns them into a second-order linear recurrence in n_1, whose characteristic roots fix the geometric structure of the solution in the priority-class index.

Proof. We start from the Model-A balance equations, obtained by setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the interior equation (4.4) and the x-boundary equation (4.6):

$$\begin{aligned}
(\lambda_1+\lambda_2+\mu)\,\pi(n_1,n_2) &= \mu\,\pi(n_1+1,n_2) + \lambda_1\,\pi(n_1-1,n_2) + \lambda_2\,\pi(n_1,n_2-1), \quad && n_1 \geq 1,\ n_2 \geq 1, \\
(\lambda_1+\lambda_2+\mu)\,\pi(n_1,0) &= \mu\,\pi(n_1+1,0) + \lambda_1\,\pi(n_1-1,0), && n_1 \geq 1,\ n_2 = 0.
\end{aligned} \tag{5.11}$$

Now we take transforms by summing all over their states, but only with respect to n_2, and we substitute our PPGF $\widetilde{P}(n_1, y) = \sum_{n_2=0}^{\infty} \pi(n_1, y) y^{n_2}$:

$$\begin{aligned}
&(\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2} \\
&= \quad \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2} + \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2} + \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2}
\end{aligned}$$

$$
\begin{aligned}
LHS_1 \quad &= \quad RHS_1 + RHS_2 + RHS_3 \\
LHS_1 & \\
&= (\lambda_1+\lambda_2+\mu)\sum_{n_2=1}^{\infty}\pi(n_1,n_2)y^{n_2} \\
&= (\lambda_1+\lambda_2+\mu)\left[\sum_{n_2=0}^{\infty}\pi(n_1,n_2)y^{n_2}-\pi(n_1,0)y^0\right] \\
&= (\lambda_1+\lambda_2+\mu)\left[\widetilde{P}(n_1,y)-\pi(n_1,0)\right] \\
RHS_1 & \\
&= \mu\sum_{n_2=1}^{\infty}\pi(n_1+1,n_2)y^{n_2} \\
&= \mu\left[\sum_{n_2=0}^{\infty}\pi(n_1+1,n_2)y^{n_2}-\pi(n_1+1,0)y^0\right] \\
&= \mu\left[\widetilde{P}(n_1+1,y)-\pi(n_1+1,0)\right] \\
RHS_2 & \\
&= \lambda_1\sum_{n_2=1}^{\infty}\pi(n_1-1,n_2)y^{n_2} \\
&= \lambda_1\left[\sum_{n_2=0}^{\infty}\pi(n_1-1,n_2)y^{n_2}-\pi(n_1-1,0)y^0\right] \\
&= \lambda_1\left[\widetilde{P}(n_1-1,y)-\pi(n_1-1,0)\right] \\
RHS_3 & \\
&= \lambda_2\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2} \\
&= \lambda_2 y\sum_{n_2=1}^{\infty}\pi(n_1,n_2-1)y^{n_2-1} \\
&= \lambda_2 y\sum_{m=0}^{\infty}\pi(n_1,m)y^{m} \\
&= \lambda_2 y\,\widetilde{P}(n_1,y) \\
(\lambda_1+\lambda_2+\mu)\big[\widetilde{P}(n_1,y)-\pi(n_1,0)\big] & \\
&= \mu\big[\widetilde{P}(n_1+1,y)-\pi(n_1+1,0)\big] \\
&\quad + \lambda_1\big[\widetilde{P}(n_1-1,y)-\pi(n_1-1,0)\big] \\
&\quad + \lambda_2 y\,\widetilde{P}(n_1,y)
\end{aligned}
$$

$$
\begin{aligned}
&\big[\mu+\lambda_1+\lambda_2(1-y)\big]\widetilde{P}(n_1,y)-\mu\widetilde{P}(n_1+1,y)-\lambda_1\widetilde{P}(n_1-1,y) \\
&\quad = (\lambda_1+\lambda_2+\mu)\pi(n_1,0)-\mu\pi(n_1+1,0)-\lambda_1\pi(n_1-1,0)
\end{aligned} \tag{5.12}
$$

By the x-boundary balance equation (5.11), the right-hand side of Equation (5.12) vanishes completely, giving the homogeneous recurrence

$$
\left[\mu+\lambda_1+\lambda_2(1-y)\right]\widetilde{P}(n_1,y)=\mu\widetilde{P}(n_1+1,y)+\lambda_1\widetilde{P}(n_1-1,y), \qquad n_1\geq 1. \tag{5.13}
$$

Characteristic roots and mode selection. Equation (5.13) is a *second-order linear recurrence* in n_1 with constant coefficients, so its general solution is a linear combination of two geometric sequences $[\widetilde{x}^{\pm}(y)]^{n_1}$, where $\widetilde{x}^{\pm}(y)$ solve the characteristic equation

$$
\mu\,\widetilde{x}^2-\left[\mu+\lambda_1+\lambda_2(1-y)\right]\widetilde{x}+\lambda_1=0, \tag{5.14}
$$

that is,

$$
\widetilde{x}^{\pm}(y)=\frac{\big(\mu+\lambda_1+\lambda_2(1-y)\big)\pm\sqrt{\big(\mu+\lambda_1+\lambda_2(1-y)\big)^2-4\lambda_1\mu}}{2\mu}.
$$

The *characteristic polynomial* (Equation (5.14)) is the reciprocal to that of the bivariate PGF approaches (i.e. the reciprocal of $\lambda_1 x^2 - [\mu + \lambda_1 + \lambda_2(1-y)]x + \mu$), thus its roots are the reciprocals of that kernel's roots: the positive root is the reciprocal of the negative one ($\widetilde{x}^+(y) = 1/x^*(y)$), and conversely $\widetilde{x}^-(y) = 1/x^+(y)$. By Vieta's formulas, $\widetilde{x}^-(y)\,\widetilde{x}^+(y) = \lambda_1/\mu = \rho_1 < 1$. Recall from §5.1.1 that $x^*(y) < 1 < x^+(y)$ for $y \in [0,1)$. We select the negative-sign root because the sequence $\{\widetilde{P}(n_1, y)\}_{n_1 \geq 0}$ must be summable:

$$\sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y) = P(1, y) \leq P(1, 1) = 1 - \pi_0 = \rho < \infty.$$

At $y = 1$, we have

$$\widetilde{x}^-(1) = \rho_1 \quad \text{and} \quad \widetilde{x}^+(1) = 1,$$

the latter making the geometric sequence $[\widetilde{x}^+(y)]^{n_1}$ non-summable. We therefore define $\widetilde{x}^*(y) := \widetilde{x}^-(y)$,

$$\widetilde{x}^*(y) := \widetilde{x}^-(y) = \frac{\big(\mu + \lambda_1 + \lambda_2(1-y)\big) - \sqrt{\big(\mu + \lambda_1 + \lambda_2(1-y)\big)^2 - 4\lambda_1\mu}}{2\mu} = \rho_1\, x^*(y). \tag{5.15}$$

The last identity holds because $\widetilde{x}^*$ and x^* share the same numerator and differ only in the denominator (2μ versus $2\lambda_1$). Next,

$$\widetilde{P}(n_1, y) = \widetilde{P}(0, y)\,[\widetilde{x}^*(y)]^{n_1}, \qquad n_1 \geq 0, \tag{5.16}$$

where the relation also holds at $n_1 = 0$ trivially. Recall that at $y = 1$, we have $\widetilde{x}^*(1) = \lambda_1/\mu = \rho_1$. Thus, $\widetilde{P}(n_1, 1) = \widetilde{P}(0, 1)\,\rho_1^{n_1}$ and $\widetilde{P}(0, 1) = P_y(1) = \rho(1 - \rho_1)$. The constant $\widetilde{P}(0, y)$ is resolved by its definition

$$\widetilde{P}(0, y) = \sum_{n_2=0}^{\infty} \pi(0, n_2)\, y^{n_2} = P(0, y) = P_y(y),$$

as determined in the bivariate PGF approaches. Summing the geometric law (Equation (5.16)) over $n_1 \geq 0$ for $x \in (0, 1]$ recovers

$$P(x, y) = \sum_{n_1=0}^{\infty} \widetilde{P}(n_1, y)\, x^{n_1} = P_y(y) \sum_{n_1=0}^{\infty} \big[\widetilde{x}^*(y)\, x\big]^{n_1} = \frac{P_y(y)}{1 - \widetilde{x}^*(y)\, x}. \tag{5.17}$$

To bring this to the form of Theorem 2, we define the kernel of the fundamental equation, divided by μ:

$$K(x, y) := (1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2 = -y\rho_1\big(x - x^*(y)\big)\big(x - x^+(y)\big), \tag{5.18}$$

where $x^*(y)$ and $x^+(y)$ are the roots found in §5.1.1. By Vieta's product formula, $x^*(y)\, x^+(y) = \mu/\lambda_1 = 1/\rho_1$. This, together with $\widetilde{x}^*(y)$ (Equation (5.15)), gives $\widetilde{x}^*(y) = \rho_1 x^*(y) = 1/x^+(y)$. Using the factorisation of $K(x, y)$ (5.18), the denominator of the reconstructed PGF (5.17) becomes

$$1 - \widetilde{x}^*(y)\, x = 1 - \frac{x}{x^+(y)} = \frac{x^+(y) - x}{x^+(y)} = \frac{K(x, y)}{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}.$$

Substituting this into our reconstructed $P(x, y)$ (Equation (5.17)), using $P_y(y) = \frac{x^*(y)(1-y)}{x^*(y)-y}\rho(1-\rho)$ and also noting that $\rho_1\, x^*(y)\, x^+(y) = 1$, we obtain

$$\begin{aligned}
P(x, y) &= \frac{x^*(y)(1-y)}{x^*(y) - y}\,\rho(1-\rho) \cdot \frac{y\rho_1\, x^+(y)\big(x - x^*(y)\big)}{K(x, y)} \\
&= \underbrace{\rho_1\, x^*(y)\, x^+(y)}_{=1}\, \frac{1}{K(x, y)}\, y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\,\rho(1-\rho) \\
&= \big[(1 + \rho_1 + \rho_2)xy - y - \rho_1 x^2 y - \rho_2 x y^2\big]^{-1} \cdot y(1-y)\, \frac{x - x^*(y)}{x^*(y) - y}\,\rho(1-\rho),
\end{aligned}$$

which recovers Equation (5.3) from Theorem 2 and concludes the proof. □

5.2 Analysis of Model-A on $\widetilde{S}$: scope and limitations of the partial-PGF recursion

Setting $\gamma_1 = \gamma_2 = \theta_1 = \theta_2 = 0$ in the $\widetilde{S}$ balance equations (4.9)–(4.13) and forming the PPGF $\widetilde{P}(y,n) = \sum_{n_2=0}^{n} \widetilde{\pi}(n_2,n)\, y^{n_2}$ yields the inhomogeneous recurrence:

$$\begin{aligned} &[\lambda_1+\lambda_2+\mu]\, \widetilde{P}(y,n) \\ &\quad= \quad (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1) + \mu y^n(1-y)\widetilde{\pi}(n+1,n+1) \end{aligned} \tag{5.19}$$

The forcing term in Equation (5.19) is controlled by the diagonal probabilities $\widetilde{\pi}(n+1,n+1) = \pi(0,n+1)$, whose decay we first establish from the exact S-solution.

Lemma 3 (Geometric decay of the boundary probabilities)**.** *For Model-A the boundary probabilities $\pi(0,n)$ satisfy $\pi(0,n) = O(r^{-n})$ for every $r < 1/\rho$; equivalently they decay geometrically with rate ρ.*

Proof. By definition $P_y(y) = P(0,y) = \sum_{n\geq 0} \pi(0,n)\, y^n$ is the generating function of $\{\pi(0,n)\}$, and from Equation (5.8) with $\pi(0,0) = \rho(1-\rho)$,

$$P_y(y) = \rho(1-\rho)\, \frac{x^*(y)(1-y)}{x^*(y)-y}.$$

Evaluating the kernel quadratic (5.4) at $x = y$ gives $\lambda_1 y^2 - [\mu+\lambda_1+\lambda_2(1-y)]y + \mu = (\lambda y-\mu)(y-1)$ with $\lambda = \lambda_1+\lambda_2$, so $x^*(y) = y$ holds only at $y = 1$ and $y = \mu/\lambda = 1/\rho$. At $y = 1$ the simple zeros of $(1-y)$ and $(x^*(y)-y)$ cancel and P_y is analytic with $P_y(1) = \rho$; the branch point of x^* lies beyond $1/\rho$. Hence the singularity of P_y nearest the origin is the simple pole at $y = 1/\rho > 1$, so P_y is analytic on the disc $|y| < 1/\rho$. By the Cauchy coefficient estimates, for every $r < 1/\rho$ there is $C_r < \infty$ with $\pi(0,n) \leq C_r\, r^{-n}$, i.e. geometric decay at rate ρ. $\square$

Approximation 1 (Heuristic PPGF on $\widetilde{S}$)**.** *Neglecting the boundary-diagonal forcing term $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ in Equation (5.19) —which is exponentially small by Lemma 3— the homogeneous recurrence admits the geometric solution*

$$\widetilde{P}(y,n) \;\approx\; \rho(1-\rho)\, [\widetilde{y}^*(y)]^n,$$

where

$$\widetilde{y}^*(y) = \frac{(\lambda+\mu) - \sqrt{(\lambda+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}, \qquad \lambda = \lambda_1+\lambda_2.$$

At $y = 1$ this recovers $\widetilde{P}(1,n) = (1-\rho)\rho^{n+1}$, consistent with the $M/M/1(\lambda,\mu)$ steady-state marginal for n customers waiting.

Justification of Approximation 1. The inhomogeneous term $\mu y^n(1-y)\widetilde{\pi}(n+1,n+1)$ prevents the direct application of a geometric-sequence ansatz. The diagonal probabilities $\widetilde{\pi}(n,n) = \pi(0,n)$ correspond to states where all n waiting customers are class 2, and by Lemma 3 they decay geometrically in n at rate ρ; the forcing term is therefore exponentially small, and neglecting it yields an approximation whose accuracy is quantified below. The resulting homogeneous recurrence is

$$(\lambda_1+\lambda_2+\mu)\widetilde{P}(y,n) = (\lambda_1+\lambda_2 y)\widetilde{P}(y,n-1) + \mu\widetilde{P}(y,n+1).$$

The characteristic polynomial of this recurrence is

$$\mu[\widetilde{y}(y)]^2 - (\lambda_1+\lambda_2+\mu)\widetilde{y}(y) + (\lambda_1+\lambda_2 y) = 0,$$

with roots

$$\widetilde{y}^{\pm}(y) = \frac{(\lambda_1+\lambda_2+\mu) \pm \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}.$$

At $y = 1$ the roots are $\widetilde{y}_+(1) = 1$ and $\widetilde{y}_-(1) = \rho$. The root $\widetilde{y}_+ = 1$ would produce a non-summable geometric series as $n \to \infty$, so only the negative-sign root is admissible:

$$\widetilde{y}^*(y) = \frac{(\lambda_1+\lambda_2+\mu) - \sqrt{(\lambda_1+\lambda_2+\mu)^2 - 4\mu(\lambda_1+\lambda_2 y)}}{2\mu}.$$

The geometric ansatz gives $\widetilde{P}(y,n) = \widetilde{P}(y,0)\cdot[\widetilde{y}^*(y)]^n$. Using $\widetilde{P}(y,0) = \widetilde{\pi}(0,0) = \pi(0,0) = \rho(1-\rho)$ (Corollary 2):

$$\widetilde{P}(y,n) = \rho(1-\rho)\cdot[\widetilde{y}^*(y)]^n.$$

Since n counts customers *waiting* (the customer in service is not counted), $n+1$ is the total number in the system, so evaluating at $y=1$:

$$\widetilde{P}(1,n) = \rho(1-\rho)\cdot\rho^n = (1-\rho)\rho^{n+1}, \qquad \square$$

which matches the $M/M/1(\lambda,\mu)$ stationary distribution for $n+1$ customers in the system [AR02].

Failure mechanism. The neglected forcing $\mu y^n(1-y)\,\widetilde{\pi}(n+1,n+1)$ feeds in the diagonal states, in which all waiting customers are class 2. Using $\widetilde{\pi}(n,n) = \pi(0,n)$ and the rate proved in Lemma 3, this term decays in n like ρ^n (modulated by y^n), whereas the homogeneous solution decays like $[\widetilde{y}^*(y)]^n$. The approximation is therefore accurate exactly when the forcing decays faster, i.e. $\rho < \widetilde{y}^*(y)$ on the relevant range of y; this holds when the priority class carries most of the load (ρ_1/ρ large, ρ moderate), precisely the empirical validity region of the approximation.

The obstruction is structural and recurs throughout this thesis: an unknown boundary/diagonal trace —here the sequence $\pi(0,n)$— couples into an otherwise solvable recursion, exactly as the unknown diagonal $P(z,z)$ enters the Volterra forcing of Models B and X (§6 and Remark 3). A rigorous closure would retain the forcing and solve Equation (5.19) by variation of constants in n, which reintroduces the very diagonal sequence $\{\pi(0,n)\}$ as an unknown to be determined self-consistently; see §12.5.

6 Analysis of Model-B ($\gamma_1,\gamma_2>0$, $\theta_1=\theta_2=0$)

In this section, we derive a general solution to the fundamental equation of Model-B. However, this is provided in terms of the unknown boundary function $P_y(y)$, which requires more advanced mathematical tools and is thus left open, but included for illustrative purposes to conclude that having derivative terms in y in our fundamental equation is a non-trivial obstacle. So, this section not only shows the difficulty of Model-B but also justifies not carrying out an analogous derivation for Model-X. Moreover, since jockeying breaks our Poisson arrivals and renewal structure, the probabilistic argument explained in § 5.1.2 is no longer available.

Model-B introduces bidirectional jockeying ($\gamma_1,\gamma_2>0$) and no abandonment ($\theta_1=\theta_2=0$). Figure 5 shows the queue layout.

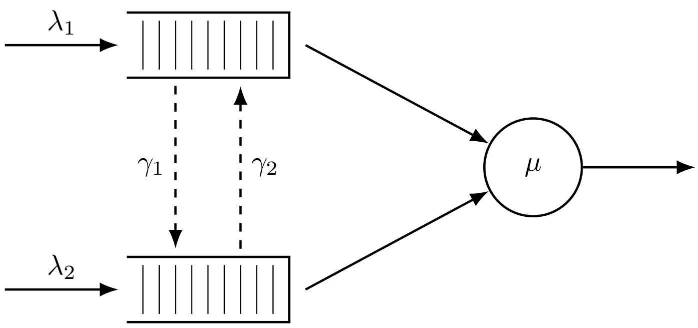

Figure 5: Queue layout for Model-B. Dashed arrows indicate bidirectional jockeying: γ_1 moves a class-1 customer down to Queue 2 and γ_2 moves a class-2 customer up to Queue 1. Both transfers conserve the total n.

Stability. $\rho = (\lambda_1+\lambda_2)/\mu < 1$. Jockeying preserves the total customer count and does not affect this condition, as the arriving customers can also be re-allocated.

Setting $\theta_1=\theta_2=0$ in the general fundamental equation (4.15) (Lemma 1) removes both abandonment factors $(x-1)$ and $(y-1)$ and leaves the *Model-B fundamental equation*

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\right] P(x,y)\\ &\quad + \gamma_1\, xy(x-y)\,\frac{\partial P}{\partial x}(x,y) + \gamma_2\, xy(y-x)\,\frac{\partial P}{\partial y}(x,y)\\ &\quad = \mu(x-y)\,P_y(y) - \mu\, x(1-y)\,\pi(0,0).\end{aligned} \tag{6.1}$$

This equation governs the entire analysis of the section, and we refer to it throughout in place of the general equation (4.15). Its only derivative terms are the two jockeying convection terms, both carrying the diagonal factor $(x-y)$ that makes the Method of Characteristics applicable.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models. Since Model-B is not solved, however, no theorem is attached to it.

Corollary 3. *The empty-system probability π_0 and the empty-queues probability $\pi(0,0)$ of Model-B coincide with those of the Model-A baseline,*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho), \tag{6.2}$$

and the diagonal of the PGF is $P(z,z) = \pi(0,0)/(1-\rho z)$.

The proof follows immediately by setting $x=y=z$ in the Model-B fundamental equation (6.1): both jockeying terms, which carry the factor $(x-y)$, vanish, dragging the term $\mu(x-y)P_y(y)$ with them. The surviving diagonal equation is therefore *identical* to that of Model-A with $x=y=z$, so Corollary 3 is the no-abandonment instance of Theorem 1: jockeying conserves the total count, leaving an $M/M/1$ process of mean busy period $\mathbb{E}[B] = (\mu-\lambda)^{-1}$ (Corollary 1).

For Model-B we do not provide a solution; however, working out the problem exposes the precise analytical obstacle, which we record as a claim. For the sake of readability, we will *not* define here the auxiliary functions; they are introduced throughout the proof. The following claim is made:

Claim 1. *Assume stability $\rho<1$. Along the characteristic lines $\gamma_2 x + \gamma_1 y = C_1$ of the Model-B fundamental equation (6.1), parametrised by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$, the bivariate PGF of Model-B admits the following integral representation:*

$$P\big(x,y(x)\big) \;=\; P_h\big(x,y(x)\big)\int_0^x \frac{h\big(t;C_1\big)}{P_h\big(t,y(t)\big)}\,dt, \tag{6.3}$$

where P_h is the homogeneous solution and the forcing function h depends linearly on the unknown boundary trace $P_y(y) = P(0,y)$. Evaluating Equation (6.3) along the diagonal $x=y=z$, where $P(z,z) = \pi(0,0)/(1-\rho z)$ is known from Corollary 3, reduces the determination of P_y to a Volterra integral equation of the first kind

$$\mathcal{F}(z) \;=\; \int_0^z \mathcal{K}(z,t)\,P_y\big(y(t;z)\big)\,dt. \tag{6.4}$$

This model is therefore left open.

Proof of Claim 1. The proof consists of five steps: (i) homogeneous solution via the Method of Characteristics, (ii) sign analysis of the exponent function E, (iii) particular solution via the Method of Variation of Parameters, (iv) determining the arbitrary function from the boundary condition, and (v) evaluation at $x=y$ yielding a Volterra-type equation.

Step (i): Homogeneous solution via the Method of Characteristics.

The Model-B fundamental equation (6.1) is a first-order linear PDE in $P(x,y)$. Setting the right-hand side to zero and dividing by y yields the homogeneous PDE

$$\gamma_1 x(x-y)\,\frac{\partial P}{\partial x} - \gamma_2 x(x-y)\,\frac{\partial P}{\partial y} = -\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P.$$

Following the Method of Characteristics (§3.6.2), we associate to this PDE the characteristic system

$$\frac{dx}{\gamma_1 x(x-y)} = \frac{dy}{-\gamma_2 x(x-y)} = \frac{dP}{-\big[(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy\big]P}.$$

Taking the ratio of the first two differentials, the $x(x-y)$ factors cancel, giving $dx/\gamma_1 = dy/(-\gamma_2)$. Integrating yields our characteristic constant:

$$C_1 = \gamma_2 x + \gamma_1 y.$$

The characteristic curves are therefore the straight lines of slope $-\gamma_2/\gamma_1$ in the (x,y)-plane. We parametrise y by setting $y(x) = (C_1-\gamma_2 x)/\gamma_1$. To extract the equation for P along this curve, we need to define our coefficient $A(x,y)$:

$$A(x,y) := (\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy. \tag{6.5}$$

Substituting $y = (C_1 - \gamma_2 x)/\gamma_1$ so that $\lambda_2 xy = (\lambda_2 C_1/\gamma_1)x - (\lambda_2\gamma_2/\gamma_1)x^2$, we obtain, collecting by powers of x,

$$\begin{aligned} A\big(x, y(x)\big) &= (\lambda_1 + \lambda_2 + \mu)x - \mu - \lambda_1 x^2 - \frac{\lambda_2 C_1}{\gamma_1}\, x + \frac{\lambda_2\gamma_2}{\gamma_1}\, x^2 \\ &= -\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1}\, x^2 + \frac{\gamma_1(\lambda_1 + \lambda_2 + \mu) - \lambda_2 C_1}{\gamma_1}\, x - \mu. \end{aligned}$$

Similarly, $x - y(x) = [(\gamma_1 + \gamma_2)x - C_1]/\gamma_1$, so $\gamma_1 x(x - y(x)) = x[(\gamma_1 + \gamma_2)x - C_1]$. The dP/P equation from the characteristic system therefore now reads as follows:

$$\frac{dP}{P} = \frac{-A(x, y(x))}{\gamma_1 x(x - y(x))}\, dx = \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1 + \lambda_2 + \mu) - \lambda_2 C_1\big]x + \gamma_1\mu}{\gamma_1 x\big[(\gamma_1 + \gamma_2)x - C_1\big]}\, dx.$$

To integrate this, we decompose the integrand via partial fractions. Writing the integrand as $\frac{1}{\gamma_1} I(x)\, dx$,

$$I(x) = K + \frac{M}{x} + \frac{N(C_1)}{(\gamma_1 + \gamma_2)x - C_1};$$

satisfying

$$\begin{aligned} &K(\gamma_1 + \gamma_2)x^2 + \big[M(\gamma_1 + \gamma_2) + N - KC_1\big]x - MC_1 \\ &= \quad (\gamma_1\lambda_1 - \lambda_2\gamma_2)x^2 - \big[\gamma_1(\lambda_1 + \lambda_2 + \mu) - \lambda_2 C_1\big]x + \gamma_1\mu. \end{aligned}$$

The x^2 and x^0 coefficient matches give K and M immediately; the x^1 match determines N after simplification:

$$K = \frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1 + \gamma_2}, \qquad M = -\frac{\gamma_1\mu}{C_1}, \qquad N(C_1) = \gamma_1\left[\frac{\lambda_1 + \lambda_2}{\gamma_1 + \gamma_2} C_1 - (\lambda_1 + \lambda_2 + \mu) + \frac{\mu(\gamma_1 + \gamma_2)}{C_1}\right]. \tag{6.6}$$

For N, the x^1 coefficient match gives

$$\begin{aligned} N &= -\gamma_1(\lambda_1 + \lambda_2 + \mu) + \lambda_2 C_1 + KC_1 - M(\gamma_1 + \gamma_2) \\ &= -\gamma_1(\lambda_1 + \lambda_2 + \mu) + \lambda_2 C_1 + \frac{(\gamma_1\lambda_1 - \lambda_2\gamma_2)\, C_1}{\gamma_1 + \gamma_2} + \frac{\gamma_1\mu(\gamma_1 + \gamma_2)}{C_1} \\ &= -\gamma_1(\lambda + \mu) + C_1\, \frac{\lambda_2(\gamma_1 + \gamma_2) + \gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1 + \gamma_2} + \frac{\gamma_1\mu(\gamma_1 + \gamma_2)}{C_1} \\ &= -\gamma_1(\lambda + \mu) + \frac{\gamma_1\lambda}{\gamma_1 + \gamma_2}\, C_1 + \frac{\gamma_1\mu(\gamma_1 + \gamma_2)}{C_1}; \end{aligned}$$

factoring γ_1 out recovers $N(C_1)$ from (6.6).

Integrating $dP/P = (1/\gamma_1)\, I(x)\, dx$ term by term gives

$$\ln P = \frac{K}{\gamma_1} x - \frac{\mu}{C_1} \ln x + \frac{N(C_1)}{\gamma_1(\gamma_1 + \gamma_2)} \ln\big[(\gamma_1 + \gamma_2)x - C_1\big] + \ln f(C_1), \tag{6.7}$$

where $f(C_1)$ is an arbitrary function of the characteristic constant. Now note that

$$(\gamma_1 + \gamma_2)x - C_1 = (\gamma_1 + \gamma_2)x - (\gamma_2 x + \gamma_1 y) = \gamma_1(x - y),$$

so $\ln[(\gamma_1 + \gamma_2)x - C_1] = \ln\gamma_1 + \ln(x - y)$.

Exponentiating Equation (6.7) term by term,

$$P \;=\; f(C_1)\, \exp\!\Big(\tfrac{K}{\gamma_1} x\Big) \cdot x^{-\mu/C_1} \cdot \gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}\, (x - y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))};$$

the factor $\gamma_1^{N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}$ depends on x and y only through C_1 and can be absorbed, together with f, into a redefined arbitrary function $F(C_1)$:

$$P \;=\; F(C_1) \cdot \exp\!\Big(\tfrac{K}{\gamma_1} x\Big) \cdot x^{-\mu/C_1} \cdot (x - y)^{\,N(C_1)/(\gamma_1(\gamma_1+\gamma_2))}.$$

Reading off the exponent of $(x - y)$, with $C_1 = \gamma_2 x + \gamma_1 y$, defines

$$E(x, y) := \frac{N(C_1)}{\gamma_1(\gamma_1 + \gamma_2)} = \frac{\lambda_1 + \lambda_2}{(\gamma_1 + \gamma_2)^2}(\gamma_2 x + \gamma_1 y) - \frac{\lambda_1 + \lambda_2 + \mu}{\gamma_1 + \gamma_2} + \frac{\mu}{\gamma_2 x + \gamma_1 y}. \tag{6.8}$$

The general homogeneous solution is therefore $F(C_1)\,P_h(x,y)$, with F an arbitrary, differentiable function of the characteristic constant C_1 and P_h the particular homogeneous solution

$$P_h(x,y) = \exp\left(\frac{\gamma_1\lambda_1 - \lambda_2\gamma_2}{\gamma_1(\gamma_1+\gamma_2)}x\right)\cdot x^{-\mu/(\gamma_2 x+\gamma_1 y)}\cdot (x-y)^{E(x,y)} \tag{6.9}$$

that will serve as the integrating factor for variation of parameters in Step (iii). We will see in Step (iv) that F must vanish.

Step (ii): The exponent $E(x,y)$ is strictly positive in $(0,1)^2$.

Recall $\lambda = \lambda_1 + \lambda_2$. Along any fixed characteristic, $C_1 = \gamma_2 x + \gamma_1 y$ is constant, so the exponent function (6.8) depends only on C_1, which ranges over $(0, \gamma_1+\gamma_2)$ as $(x,y) \in (0,1)^2$:

$$E(C_1) = \frac{\lambda}{(\gamma_1+\gamma_2)^2}\,C_1 - \frac{\lambda+\mu}{\gamma_1+\gamma_2} + \frac{\mu}{C_1}.$$

Showing that $E(C_1) > 0$ on $C_1 \in (0,\gamma_1+\gamma_2)$, follows from two facts. First, E vanishes at the right endpoint,

$$E(\gamma_1+\gamma_2) = \frac{\lambda - (\lambda+\mu)+\mu}{\gamma_1+\gamma_2} = 0.$$

Second, E is strictly decreasing in $(0,\gamma_1+\gamma_2]$, using $u \le \gamma_1+\gamma_2$ and $\mu > \lambda$ (i.e. $\rho < 1$):

$$\begin{aligned} E'(u) &= \frac{\lambda}{(\gamma_1+\gamma_2)^2} - \frac{\mu}{u^2} \\ &\le \frac{\lambda}{(\gamma_1+\gamma_2)^2} - \frac{\mu}{(\gamma_1+\gamma_2)^2} = \frac{\lambda-\mu}{(\gamma_1+\gamma_2)^2} < 0, \end{aligned}$$

Since E decreases strictly to 0 at its right endpoint, it is strictly positive at every point to its left, so:

$$E(C_1) > E(\gamma_1+\gamma_2) = 0 \qquad \text{for all } C_1 \in (0,\gamma_1+\gamma_2),$$

that is, $E(x,y) > 0$ for all $(x,y) \in (0,1)^2$. For this proof, the crucial boundary behaviour is that $E \to 0$ as $(x,y) \to (1,1)$; the blow-up caused by μ/C_1 as $C_1 \to 0^+$ is irrelevant.

Step (iii): Particular solution via variation of parameters.

This step consists in upgrading the homogeneous solution P_h (Equation (6.9)) with its inhomogeneous part, along the characteristic. We start by rewriting our fundamental equation for Model-B (Equation (6.1)). The coefficient of P on the LHS can be written as $yA(x,y)$ —where $A(x,y)$ is defined as in Step (i) (Equation (6.5))—; and we define the RHS of the fundamental equation as follows:

$$G(x,y) := \mu(x-y)\,P_y(y) - \mu\,x(1-y)\,\pi(0,0); \tag{6.10}$$

this forcing involves the unknown boundary trace $P_y(y) = P(0,y)$ together with the known constant $\pi(0,0) = \rho(1-\rho)$ of Corollary 3. Separating the terms that involve derivatives and the ones that do not, we have

$$\gamma_1 xy(x-y)\,\partial_x P + \gamma_2 xy(y-x)\,\partial_y P = -\,y\,A(x,y)\,P + G(x,y),$$

which is the convection form (3.14) of the Method of Characteristics. So, along the characteristic $C_1 = \gamma_2 x + \gamma_1 y$ found in Step (i) (parametrised by $y(x) = (C_1 - \gamma_2 x)/\gamma_1$), §3.6.2 reduces it to a first-order linear ODE of the form (3.17),

$$\frac{dP}{dx} = f(x;C_1)\,P + h(x;C_1). \tag{6.11}$$

The homogeneous part $\frac{dP}{dx} = f(x;C_1)\,P$ is P_h (Equation (6.9)), so there is no need to write f out; and its forcing function is the right-hand side divided by the coefficient of $\partial_x P$,

$$h(x;C_1) = \frac{G\big(x,y(x)\big)}{\gamma_1\,x\,y(x)\,\big(x-y(x)\big)}. \tag{6.12}$$

Equation (6.11) is an instance of Equation (3.17) with the known homogeneous solution P_h. So the variation-of-parameters formula (3.18) of §3.6.3 delivers its general solution, with the arbitrary constant becoming a function $F(C_1)$ of the characteristic label:

$$P(x,y(x)) = P_h(x,y(x))\left[F(C_1) + \int_{x_0}^{x}\frac{h(t;C_1)}{P_h(t,y(t))}\,dt\right]. \tag{6.13}$$

Step (iv): Pinning the free function from the boundary condition at $x = 0$.

Along any characteristic with constant $C_1 > 0$, the endpoint at $x = 0$ is $y(0) = C_1/\gamma_1$, giving

$$P\left(0, \frac{C_1}{\gamma_1}\right) = P_y\left(\frac{C_1}{\gamma_1}\right),$$

which is finite, being a value of the boundary generating function P_y. However, P_h (Equation (6.9)) contains the factor $x^{-\mu/C_1} \to +\infty$ as $x \to 0^+$ (since $\mu/C_1 > 0$ for $C_1 > 0$), while the integral in Equation (6.13) vanishes as $x_0 \to 0$, since the integration interval shrinks. Therefore

$$\lim_{x\to 0^+} P(x, y(x)) = \lim_{x\to 0^+} P_h(x, y(x)) \cdot F(C_1) = +\infty \cdot F(C_1),$$

which can only be finite if $F(C_1) = 0$. Setting $x_0 = 0$ and $F(C_1) = 0$ in Equation (6.13) yields the integral representation of Claim 1:

$$P(x, y(x)) = P_h(x, y(x)) \int_0^x \frac{h(t; C_1)}{P_h(t, y(t))}\, dt. \tag{6.14}$$

Step (v): Reduction to the Volterra equation.

By Corollary 3 the empty-state probabilities are $\pi_0 = 1 - \rho$ and $\pi(0, 0) = \rho(1 - \rho)$, with diagonal value $P(z, z) = \pi(0, 0)/(1 - \rho z)$. We plug those, along with our inhomogeneous part G (Equation (6.10)) into our forcing function (Equation (6.12))

$$\begin{aligned} h(t; C_1) &= \frac{\mu(t - y(t))\, P_y(y(t)) - \mu t(1 - y(t))\, \pi(0, 0)}{\gamma_1\, t\, y(t)\, (t - y(t))} \\ &= \frac{\mu\, P_y(y(t))}{\gamma_1\, t\, y(t)} - \frac{\mu(1 - y(t))\, \pi(0, 0)}{\gamma_1\, y(t)\, (t - y(t))}. \end{aligned} \tag{6.15}$$

We will evaluate Equation (6.14) along the diagonal —i.e. $x = y = z$, with points of the form (z, z)—, which has $C_1 = (\gamma_1 + \gamma_2)z$ and thus the following parametrisation, for fixed z:

$$y(t; z) = \frac{(\gamma_1 + \gamma_2)z - \gamma_2 t}{\gamma_1}, \tag{6.16}$$

so $y(0; z) = (\gamma_1 + \gamma_2)z/\gamma_1 > z$ and $y(z; z) = z$. Thus $y(t; z) > t$ for all $t \in [0, z)$, which implies $t - y(t; z) < 0$ inside the integration interval. Note the change of sign of the second summand when writing $t - y(t; z) = -(y(t; z) - t)$ in the denominator of Equation (6.15):

$$h\big(t;\, (\gamma_1 + \gamma_2)z\big) = \frac{\mu}{\gamma_1\, t\, y(t; z)}\, P_y\big(y(t; z)\big) + \frac{\mu\, (1 - y(t; z))}{\gamma_1\, y(t; z)\, (y(t; z) - t)}\, \pi(0, 0). \tag{6.17}$$

From Step (ii), $E > 0$, so the factor $(x - y(x; z))^E$ in P_h vanishes as $x \to z^-$, giving $P_h\big(x, y(x; z)\big) \to 0$. At the same time, the integrand in Equation (6.14) has a singularity near $t = z$: the forcing function (6.17) contributes with a $(y(t; z) - t)^{-1}$ factor; and the denominator $P_h\big(t, y(t; z)\big)$, with a $(y(t; z) - t)^{-E}$. So together $h/P_h \sim (y(t; z) - t)^{-1-E}$; since $-1 - E < -1$, the integral diverges as $x \to z^-$. Nevertheless, the product of the two factors is finite: by Step (iv), $P_h\big(x, y(x; z)\big) \int_0^x \frac{h}{P_h}\, dt$ equals the PGF value $P\big(x, y(x; z)\big)$, which converges to $P(z, z)$ as $x \to z^-$. This behaviour aligns with the fact that we are working in the context of PGFs, which are always bounded. By Corollary 3 that value is $\pi(0, 0)/(1 - \rho z)$, so the $0 \cdot \infty$ limit resolves to

$$P(z, z) = \frac{\pi(0, 0)}{1 - \rho z} = \lim_{x\to z^-} P_h\big(x, y(x; z)\big) \int_0^x \frac{h\big(t; (\gamma_1 + \gamma_2)z\big)}{P_h\big(t, y(t; z)\big)}\, dt.$$

Substituting Equation (6.17) and separating the term containing the unknown boundary function P_y from the known remainder, this limit splits into

$$\begin{aligned} \frac{\pi(0, 0)}{1 - \rho z} &= \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x; z)\big)}{\gamma_1} \int_0^x \frac{P_y\big(y(t; z)\big)}{t \cdot y(t; z) \cdot P_h\big(t, y(t; z)\big)}\, dt \\ &\quad + \lim_{x\to z^-} \frac{\mu\, P_h\big(x, y(x; z)\big)\, \pi(0, 0)}{\gamma_1} \int_0^x \frac{1 - y(t; z)}{y(t; z) \cdot (y(t; z) - t) \cdot P_h\big(t, y(t; z)\big)}\, dt. \end{aligned} \tag{6.18}$$

We evaluate the two limits in Equation (6.18) separately. Three facts make the diagonal characteristic exactly solvable.

First, its geometry is linear:

$$y(t;z) - t = \frac{(\gamma_1+\gamma_2)z - \gamma_2 t}{\gamma_1} - t = \frac{\gamma_1+\gamma_2}{\gamma_1}\,(z-t). \tag{6.19}$$

Second, the exponent of Step (ii) is constant along the characteristic ($C_1 = (\gamma_1+\gamma_2)z$) and can be factored as follows:

$$E_z := E\big((\gamma_1+\gamma_2)z\big) = \frac{\lambda z^2 - (\lambda+\mu)z + \mu}{(\gamma_1+\gamma_2)z} = \frac{(\mu-\lambda z)(1-z)}{(\gamma_1+\gamma_2)z} > 0, \qquad z \in (0,1).$$

Third, every factor of P_h in Equation (6.9) except the exponential, the power of x, and $(x-y)^E$ depends on (x,y) only through C_1, which is fixed on the characteristic. The ratio therefore splits nicely into a smooth part and a single power:

$$\frac{P_h\big(x, y(x;z)\big)}{P_h\big(t, y(t;z)\big)} = R(x,t)\left(\frac{z-x}{z-t}\right)^{E_z}, \qquad R(x,t) := e^{\frac{K}{\gamma_1}(x-t)}\left(\frac{t}{x}\right)^{\mu/((\gamma_1+\gamma_2)z)}, \tag{6.20}$$

with R smooth and $R(z,z)=1$.

We apply that to the *second* limit of Equation (6.18). Substituting Equations (6.20) and (6.19), its integrand near $t=z$ is $w(z)(z-t)^{-1-E_z}$ with $w(z) = \mu\pi(0,0)(1-z)/((\gamma_1+\gamma_2)z)$. The two singular factors then balance exactly: the prefactor vanishes like $(z-x)^{E_z}$ while the truncated integral diverges like $(z-x)^{-E_z}/E_z$. Hence their product converges to $w(z)/E_z$:

$$\begin{aligned}&\lim_{x\to z^-}(z-x)^{E_z}\int^x w(z)(z-t)^{-1-E_z}\,dt\\ &= \frac{w(z)}{E_z} = \frac{\mu\pi(0,0)(1-z)}{(\gamma_1+\gamma_2)z}\cdot\frac{(\gamma_1+\gamma_2)z}{(\mu-\lambda z)(1-z)} = \frac{\mu\pi(0,0)}{\mu-\lambda z} = \frac{\pi(0,0)}{1-\rho z}.\end{aligned}$$

The *first* limit vanishes: its integrand behaves as $(z-t)^{-E_z}$, one power less singular than the second limit's $(z-t)^{-1-E_z}$, and is killed by the vanishing prefactor $(z-x)^{E_z}$. The *second* limit reproduces the full diagonal value from $\pi(0,0)$. Thus the diagonal fixes no information about P_y at leading order; determining P_y requires the next order in $(z-x)$, which is the open problem.

Collecting the P_y-term of Equation (6.18) into a kernel and the remaining $\pi(0,0)$-term into a known datum gives

$$\mathcal{K}(z,t) := \frac{\mu}{\gamma_1}\,\frac{R(z,t)}{t\,y(t;z)}\,(z-t)^{-E_z}, \tag{6.21}$$

$$\mathcal{F}(z) := \frac{\pi(0,0)}{1-\rho z} - \frac{\mu\,\pi(0,0)}{\gamma_1}\int_0^z \frac{R(z,t)\,\big(1-y(t;z)\big)}{y(t;z)\,\big(y(t;z)-t\big)}\,(z-t)^{-E_z}\,dt, \tag{6.22}$$

so that the diagonal identity (6.18) matches the first-kind Volterra equation (6.4), which proves the claim. Both integrals are singular at the endpoint $t=z$ —their integrands behave as $(z-t)^{-E_z}$ and $(z-t)^{-1-E_z}$— and must be read in a regularised, finite-part sense; carrying out that regularisation rigorously is precisely the analytical obstruction that leaves Model-B open. □

Model-B is beyond the scope of this thesis and, therefore, left open. Its role is to illustrate the mathematical obstacle —the Volterra-type integral equation. Model-B_2 and Model-C_2 avoid such a problem by shutting down all derivative terms in y. This allows us to take our boundary function $P_y(y)$ outside the integral, as we would only integrate with respect to x, as opposed to $y(x)$.

Remark 3 (The full Model-X is *a fortiori* intractable). Model-B already shows the central problem of taking this route. As one can imagine, reinstating the abandonment mechanism —$\theta_1,\theta_2>0$— does not help simplify the problem; quite the opposite. It deepens the difficulty in two ways: first, the characteristics cease to be straight lines; second, we lose the diagonal anchor —Corollary 3 no longer holds—, so the diagonal $P(z,z)$ is no longer known in closed form.

7 Analysis of Model-B_2 ($\gamma_1>0,\ \gamma_2=0,\ \theta_1=\theta_2=0$)

Model-B_2 keeps one-way jockeying from class-1 to class-2 ($\gamma_1>0,\ \gamma_2=0$) and no abandonment ($\theta_1=\theta_2=0$). This model is more tractable than Model-B, as all the terms with derivatives in

y disappear, leaving us a first-order linear ODE in x, for each fixed $y \in (0, 1]$. We will solve this using the integrating factor (§3.6.1).

Stability. As in Models A and B, $\rho = (\lambda_1+\lambda_2)/\mu < 1$. Jockeying conserves the total customer count and does not affect the stability condition. Figure 6 shows the queue layout.

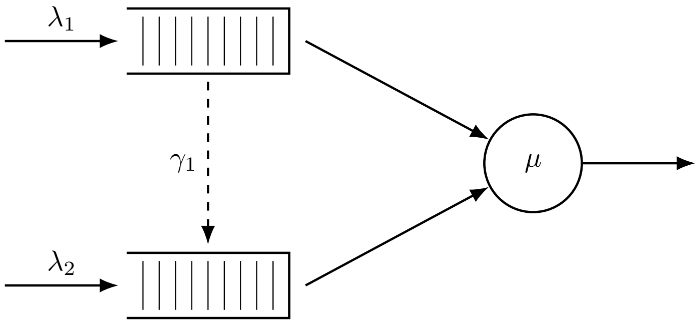

Figure 6: Queue layout for Model-B_2. Only the $1 \to 2$ jockeying direction is active (dashed, downward): class-1 customers may defect to Queue 2, but no customer can move in the reverse direction ($\gamma_2 = 0$).

7.1 Reduced fundamental equation

Setting $\gamma_2 = 0$ and $\theta_1 = \theta_2 = 0$ in the general fundamental equation (4.15) (Lemma 1) annihilates the derivative terms in y, leaving:

$$\begin{aligned}\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2\big] P + \gamma_1 xy(x-y)\,\frac{\partial P}{\partial x}\\ = \mu(x-y)\,P_y(y) - \mu x(1-y)\,\pi(0,0).\end{aligned} \tag{7.1}$$

For each fixed $y \in (0, 1]$ this is a first-order linear ODE in x; the entire analysis of this section is built on it.

Theorem 3. *Consider the two-class non-preemptive priority queueing system of Model-B_2 with per-customer jockeying rate $\gamma_1 > 0$ from Queue 1 to Queue 2 ($\gamma_2 = 0$, $\theta_1 = \theta_2 = 0$). Under the stability condition $\rho = \lambda/\mu < 1$, $\lambda = \lambda_1 + \lambda_2$, the boundary generating function is*

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu - \lambda y} \cdot \frac{{}_1F_1\big(\alpha(y)+1;\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\, b^\star(y);\, -\lambda_1 y/\gamma_1\big)}, \tag{7.2}$$

and the bivariate PGF, for $0 \le x \le y$, is

$$P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\, x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{7.3}$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function, the exponents are

$$\alpha(y) = \frac{\mu}{\gamma_1 y}, \qquad \beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y}, \qquad b^\star(y) = \alpha + \beta + 1 = \frac{\mu + \lambda(1-y) + \gamma_1}{\gamma_1}, \tag{7.4}$$

the auxiliary integrals are

$$\mathcal{H}_1(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha-1}(y-t)^{\beta}\, dt, \tag{7.5}$$

$$\mathcal{H}_2(x,y) = \int_0^x e^{-\lambda_1 t/\gamma_1}\, t^{\alpha}(y-t)^{\beta-1}\, dt, \tag{7.6}$$

and the empty-state probability $\pi(0,0) = \rho(1-\rho)$ is given by Corollary 4 below. On $y < x \le 1$ the same expression (7.3) holds with $\mathcal{H}_1, \mathcal{H}_2$ replaced by their finite-part continuations; that the right-hand side is genuinely analytic across the interior point $x = y$ is the content of the smoothness condition established in Step (ii) of the proof.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models.

Corollary 4. *The empty-system probability π_0 (server idle, no one waiting) and the both-queues-empty probability $\pi(0,0)$ (server busy, $N_1 = N_2 = 0$) of Model-B_2 are*

$$\pi_0 = 1-\rho, \qquad \pi(0,0) = \rho(1-\rho),$$

identical to the Model-A baseline.

As in Model-B, the proof follows trivially by setting $x = y = z$ in our fundamental equation (Equation (7.1)). The surviving equation is identical to the homologous one in Model-A, so Corollary 4 is the no-abandonment instance of Theorem 1, with the $M/M/1$ mean busy period $\mathbb{E}[B] = (\mu-\lambda)^{-1}$ (Corollary 1).

The proof of Theorem 3 consists of three steps: (i) obtaining the standard form and the integration factor, (ii) determining $P_y(y)$ by analyticity in $x = y$, and (iii) recovering $P(x,y)$.

Proof of Theorem 3. **Step (i): Standard form and the integrating factor.**

Fix $y \in (0,1]$, cancel a factor y on both sides and divide the reduced fundamental equation (7.1) by the derivative term in x. The equation takes standard form $\frac{dP}{dx} + p(x;y)\,P = q(x;y)$ of §3.6.1 (Equation (3.12)), with

$$p(x;y) = \frac{(\lambda_1+\lambda_2+\mu)x - \mu - \lambda_1 x^2 - \lambda_2 xy}{\gamma_1 x(x-y)}, \tag{7.7}$$

$$q(x;y) = \frac{\mu\,P_y(y)}{\gamma_1 xy} - \frac{\mu(1-y)\,\pi(0,0)}{\gamma_1 y(x-y)}. \tag{7.8}$$

Write the numerator of p as a polynomial in x,

$$N(x) = -\lambda_1 x^2 + \big[(\lambda_1+\lambda_2+\mu) - \lambda_2 y\big]x - \mu = -\lambda_1\big(x - x^*(y)\big)\big(x - x^+(y)\big),$$

where the roots $x^*(y), x^+(y)$ are those found for the kernel in Model-A (Equation (5.7)), with $\lambda_1 x^* x^+ = \mu$ following from Vieta's formula. The Partial Fraction Decomposition of $p = N(x)/\big[\gamma_1 x(x-y)\big]$ reads as follows:

$$\begin{aligned}
A &= -\lambda_1,\\
B &= \frac{N(0)}{0-y} = \frac{-\mu}{-y} = \frac{\mu}{y},\\
C &= \frac{N(y)}{y} = \frac{-(\lambda y-\mu)(y-1)}{y} = \frac{(1-y)(\lambda y-\mu)}{y},
\end{aligned}$$

with $N(y) = -\lambda y^2 + (\lambda+\mu)y - \mu = -(\lambda y - \mu)(y-1)$. Thus

$$p(x;y) = \frac{1}{\gamma_1}\left[-\lambda_1 + \frac{\mu/y}{x} + \frac{(1-y)(\lambda y-\mu)/y}{x-y}\right]. \tag{7.9}$$

Since we will have a factor $(x-y)$ due to jockeying, we work on the region $0 \le x \le y$, where $x - y \le 0$; the complementary region $y < x \le 1$ is recovered later. On this region, a term $\beta \ln|x-y|$ becomes simply $\beta\ln(y-x)$, so integrating the partial fraction (7.9) yields, up to a constant,

$$\begin{aligned}
\int^x p(t;y)\,dt &= \frac{1}{\gamma_1}\Big[-\lambda_1 x + \frac{\mu}{y}\ln x + \frac{(1-y)(\lambda y-\mu)}{y}\,\ln(y-x)\Big]\\
&= -\frac{\lambda_1}{\gamma_1}\,x + \alpha(y)\ln x + \beta(y)\ln(y-x),
\end{aligned}$$

with the exponents $\alpha(y) = \mu/(\gamma_1 y) > 0$ and $\beta(y) = (1-y)(\lambda y - \mu)/(\gamma_1 y)$of Equation (7.4). Exponentiating, and dropping the multiplicative constant to which an integrating factor is indifferent, gives our integrating factor

$$\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}\, x^{\alpha(y)}\,(y-x)^{\beta(y)},$$

real-valued in our region because we take the base $(y-x) > 0$ rather than $(x-y)$; its reciprocal $1/\mu_I$ is the homogeneous solution, playing here the role that P_h played for Model-B (§3.6.3). It indeed solves $\mu_I'/\mu_I = p$:

$$\frac{\mu_I'(x;y)}{\mu_I(x;y)} = -\frac{\lambda_1}{\gamma_1} + \frac{\alpha}{x} + \beta\,\frac{d}{dx}\ln(y-x)$$

$$= -\frac{\lambda_1}{\gamma_1} + \frac{\alpha}{x} + \frac{\beta}{x-y} = p(x;y).$$

The general solution of §3.6.1 (Equation (3.13)) now applies, and its arbitrary function is fixed immediately: since $\alpha > 0$ we have $\mu_I \to 0$ as $x \to 0^+$, so the boundary contribution $P(0,y)\,\mu_I(0;y)$ vanishes and integrating $\frac{d}{dx}[P\,\mu_I] = q\,\mu_I$ over $(0,x]$ leaves

$$P(x,y)\,\mu_I(x;y) = \int_0^x q(t;y)\,\mu_I(t;y)\,dt.$$

Multiplying q from Equation (7.8) by μ_I and applying the identity $(y-t)^\beta/(t-y) = -(y-t)^{\beta-1}$ to the second term, the integrand separates into two pieces,

$$q(t;y)\,\mu_I(t;y) = \frac{\mu}{\gamma_1 y}\Big[P_y(y)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha-1}(y-t)^\beta + (1-y)\pi(0,0)\,e^{-\lambda_1 t/\gamma_1}\,t^{\alpha}(y-t)^{\beta-1}\Big],$$

With the auxiliary integrals $\mathcal{H}_1, \mathcal{H}_2$ of Equations (7.5)–(7.6) and dividing through by $\mu_I(x;y) = e^{-\lambda_1 x/\gamma_1}x^\alpha(y-x)^\beta$, the solution on the principal region $0 \le x \le y$ is

$$P(x,y) = \frac{\mu\,e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Big[P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\Big], \tag{7.10}$$

which still contains the unknown boundary function $P_y(y)$; Step (ii) determines it.

Step (ii): Determine $P_y(y)$ by analyticity at $x = y$.

Under stability (i.e. $\rho < 1$) $\mu - \lambda y > 0$ for $y \in (0,1)$, so the second exponent is negative:

$$\beta(y) = \frac{(1-y)(\lambda y - \mu)}{\gamma_1 y} < 0, \qquad 0 < y < 1. \tag{7.11}$$

So the prefactor $(y-x)^{-\beta} = (y-x)^{|\beta|}$ in Equation (7.10) *vanishes* as $x \to y^-$, while $\mathcal{H}_1, \mathcal{H}_2$ may *diverge*: the product is an indeterminate $0 \cdot \infty$ form, whose resolution fixes $P_y(y)$.

On the diagonal the value is known outright. Setting $x = y = z$ in the fundamental equation (7.1) drops the jockeying and boundary terms (each containing a factor $x - y$) and leaves Model A's balance, so

$$P(z,z) = \frac{\rho(1-\rho)}{1-\rho z}, \tag{7.12}$$

the diagonal value of Corollary 2.

Smoothness condition and $P_y(y)$. Pull the analytic, non-vanishing prefactor of Equation (7.10) into $\Psi(x) = \mu\,e^{\lambda_1 x/\gamma_1}x^{-\alpha}/(\gamma_1 y)$, so that

$$P(x,y) = \Psi(x)\,(y-x)^{|\beta|}\big[\,P_y(y)\,\mathcal{H}_1(x,y) + (1-y)\pi(0,0)\,\mathcal{H}_2(x,y)\,\big], \qquad |\beta| = -\beta > 0.$$

We need only the leading behaviour of the two integrals as $x \to y^-$, not a full expansion. Each splits into its endpoint value $\mathcal{H}_i(y,y) = \int_0^y (\cdots)\,dt$ and a remainder set by the integrand near $t = y$ —$\mathcal{H}_1$ carries $(y-t)^\beta$, $\mathcal{H}_2$ carries $(y-t)^{\beta-1}$— so that, with constants c_1, c_2 whose values we never need,

$$\mathcal{H}_1(x,y) = \mathcal{H}_1(y,y) + c_1\,(y-x)^{\beta+1} + \cdots, \qquad \mathcal{H}_2(x,y) = \mathcal{H}_2(y,y) + c_2\,(y-x)^{\beta} + \cdots.$$

Multiplied by $(y-x)^{-\beta}$, the c_1, c_2 remainders turn into a smooth $O(y-x)$ term and the constant regular value (7.12); only the endpoint values retain the bare, non-integer power $(y-x)^{|\beta|}$ ($|\beta| \notin \mathbb{Z}$ generically). Since $P(\cdot,y)$ is a power series in x —hence smooth at the interior point $x = y < 1$, whereas that power is not— its coefficient must vanish:

$$P_y(y)\,\mathcal{H}_1(y,y) + (1-y)\,\pi(0,0)\,\mathcal{H}_2(y,y) = 0, \tag{7.13}$$

the endpoint integrals being read, since $\beta < 0$, as the Euler–Beta continuations found next. To evaluate these endpoint integrals, substitute $t = ys$ ($dt = y\,ds$) in Equations (7.5)–(7.6), which rescales the two integrands and pulls out a common $y^{\alpha+\beta}$,

$$t^{\alpha-1}(y-t)^\beta\,dt = y^{\alpha+\beta}\,s^{\alpha-1}(1-s)^\beta\,ds,$$
$$t^{\alpha}(y-t)^{\beta-1}\,dt = y^{\alpha+\beta}\,s^{\alpha}(1-s)^{\beta-1}\,ds,$$

so that, with $c := \lambda_1 y/\gamma_1$,

$$\mathcal{H}_1(y,y) = y^{\alpha+\beta}\int_0^1 s^{\alpha-1}(1-s)^{\beta}e^{-cs}\,ds, \qquad \mathcal{H}_2(y,y) = y^{\alpha+\beta}\int_0^1 s^{\alpha}(1-s)^{\beta-1}e^{-cs}\,ds.$$

These are Euler integrals (3.10), $\int_0^1 t^{a-1}(1-t)^{b-1}e^{zt}\,dt = B(a,b)\,{}_1F_1(a;a+b;z)$, with $(a,b,z) = (\alpha,\beta{+}1,-c)$ for $\mathcal{H}_1$ and $(\alpha{+}1,\beta,-c)$ for $\mathcal{H}_2$, so $a{+}b = b^\star(y)$ in both. As $\beta<0$ by Equation (7.11), the right-hand side is read as its analytic continuation in b beyond the convergence range $b>0$, giving

$$\mathcal{H}_1(y,y) = B(\alpha,\,\beta+1)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big), \tag{7.14}$$

$$\mathcal{H}_2(y,y) = B(\alpha+1,\,\beta)\,y^{\alpha+\beta}\,{}_1F_1\big(\alpha+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big). \tag{7.15}$$

Solving the smoothness condition (7.13) for $P_y(y)$ and inserting the endpoint values (7.14)–(7.15), the common factor $y^{\alpha+\beta}$ cancels and

$$P_y(y) = -(1-y)\,\pi(0,0)\,\frac{\mathcal{H}_2(y,y)}{\mathcal{H}_1(y,y)} = -(1-y)\,\pi(0,0)\,\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)}\,\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\lambda_1 y/\gamma_1\big)}. \tag{7.16}$$

The Beta ratio reduces, via $\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$ and $\Gamma(\beta+1)=\beta\Gamma(\beta)$, to

$$\begin{aligned}\frac{B(\alpha+1,\,\beta)}{B(\alpha,\,\beta+1)} &= \frac{\Gamma(\alpha+1)\Gamma(\beta)}{\Gamma(\alpha+\beta+1)}\cdot\frac{\Gamma(\alpha+\beta+1)}{\Gamma(\alpha)\Gamma(\beta+1)}\\ &= \frac{\Gamma(\alpha+1)}{\Gamma(\alpha)}\cdot\frac{\Gamma(\beta)}{\Gamma(\beta+1)} = \frac{\alpha}{\beta},\end{aligned} \tag{7.17}$$

and the surviving prefactor simplifies, using $\alpha/\beta = \mu/\big[(1-y)(\lambda y-\mu)\big]$, as

$$\begin{aligned}-(1-y)\,\frac{\alpha}{\beta} &= -(1-y)\cdot\frac{\mu}{(1-y)(\lambda y-\mu)}\\ &= -\frac{\mu}{\lambda y-\mu} = \frac{\mu}{\mu-\lambda y}.\end{aligned} \tag{7.18}$$

Substituting Equations (7.17)–(7.18) into Equation (7.16) gives the boundary generating function

$$P_y(y) = \frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha(y)+1;\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha(y);\,b^\star(y);\,-\lambda_1 y/\gamma_1\big)}, \tag{7.19}$$

confirming Equation (7.2). The prefactor equals the regular value $P(y,y)$ of Equation (7.12), and the Kummer ratio is the boundary correction. As $y\to 0^+$ the argument $-\lambda_1 y/\gamma_1 \to 0$, so the ratio tends to 1 and $P_y(0)=\pi(0,0)$, consistent with $P_y(0)=\sum_{n_2}\pi(0,n_2)\cdot 0^{n_2}=\pi(0,0)$.

Step (iii): Recovery of $P(x,y)$.

Substituting Equation (7.19) into the real-form expression (7.10) gives, for $0\le x\le y$,

$$\begin{aligned}P(x,y) = \frac{\mu\, e^{\lambda_1 x/\gamma_1}\,x^{-\alpha}(y-x)^{-\beta}}{\gamma_1 y}\Bigg[&\frac{\mu\,\pi(0,0)}{\mu-\lambda y}\cdot\frac{{}_1F_1\big(\alpha+1;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}{{}_1F_1\big(\alpha;\,b^\star;\,-\frac{\lambda_1 y}{\gamma_1}\big)}\cdot\mathcal{H}_1(x,y)\\ &+(1-y)\pi(0,0)\cdot\mathcal{H}_2(x,y)\Bigg],\end{aligned}$$

which is the statement of the theorem (Equation (7.3)). The smoothness condition (7.13) guarantees that this extends analytically across $x=y$, with diagonal value $P(y,y)=\rho(1-\rho)/(1-\rho y)$ as computed in Equation (7.12). On $y<x\le 1$ the same formula holds with $\mathcal{H}_1,\mathcal{H}_2$ read as their finite-part continuations; $\pi(0,0)=\rho(1-\rho)$ from Corollary 4 makes the solution fully explicit. The recovery of Model-A in the limit $\gamma_1\to 0^+$ is deferred to Remark 4 below, as the limit is singular and merits separate discussion. □

Remark 4 (The singular limit $\gamma_1\to 0^+$). The no-jockeying limit $\gamma_1\to 0^+$ is a *singular* perturbation: the coefficient $\gamma_1 xy(x-y)$ of the highest derivative $\partial_x P$ vanishes, so one must check that the convection term itself, not merely its coefficient, disappears. It does. Each P is a probability generating function, so $|P|\le 1$ on the closed unit polydisc uniformly in γ_1, and Cauchy's estimate then bounds $\partial_x P$ uniformly on every compact subset of the open polydisc; hence $\gamma_1 xy(x-y)\,\partial_x P\to 0$ there.

The fundamental equation (7.1) therefore degenerates to the algebraic Model-A equation (5.2), and the uniformly bounded family $\{P\}_{\gamma_1}$, being normal, converges (in full, by uniqueness of the analytic solution) to the Model-A PGF (5.3). In particular the boundary function reduces to the Model-A value (5.8),

$$P_y(y) \xrightarrow[\gamma_1 \to 0^+]{} \pi(0,0)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}, \tag{7.20}$$

with $x^*(y)$ the Model-A kernel root (5.7); this needs no special-function asymptotics.

The same limit can be read off the closed form (7.19). Its prefactor $\mu\,\pi(0,0)/(\mu-\lambda y) = \pi(0,0)/(1-\rho y)$ is γ_1-independent, so Equation (7.20) is equivalent to the Kummer ratio tending to

$$\frac{{}_1F_1\big(\alpha+1;\, b^\star;\, -\lambda_1 y/\gamma_1\big)}{{}_1F_1\big(\alpha;\, b^\star;\, -\lambda_1 y/\gamma_1\big)} \xrightarrow[\gamma_1 \to 0^+]{} (1-\rho y)\,\frac{x^*(y)\,(1-y)}{x^*(y)-y}. \tag{7.21}$$

The factor $(1-\rho y)$ shows the ratio does *not* tend to the bare $x^*(y)(1-y)/(x^*(y)-y)$; the discrepancy is exactly the prefactor. This matches the first-parameter three-term recurrence for ${}_1F_1$ [DLM, §13.3]: as $\gamma_1 \to 0^+$ the parameters α, $b^\star$ and $c = \lambda_1 y/\gamma_1$ all grow like $1/\gamma_1$, and the recurrence reduces at leading order to a quadratic whose admissible root is Equation (7.21).

8 Analysis of Model-C_2 ($\theta_1 > 0,\ \gamma_1 = \gamma_2 = \theta_2 = 0$)

In this section we derive $P(x,y)$ for the model in which only class-1 customers abandon, with no class-2 abandonment and no jockeying. Figure 7 shows the queue layout.

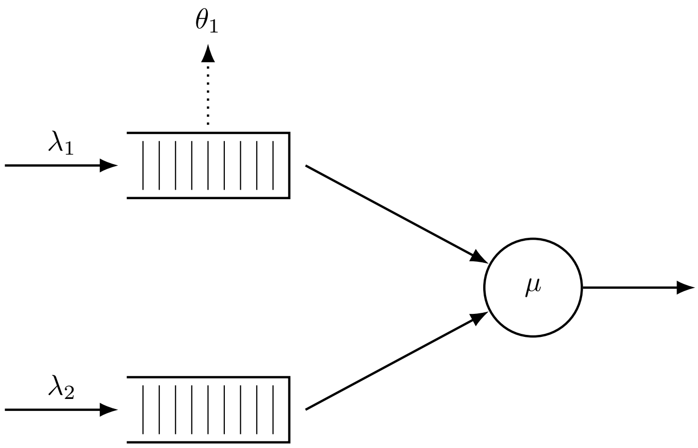

Figure 7: Queue layout for Model-C_2. Class-1 customers abandon the system at rate θ_1 per waiting customer (dotted arrow, upward from Queue 1). Class-2 never abandons ($\theta_2 = 0$). There is no jockeying.

This choice of parameters is motivated by two key aspects of the resulting model: first, class-2 still sees Poisson arrivals, so the probabilistic argument of Model-A carries over; second, substituting these values into the general fundamental equation (4.15) leaves a first-order linear ODE in x with no y-derivatives, so the boundary function $P_y(y)$ comes out of the integral, sparing us the Volterra equation that left Model-B open (§6).

Stability. Unlike the models without abandonment, stability here is non-trivial and rests on two key observations that also drive the analysis below. First, the class queue 1 alone is an $M/M/1{+}M$ (Erlang-A) subsystem with rates λ_1, μ, θ_1, which is stable for *every* load (§3.3); its busy period B_C is always well defined. Second, class-2 sees an $M/G/1$ queue whose service time is one such class-1 busy period B_C (made precise in §8.3). The only constraint on stability comes from this second queue, which is stable exactly when its utilisation stays below one:

$$\lambda_2\,\mathbb{E}[B_C] \;<\; 1 \qquad \Longleftrightarrow \qquad \rho_2 \cdot {}_1F_1\Big(1;\, \tfrac{\mu}{\theta_1}+1;\, \tfrac{\lambda_1}{\theta_1}\Big) \;<\; 1. \tag{8.1}$$

8.1 Reduced fundamental equation

Setting $\theta_1 > 0$ and $\gamma_1 = \gamma_2 = \theta_2 = 0$ in the general fundamental equation (4.15) clears most of the derivative terms, with the class-1 abandonment term as the only survivor. What remains is, for each fixed y, a first-order linear ODE in x:

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1 x^2 y-\lambda_2 x y^2\right] P(x,y)+\theta_1\, xy(x-1)\,\frac{\partial}{\partial x}P(x,y)\\ &\quad=\quad \mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned}\tag{8.2}$$

Equation (8.2) has two unknowns: $P(x,y)$ and its boundary trace $P_y(y)$.

The following theorem states the closed form. Its proof, given immediately afterwards in §8.2, is purely analytical; §8.3 then re-derives the boundary function $P_y(y)$ probabilistically as an independent method to determine the boundary function.

Theorem 4. *Consider the two-class non-preemptive priority system of Model-C_2 ($\theta_1>0$, $\gamma_1=\gamma_2=\theta_2=0$): Poisson arrivals at rates λ_1,λ_2, common exponential service at rate μ, and per-customer abandonment of* waiting *class-1 customers at rate θ_1. Under the stability condition (8.1), the probability generating function is*

$$\begin{aligned}P(x,y)\;&=\;\frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\mu/\theta_1}(1-x)^{\lambda_2(1-y)/\theta_1}\big(\widetilde{B}_C(\lambda_2(1-y))-y\big)}\\ &\quad\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y)-\big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big],\end{aligned}$$

where

$$\begin{aligned}\mathcal{H}_1(x,y)\;&=\;\int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\mu/\theta_1-1}(1-t)^{\,\lambda_2(1-y)/\theta_1}\,dt,\\ \mathcal{H}_2(x,y)\;&=\;\int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\mu/\theta_1}(1-t)^{\,\lambda_2(1-y)/\theta_1-1}\,dt,\end{aligned}$$

$\widetilde{B}_C(s)$ is the LST of the busy period of the class-1 $M/M/1{+}M$ subsystem given by Equation (3.3), and $\pi(0,0)$ is given by Corollary 5 below.

The following corollary is invoked:

Corollary 5. *The limiting probability $\pi(0,0)$ of the busy state with both queues empty, and the empty-system probability π_0, are*

$$\pi(0,0)\;=\;\frac{(\lambda_1+\lambda_2)\big(1-\lambda_2\,\mathbb{E}[B_C]\big)}{\mu\,\big(1+\lambda_1\,\mathbb{E}[B_C]\big)}\;=\;(\rho_1+\rho_2)\,\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]},\tag{8.3}$$

$$\pi_0\;=\;\frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]},\tag{8.4}$$

where $\mathbb{E}[B_C]$ is the mean class-1 busy period (3.4). As $\theta_1\to 0^+$, $\mathbb{E}[B_C]\to(\mu-\lambda_1)^{-1}$ and these recover the Model-A baseline $\pi(0,0)=\rho(1-\rho)$, $\pi_0=1-\rho$.

8.2 Analytical derivation of $P(x,y)$

Proof of Theorem 4. **Step (i): standard form and integrating factor.**

The surviving abandonment factor $(x-1)$ places the operative singularity on the boundary $x=1$. Dividing Equation (8.2) by the coefficient $\theta_1 xy(x-1)$ of the derivative puts it in the standard form (Equation (3.12)):

$$\frac{\partial P}{\partial x}+\underbrace{\frac{(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy}{\theta_1\,x(x-1)}}_{p(x;\,y)}\,P\;=\;\underbrace{\frac{\mu(x-y)\,P_y(y)-\mu x(1-y)\,\pi(0,0)}{\theta_1\,xy(x-1)}}_{q(x;\,y)}.$$

The numerator of $p(x;y)$ is the quadratic $A(x,y):=(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy$, the kernel polynomial common to all seen models. Applying Partial Fraction Decomposition to $A(x;y)/(x(x-1))$ yields

$$\begin{aligned}&A=\lambda_1,\\ &B=A(0,y)/(0-1)=(-\mu)/(-1)=\mu,\\ &C=A(1,y)=\lambda_2(1-y);\end{aligned}$$

so,

$$p(x;y) \;=\; \frac{1}{\theta_1}\cdot\frac{A(x;y)}{x(x-1)} \;=\; \frac{1}{\theta_1}\left[-\lambda_1+\frac{\mu}{x}+\frac{\lambda_2(1-y)}{x-1}\right]. \tag{8.5}$$

Integrating term by term, $\int p\,dx = \theta_1^{-1}\big[-\lambda_1 x+\mu\ln x+\lambda_2(1-y)\ln(1-x)\big]$, which exponentiates to the integrating factor

$$\mu_I(x;y) \;=\; e^{-\lambda_1 x/\theta_1}\,x^{\alpha}\,(1-x)^{\beta(y)}, \tag{8.6}$$

with

$$\alpha \;=\; \frac{\mu}{\theta_1}, \qquad \beta(y) \;=\; \frac{\lambda_2(1-y)}{\theta_1}. \tag{8.7}$$

Both exponents are non-negative ($\alpha>0$ always; $\beta(y)\ge 0$ for $y\in[0,1]$), so μ_I vanishes at $x=0$ and at $x=1$. By the integrating-factor solution (3.13) —with the boundary term $P(0,y)\,\mu_I(0;y)$ killed by $\mu_I(0;y)=0$— we obtain

$$P(x,y) \;=\; \frac{1}{\mu_I(x;y)}\int_0^x q(t;y)\,\mu_I(t;y)\,dt. \tag{8.8}$$

Step (ii): determination of $P_y(y)$ by boundedness at $x=1$.

We determine $P_y(y)$ by imposing boundedness of $P(x,y)$ as $x\to 1$, for fixed $y\in[0,1)$: we have $\beta(y)>0$, so $\mu_I(x;y)\to 0$. Since the left-hand side $P(x,y)$ of Equation (8.8) stays bounded while $1/\mu_I(x;y)\to\infty$, the integral must vanish in the limit:

$$\int_0^1 q(t;y)\,\mu_I(t;y)\,dt \;=\; 0. \tag{8.9}$$

Substituting the explicit q and μ_I, the integrand is

$$q(t;y)\,\mu_I(t;y) \;=\; \frac{\mu\big[(t-y)\,P_y(y)-t(1-y)\,\pi(0,0)\big]}{\theta_1\,y}\cdot\frac{e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)}}{t\,(t-1)},$$

where $t^\alpha/t=t^{\alpha-1}$ and $(1-t)^{\beta(y)}/(t-1)=-(1-t)^{\beta(y)-1}$ flips the sign and lowers the exponent. The constant prefactor $-\mu/(\theta_1 y)$, nonzero for $y\in(0,1]$, divides out of Equation (8.9), leaving

$$\int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\big[(t-y)\,P_y(y)-t(1-y)\,\pi(0,0)\big]\,dt \;=\; 0.$$

Introduce the auxiliary integrals

$$\begin{aligned}\mathcal{J}_0(y) &\;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)-1}\,dt,\\ \mathcal{J}_1(y) &\;=\; \int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt.\end{aligned}$$

Distributing the bracket over $t^{\alpha-1}$, we get the term $t\cdot t^{\alpha-1}=t^\alpha$ in $\mathcal{J}_1$ and $t^{\alpha-1}$ in $\mathcal{J}_0$, becoming

$$\begin{aligned}0 &\;=\; P_y(y)\int_0^1 e^{-\lambda_1 t/\theta_1}\big(t^\alpha-y\,t^{\alpha-1}\big)(1-t)^{\beta(y)-1}\,dt \;-\; (1-y)\,\pi(0,0)\int_0^1 e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\,dt\\ &\;=\; P_y(y)\big[\mathcal{J}_1(y)-y\,\mathcal{J}_0(y)\big] \;-\; (1-y)\,\pi(0,0)\,\mathcal{J}_1(y),\end{aligned}$$

so

$$P_y(y) \;=\; \pi(0,0)\,(1-y)\,\frac{\mathcal{J}_1(y)}{\mathcal{J}_1(y)-y\,\mathcal{J}_0(y)}. \tag{8.10}$$

It remains to evaluate the ratio $\mathcal{J}_1/\mathcal{J}_0$. Setting $a=\alpha=\mu/\theta_1$, $b=s/\theta_1$, $c=\lambda_1/\theta_1$, the Euler representation (3.10) writes each $\mathcal{J}_i$ as a Beta multiple of a ${}_1F_1$ value, and the closed-form summation (3.7) of §3.3 identifies their ratio with the busy-period LST:

$$\widetilde{B}_C(s) \;=\; \frac{\displaystyle\int_0^1 t^{a}(1-t)^{b-1}e^{-ct}\,dt}{\displaystyle\int_0^1 t^{a-1}(1-t)^{b-1}e^{-ct}\,dt} \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}. \tag{8.11}$$

Taking $s = \lambda_2(1-y)$ —so that $b = \beta(y)$ matches the exponent in Equation (8.7)— gives $\widetilde{B}_C(\lambda_2(1-y)) = \mathcal{J}_1(y)/\mathcal{J}_0(y)$. Dividing the numerator and denominator of Equation (8.10) by $\mathcal{J}_0(y)$ yields the boundary generating function

$$P_y(y) \;=\; \pi(0,0)\,\frac{(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y)) - y}. \tag{8.12}$$

This reaches $P_y(y)$ analytically through the Beta-integral (Equation (8.11)) alone, without requiring the probabilistic argument in §8.3.

Step (iii): recovery of $P(x,y)$.

We return to the ODE (Equation (8.2)) and substitute $P_y(y)$ (Equation (8.12)) into its rhs. Writing $\zeta(y) \equiv \widetilde{B}_C(\lambda_2(1-y))$ for brevity,

$$\begin{aligned}
\text{RHS} \;&=\; \mu(x-y)\,\pi(0,0)\,\frac{(1-y)\,\zeta(y)}{\zeta(y)-y} \;-\; \mu x(1-y)\,\pi(0,0) \\
&=\; \mu\,\pi(0,0)\,(1-y)\left[\frac{(x-y)\,\zeta(y)}{\zeta(y)-y} - x\right] \\
&=\; \mu\,\pi(0,0)\,(1-y)\,\frac{(x-y)\,\zeta(y) - x\big[\zeta(y)-y\big]}{\zeta(y)-y} \\
&=\; \mu\,\pi(0,0)\,y(1-y)\,\frac{x-\zeta(y)}{\zeta(y)-y},
\end{aligned}$$

which has the same form as the RHS of Model-A, now with the more involved service time B_C. With $p(x;y)$ and $\mu_I(x;y)$ as in Equations (8.5) and (8.6), the standard-form source becomes

$$q(x;y) \;=\; \frac{\mu\,\pi(0,0)\,(1-y)\,\big[x-\zeta(y)\big]}{\theta_1\,x(x-1)\,\big[\zeta(y)-y\big]},$$

and Equation (8.8) reads $P(x,y) = \mu_I(x;y)^{-1}\int_0^x q(t;y)\,\mu_I(t;y)\,dt$. To carry out the integral, use the partial fraction

$$\frac{t-\zeta(y)}{t(t-1)} \;=\; \frac{\zeta(y)(t-1) + t\big(1-\zeta(y)\big)}{t(t-1)} \;=\; \frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1},$$

and define the auxiliary integrals

$$\begin{aligned}
\mathcal{H}_1(x,y) \;&=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha-1}(1-t)^{\,\beta(y)}\,dt, \\
\mathcal{H}_2(x,y) \;&=\; \int_0^x e^{-\lambda_1 t/\theta_1}\,t^{\,\alpha}(1-t)^{\,\beta(y)-1}\,dt,
\end{aligned}$$

with $\alpha, \beta(y)$ as in Equation (8.7). They are related to the full integrals of Step-(ii) in the following way: $\mathcal{H}_2(1,y) = \mathcal{J}_1(y)$ exactly, and $\mathcal{H}_1(1,y) \neq \mathcal{J}_0(y)$ (its $(1-t)$ exponent is $\beta(y)$, not $\beta(y)-1$).

Multiplying $q(t;y)$ by $\mu_I(t;y) = e^{-\lambda_1 t/\theta_1} t^{\alpha}(1-t)^{\beta(y)}$ and inserting the partial fraction, the integrand splits into the two pieces defining $\mathcal{H}_1, \mathcal{H}_2$:

$$\begin{aligned}
q(t;y)\,\mu_I(t;y) \;&=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\left[\frac{\zeta(y)}{t} + \frac{1-\zeta(y)}{t-1}\right] e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)} \\
&=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha-1}(1-t)^{\beta(y)} - \big(1-\zeta(y)\big)\,e^{-\lambda_1 t/\theta_1}\,t^{\alpha}(1-t)^{\beta(y)-1}\Big],
\end{aligned}$$

the second term again using $(1-t)^{\beta(y)}/(t-1) = -(1-t)^{\beta(y)-1}$, which produces both the minus sign and the exponent shift on $\mathcal{H}_2$. Since $\pi(0,0)$ and $\zeta(y)$ leave the integral,

$$\int_0^x q(t;y)\,\mu_I(t;y)\,dt \;=\; \frac{\mu\,\pi(0,0)\,(1-y)}{\theta_1\big(\zeta(y)-y\big)}\Big[\zeta(y)\,\mathcal{H}_1(x,y) \;-\; \big(1-\zeta(y)\big)\,\mathcal{H}_2(x,y)\Big].$$

Dividing by $\mu_I(x;y)$ and restoring $\zeta(y) = \widetilde{B}_C(\lambda_2(1-y))$, we arrive at the closed form

$$\begin{aligned}
P(x,y) \;&=\; \frac{e^{\lambda_1 x/\theta_1}\,\mu\,\pi(0,0)\,(1-y)}{\theta_1\,x^{\alpha}(1-x)^{\beta(y)}\big(\widetilde{B}_C(\lambda_2(1-y)) - y\big)} \\
&\quad\times\Big[\widetilde{B}_C(\lambda_2(1-y))\,\mathcal{H}_1(x,y) - \big(1-\widetilde{B}_C(\lambda_2(1-y))\big)\,\mathcal{H}_2(x,y)\Big],
\end{aligned} \tag{8.13}$$

which is the expression in the statement of the theorem. □

The non-trivial proof for the corollary invoked in Theorem 4 comes next

Proof of Corollary 5. The idle system state, with limiting probability π_0, only communicates to the busy-server-empty-queues state with probability $\pi(0,0)$. No abandonments can take place in either state, as the queues are empty in both cases. By the cut identity of Theorem 1 (Equation (4.8)),

$$(\lambda_1+\lambda_2)\,\pi_0 \;=\; \mu\,\pi(0,0), \tag{8.14}$$

so $\pi(0,0) = (\rho_1+\rho_2)\,\pi_0$. We again use the $M/G/1$ structure identified in §8.3. The long-run fraction of time this queue is busy is $\rho_B = \lambda_2\,\mathbb{E}[B_C]$, and "idle" refers exclusively to no class-2 customers being present. So

$$\mathbb{P}(\text{no class-2 present}) \;=\; 1-\lambda_2\,\mathbb{E}[B_C].$$

Conditional on not having class-2 customers, the priority subsystem evolves as the $M/M/1+M$ queue (λ_1,μ,θ_1) defining B_C, which alternates idle and busy periods in a renewal structure, whose regenerations are the moments it empties. An idle period lasts until the next class-1 arrival takes place, i.e. at an exponential rate λ_1. We have

$$\mathbb{P}(\text{class-1 absent} \mid \text{no class-2 present}) \;=\; \frac{1/\lambda_1}{1/\lambda_1+\mathbb{E}[B_C]} \;=\; \frac{1}{1+\lambda_1\,\mathbb{E}[B_C]}.$$

The fully idle event is the intersection of $A = \{\text{class-1 absent}\}$ and $B = \{\text{no class-2 present}\}$, so $\mathbb{P}(A\cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$ (not independence),

$$\pi_0 \;=\; \big(1-\lambda_2\,\mathbb{E}[B_C]\big)\cdot\frac{1}{1+\lambda_1\,\mathbb{E}[B_C]} \;=\; \frac{1-\lambda_2\,\mathbb{E}[B_C]}{1+\lambda_1\,\mathbb{E}[B_C]},$$

which is Equation (8.4); substituting into Equation (8.14) gives Equation (8.3). Equivalently, this is the renewal–reward form $\pi_0 = 1/\big(1+\lambda\,\mathbb{E}[B]\big)$ of Theorem 1 with whole-system mean busy period $\mathbb{E}[B] = \mathbb{E}[B_C]/\big(1-\lambda_2\,\mathbb{E}[B_C]\big)$ —the mean busy period of the $M/G/1$ queue in which each class-2 arrival (rate λ_2) carries a service requirement of one class-1 busy period B_C. □

8.3 Alternative derivation of $P_y(y)$: the $M/G/1$ / Pollaczek–Khinchine route

The boundary function $P_y(y)$ admits a second derivation, using the same probabilistic argument seen in §5.1.2 that does not require imposing boundedness. By the *law of total expectation*, splitting $P_y(y) = \mathbb{E}[\mathbf{1}\{\text{busy}, N_1 = 0\}\,y^{N_2}]$ over the event $\{\text{busy}, N_1 = 0\}$ factors it into the mass of that event and the conditional PGF of N_2 given it:

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1=0)\,\cdot\,\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1=0\big]\,. \tag{8.15}$$

We adopt the point of view of a class-2 customer while the server is busy. Unlike jockeying, class-1 customers abandon the system, so the class-2 Poisson arrival stream still holds. The class-2 *effective service time* is therefore one full busy period B_C of the class-1 subsystem —an $M/M/1{+}M$ queue with rates λ_1,μ,θ_1, rather than the pure $M/M/1$ of Model-A. So class-2 sees an $M/G/1$ queue with Poisson input λ_2 and service distributed as B_C, whose LST $\widetilde{B}_C(s)$ and mean $\mathbb{E}[B_C]$ come from the first-step analysis of §3.3 (Equations (3.3) and (3.4)). This $M/G/1$ is stable if and only if its utilisation $\lambda_2\,\mathbb{E}[B_C] < 1$ (Equation (8.1)), a condition strictly weaker than the no-abandonment requirement $\rho = \rho_1+\rho_2 < 1$.

Applying the Pollaczek–Khinchine formula (3.8) with the identifications $\lambda \mapsto \lambda_2$, $\widetilde{B} \mapsto \widetilde{B}_C$, $\rho_B = \lambda_2\,\mathbb{E}[B_C]$ gives the PGF of the stationary number of class-2 customers in that $M/G/1$:

$$L_2(y) \;=\; \frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}.$$

Since class-2 arrivals are Poisson, by *PASTA* (§3.2) the number of class-2 customers an arrival finds equals the time-average; conditioning on $\{\text{busy},\, N_1=0\}$ identifies that count with N_2:

$$\mathbb{E}\big[y^{N_2}\,\big|\,\text{busy},\, N_1=0\big] \;=\; L_2(y). \tag{8.16}$$

Substituting Equation (8.16) into the decomposition (8.15) gives

$$P_y(y) \;=\; \mathbb{P}(\text{busy},\, N_1=0)\,\frac{(1-\lambda_2\mathbb{E}[B_C])\,(1-y)\,\widetilde{B}_C(\lambda_2(1-y))}{\widetilde{B}_C(\lambda_2(1-y))-y}. \tag{8.17}$$

The unknown $\mathbb{P}(\text{busy}, N_1 = 0)$ is determined by $P_y(0) = \pi(0,0)$: evaluating Equation (8.17) at $y = 0$, where $\widetilde{B}_C(\lambda_2)$ cancels,

$$\begin{aligned}\pi(0,0) = P_y(0) &= \mathbb{P}(\text{busy},\ N_1 = 0)\,\frac{(1-\lambda_2\mathbb{E}[B_C])\cdot 1\cdot \widetilde{B}_C(\lambda_2)}{\widetilde{B}_C(\lambda_2)}\\ &= \mathbb{P}(\text{busy},\ N_1 = 0)\,(1-\lambda_2\mathbb{E}[B_C]),\end{aligned}$$

so $\mathbb{P}(\text{busy}, N_1 = 0) = \pi(0,0)/(1-\lambda_2\mathbb{E}[B_C])$. Substituting back, the factor $(1-\lambda_2\mathbb{E}[B_C])$ cancels and we recover exactly Equation (8.12), in agreement with the analytical route. The result also matches the probabilistic argument for Model-A (Equation (5.8)); the only difference is the effective service time.

Finally, this route shows the Kummer fraction on the probabilistic side. The busy-period LST that the Pollaczek–Khinchine formula feeds on is, from the Erlang-A analysis (§3.3, Equation (3.7)), the ratio of confluent hypergeometric functions

$$\widetilde{B}_C(s) \;=\; \frac{B(a+1,b)\,{}_1F_1(a+1;\,a+1+b;\,-c)}{B(a,b)\,{}_1F_1(a;\,a+b;\,-c)}, \qquad a=\frac{\mu}{\theta_1},\quad b=\frac{s}{\theta_1},\quad c=\frac{\lambda_1}{\theta_1}.$$

This is the very expression (8.11) that the analytical route produced as the Beta-integral ratio $\mathcal{J}_1(y)/\mathcal{J}_0(y)$ at $s = \lambda_2(1-y)$. The two derivations therefore meet at this Kummer fraction: what boundedness delivers as a ratio of Euler integrals is precisely the LST of the class-1 busy period.

9 Analysis of Model-B_2^{H} ($\gamma_1 > 0$ head-of-line, $\gamma_2 = \theta_1 = \theta_2 = 0$)

Inspired by the work of Devos et al. [DWFB20], this model —together with Model-C_2^{H}— eliminates the state-dependent jockeying rate for the class-1, only allowing the head-of-line customer from class-1 to transition at a fixed rate γ_1 whenever $n_1 \geq 1$, giving an aggregate rate $\gamma_1\,\mathbf{1}_{\{n_1\geq 1\}}$.

This significantly simplifies the model. The flat rate replaces the derivative term in x found in §7 by an algebraic factor $\gamma_1 y(x-y)$ so the fundamental equation remains algebraic and can be solved by the Kernel Method exactly as in Model-A (§5).

Stability. Jockeying conserves the total number of customers; since there is no abandonment, the chain is positive recurrent if and only if the total traffic intensity satisfies

$$\rho = \frac{\lambda_1+\lambda_2}{\mu} < 1. \tag{9.1}$$

9.1 Balance equations

The jockeying transition $(n_1, n_2) \to (n_1 - 1, n_2 + 1)$ happens at the fixed rate γ_1 for $n_1 \geq 1$. On state space S the global balance equations read

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu+\gamma_1]\,\pi(n_1,n_2)\\ &\quad= \quad \mu\,\pi(n_1+1,n_2)+\lambda_1\pi(n_1-1,n_2)+\lambda_2\pi(n_1,n_2-1)+\gamma_1\pi(n_1+1,n_2-1) \qquad n_1\geq 1,\ n_2\geq 1,\end{aligned} \tag{9.2}$$

$$[\lambda_1+\lambda_2+\mu+\gamma_1]\,\pi(n_1,0) \;=\; \mu\,\pi(n_1+1,0)+\lambda_1\pi(n_1-1,0) \qquad n_1\geq 1,\ n_2=0, \tag{9.3}$$

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu]\,\pi(0,n_2)\\ &\quad= \quad \mu\,\pi(1,n_2)+\gamma_1\pi(1,n_2-1)+\lambda_2\pi(0,n_2-1)+\mu\,\pi(0,n_2+1) \qquad n_1=0,\ n_2\geq 1,\end{aligned} \tag{9.4}$$

$$\begin{aligned}&[\lambda_1+\lambda_2+\mu]\,\pi(0,0) = (\lambda_1+\lambda_2)\pi_0+\mu\,\pi(1,0)+\mu\,\pi(0,1),\\ &(\lambda_1+\lambda_2)\pi_0 = \mu\,\pi(0,0).\end{aligned} \tag{9.5}$$

9.2 Fundamental PGF equation

Lemma 4 (Fundamental equation for Model-B_2^{H})**.** *Assume positive recurrence, so that the stationary distribution exists and $P(x,y) = \sum_{n_1,n_2\geq 0}\pi(n_1,n_2)x^{n_1}y^{n_2}$ is analytic on the closed unit polydisc. Then $P(x,y)$ and the boundary function $P_y(y) = P(0,y)$ satisfy*

$$\begin{aligned}&\left[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2+\gamma_1y(x-y)\right]P(x,y)\\ &\quad= \quad [\mu(x-y)+\gamma_1y(x-y)]\,P_y(y)-\mu x(1-y)\,\pi(0,0).\end{aligned} \tag{9.6}$$

Proof. Besides the jockeying terms, equations (9.2)–(9.5) coincide with those of Model-A. Multiplying each balance equation by $x^{n_1}y^{n_2}$, summing over its domain and multiplying through by xy, we obtain the fundamental equation for Model-A (Equation (5.2)):

$$\big[(\lambda_1+\lambda_2+\mu)xy-\mu y-\lambda_1x^2y-\lambda_2xy^2\big]\,P(x,y)=\mu(x-y)P_y(y)-\mu x(1-y)\,\pi(0,0),$$

so it remains only to collect the jockeying contribution. Its *out*-flow appears in every state with $n_1\ge 1$:

$$\gamma_1\sum_{n_1\ge1}\sum_{n_2\ge0}\pi(n_1,n_2)\,x^{n_1}y^{n_2}=\gamma_1\big[P(x,y)-P_y(y)\big],$$

while its *in*-flow enters every state with $n_2\ge 1$ from (n_1+1,n_2-1):

$$\gamma_1\sum_{n_1\ge0}\sum_{n_2\ge1}\pi(n_1+1,n_2-1)\,x^{n_1}y^{n_2}=\gamma_1\,\frac{y}{x}\sum_{m_1\ge1}\sum_{m_2\ge0}\pi(m_1,m_2)\,x^{m_1}y^{m_2}=\gamma_1\,\frac{y}{x}\big[P(x,y)-P_y(y)\big],$$

where $m_1=n_1+1,\ m_2=n_2-1$. The jockeying contribution (out minus in) is hence $\gamma_1\big(1-\frac{y}{x}\big)\big[P(x,y)-P_y(y)\big]$; multiplying by xy gives $\gamma_1 y(x-y)\big[P(x,y)-P_y(y)\big]$. Adding this to the Model-A equation and moving the P_y part to the right-hand side yields Equation (9.6). □

9.3 Closed-form solution

The following theorem is the main result of this section; its proof is based on the analytical (Kernel Method) treatment of Model-A in §5.1.1.

Theorem 5 (Closed-form PGF for Model-B_2^{H})**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1,λ_2, common exponential service rate μ, and head-of-line class-1 jockeying at rate $\gamma_1>0$ (with $\gamma_2=\theta_1=\theta_2=0$). Under the stability condition (9.1), the joint PGF $P(x,y)=\mathbb{E}[x^{N_1}y^{N_2}]$ is the rational function*

$$P(x,y)\;=\;\frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y)-y\big)\,\big(x^+(y)-x\big)},\tag{9.7}$$

where $x^*(y)$ and $x^+(y)$ are the two roots of the kernel quadratic

$$\lambda_1x^2-\big[\lambda_1+\lambda_2(1-y)+\mu+\gamma_1\big]x+(\mu+\gamma_1y)=0,\tag{9.8}$$

with $x^*(y)$ the root inside the unit disc.

The result below is stated as a corollary, so that it sits at the same hierarchy level as the corresponding results for the other models.

Corollary 6. *Under Equation (9.1), the empty-state probabilities of Model-B_2^{H} coincide with those of Model-A,*

$$\pi_0=1-\rho,\qquad \pi(0,0)=\rho(1-\rho).$$

In particular $P(1,1)=1-\pi_0=\rho$ and the diagonal generating function satisfies $P(z,z)=\pi(0,0)/(1-\rho z)$.

The proof is the no-abandonment instance of Theorem 1 (Corollary 1): head-of-line jockeying conserves the total count, so $\mathbb{E}[B]=(\mu-\lambda)^{-1}$ and the baseline values hold.

9.3.1 Analytical approach for Model-B_2^{H}

Proof of Theorem 5. We apply the *Kernel Method* to the fundamental equation (9.6). Fix $y\in(0,1]$ and seek the values of x at which the coefficient of $P(x,y)$ vanishes. Since $P(x,y)$ is a PGF it is bounded on the closed unit disc, so at any such root the right-hand side of Equation (9.6) must also vanish. Dividing the kernel by $y\neq 0$,

$$0=(\lambda_1+\lambda_2+\mu)x-\mu-\lambda_1x^2-\lambda_2xy+\gamma_1(x-y),$$

and rearranging into a quadratic in x gives exactly Equation (9.8),

$$\lambda_1x^2-\big[\lambda_1+\lambda_2(1-y)+\mu+\gamma_1\big]x+(\mu+\gamma_1y)=0,$$

whose two roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2 - 4\lambda_1(\mu + \gamma_1 y)}}{2\lambda_1}, \qquad A(y) := \lambda_1 + \lambda_2(1-y) + \mu + \gamma_1.$$

We label the root carrying the negative sign $x^*(y)$ and the other $x^+(y)$. By Vieta's formulas the product of the roots is $x^*(y)\, x^+(y) = (\mu + \gamma_1 y)/\lambda_1$.

To locate the roots relative to the unit disc we evaluate the kernel quadratic, written as $f(x) := \lambda_1 x^2 - A(y)x + (\mu + \gamma_1 y)$, at $x = 1$:

$$f(1) = \lambda_1 - \big[\lambda_1 + \lambda_2(1-y) + \mu + \gamma_1\big] + (\mu + \gamma_1 y) = -(1-y)\,(\lambda_2 + \gamma_1) < 0 \qquad \text{for } y \in [0,1).$$

We have that$x^*(y)$ is the unique root inside the unit disc, and at $y = 1$ one checks $f(1) = 0$ with $x^*(1) = 1$ and $x^+(1) = (\mu + \gamma_1)/\lambda_1 > 1$. For later use, we record $\frac{d}{dy}x^*(1)$. The cleanest route is implicit differentiation of $f(x,y) := \lambda_1 x^2 - A(y)x + (\mu + \gamma_1 y) = 0$ at $(x,y) = (1,1)$, where $f_y = \lambda_2 x + \gamma_1$ and $f_x = 2\lambda_1 x - A(y)$ evaluate to $f_y(1,1) = \lambda_2 + \gamma_1$ and $f_x(1,1) = \lambda_1 - \mu - \gamma_1$, so $\frac{d}{dy}x^*(1) = -f_y/f_x$:

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2 + \gamma_1}{\mu + \gamma_1 - \lambda_1}. \tag{9.9}$$

Unlike Model-A's Equation (5.6), the numerator carries an extra γ_1: the constant term $\mu + \gamma_1 y$ of the kernel (9.8) is itself y-dependent here.

Setting $x = x^*(y)$ in Equation (9.6) forces the right-hand side to vanish:$(x^*(y) - y)(\mu + \gamma_1 y)\, P_y(y) = \mu x^*(y)(1-y)\,\pi(0,0)$, whence the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1-y)\,\pi(0,0)}{(\mu + \gamma_1 y)\,\big(x^*(y) - y\big)}. \tag{9.10}$$

Finally, substitute Equation (9.10) and $\pi(0,0) = \rho(1-\rho)$ (Corollary 6) back into Equation (9.6). Writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$ and simplifying the right-hand side,

$$\begin{aligned}\big[\mu(x-y) + \gamma_1 y(x-y)\big] P_y(y) - \mu x(1-y)\,\pi(0,0) &= \mu(1-y)\,\pi(0,0)\left[\frac{(x-y)\,x^*(y)}{x^*(y) - y} - x\right] \\ &= \mu(1-y)\,\pi(0,0)\,\frac{y\,\big(x - x^*(y)\big)}{x^*(y) - y},\end{aligned}$$

where the bracket collapses because $(x-y)x^* - x(x^* - y) = y(x - x^*)$. Dividing by the factored kernel, the common factor $\big(x - x^*(y)\big)$ cancels out and we have

$$P(x,y) = \frac{\mu(1-y)\,\pi(0,0)\,y\,(x - x^*(y))/(x^*(y) - y)}{-\lambda_1 y\,(x - x^*(y))(x - x^+(y))} = \frac{\mu\,\rho(1-\rho)\,(1-y)}{\lambda_1\,\big(x^*(y) - y\big)\,\big(x^+(y) - x\big)},$$

which matches Equation (9.7) and concludes the proof. □

10 Analysis of Model-C_2^{H} ($\theta_1 > 0$ head-of-line, $\gamma_1 = \gamma_2 = \theta_2 = 0$)

Model-C_2^{H} is the abandonment counterpart of Model-B_2^{H}, and the head-of-line simplification of the class-1 abandonment model Model-C_2. Only the *customer at the head of Queue 1* abandons, at the fixed rate θ_1 whenever $n_1 \geq 1$. As in Model-B_2^{H}, the flat rate $\theta_1\,\mathbf{1}_{\{n_1 \geq 1\}}$ contributes an *algebraic* term rather than a derivative, and the Kernel Method applies directly. True departure breaks the conservation of the total number of customers that we had in models without abandonment.

Stability. Because only the head-of-line from class-1 abandons, the aggregate abandonment rate is bounded by θ_1 regardless of queue length as long as we have a queue, so a genuine stability condition is required. As shown in Corollary 7 below, the chain is positive recurrent if and only if $\pi_0 > 0$, that is

$$(\mu + \theta_1)(\mu - \lambda) + \lambda_1\theta_1 \;>\; 0, \qquad \lambda = \lambda_1 + \lambda_2. \tag{10.1}$$

This is *weaker* than $\rho < 1$: if $\rho < 1$ then $\mu - \lambda > 0$ and the condition holds trivially, but it can also hold with $\rho \geq 1$ provided θ_1 is large enough. Moreover, Equation (10.1) implies $\rho_C := \lambda_1/(\mu + \theta_1) < 1$. Hence $\mu + \theta_1 > \lambda_1$, i.e. $\rho_C < 1$.

10.1 Balance equations

The abandonment transition $(n_1, n_2) \to (n_1 - 1, n_2)$ occurs at the fixed rate θ_1 whenever $n_1 \geq 1$. Since both a service completion and a head abandonment from $(n_1 + 1, n_2)$ land in (n_1, n_2), the two mechanisms combine into the coefficient $\mu + \theta_1$ on the in-flows. On S,

$$\begin{aligned}
&[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, n_2) \\
&\quad = \quad (\mu + \theta_1)\pi(n_1 + 1, n_2) + \lambda_1 \pi(n_1 - 1, n_2) + \lambda_2 \pi(n_1, n_2 - 1) && n_1 \geq 1,\ n_2 \geq 1, \quad (10.2)\\
&[\lambda_1 + \lambda_2 + \mu + \theta_1]\, \pi(n_1, 0) \;=\; (\mu + \theta_1)\pi(n_1 + 1, 0) + \lambda_1 \pi(n_1 - 1, 0) && n_1 \geq 1,\ n_2 = 0,\\
&[\lambda_1 + \lambda_2 + \mu]\, \pi(0, n_2) \\
&\quad = \quad (\mu + \theta_1)\pi(1, n_2) + \lambda_2 \pi(0, n_2 - 1) + \mu\, \pi(0, n_2 + 1) && n_1 = 0,\ n_2 \geq 1,\\
&[\lambda_1 + \lambda_2 + \mu]\, \pi(0, 0) = (\lambda_1 + \lambda_2)\pi_0 + (\mu + \theta_1)\pi(1, 0) + \mu\, \pi(0, 1),\\
&(\lambda_1 + \lambda_2)\pi_0 = \mu\, \pi(0, 0). && (10.3)
\end{aligned}$$

10.2 Fundamental PGF equation

We introduce the following result:

Lemma 5 (Fundamental equation for Model-C_2^{H})**.** *Assume positive recurrence. Then the joint PGF $P(x, y)$ and the boundary function $P_y(y) = P(0, y)$ satisfy*

$$\begin{aligned}
&\big[(\lambda_1 + \lambda_2 + \mu)xy - \mu y - \lambda_1 x^2 y - \lambda_2 x y^2 + \theta_1 y(x - 1)\big]\, P(x, y) \\
&\quad = \quad [\mu(x - y) + \theta_1 y(x - 1)]\, P_y(y) - \mu x(1 - y)\, \pi(0, 0).
\end{aligned} \tag{10.4}$$

Proof. As in Lemma 4, the non-abandonment terms of Equations (10.2)–(10.3) reproduce the Model-A fundamental equation (5.2) after multiplication by $x^{n_1} y^{n_2}$, summation, and multiplication by xy. The abandonment *out*-flow appears in every state with $n_1 \geq 1$,

$$\theta_1 \sum_{n_1 \geq 1} \sum_{n_2 \geq 0} \pi(n_1, n_2)\, x^{n_1} y^{n_2} = \theta_1 \big[P(x, y) - P_y(y)\big],$$

and its *in*-flow enters every state (n_1, n_2) from $(n_1 + 1, n_2)$,

$$\theta_1 \sum_{n_1 \geq 0} \sum_{n_2 \geq 0} \pi(n_1 + 1, n_2)\, x^{n_1} y^{n_2} = \theta_1\, x^{-1} \sum_{m_1 \geq 1} \sum_{n_2 \geq 0} \pi(m_1, n_2)\, x^{m_1} y^{n_2} = \theta_1\, x^{-1} \big[P(x, y) - P_y(y)\big],$$

with $m_1 = n_1 + 1$. The net contribution is $\theta_1 \big(1 - x^{-1}\big)\big[P(x, y) - P_y(y)\big]$; multiplying by xy gives $\theta_1 y(x - 1)\big[P(x, y) - P_y(y)\big]$. Adding to the Model-$A$ equation and rearranging yields Equation (10.4). □

10.3 Closed-form solution

Theorem 6 (Closed-form PGF for Model-C_2^{H})**.** *Consider the two-class non-preemptive priority queue with Poisson arrival rates λ_1, λ_2, common exponential service rate μ, and head-of-line class-1 abandonment at rate $\theta_1 > 0$ (with $\gamma_1 = \gamma_2 = \theta_2 = 0$). Under the stability condition (10.1), the joint PGF $P(x, y) = \mathbb{E}[x^{N_1} y^{N_2}]$ is the rational function*

$$P(x, y) \;=\; \frac{\mu(\mu + \theta_1)\,(1 - y)\,\pi(0, 0)}{\lambda_1\,\big(x^{+}(y) - x\big)\,\big[(\mu + \theta_1 y)\, x^{*}(y) - y(\mu + \theta_1)\big]}, \tag{10.5}$$

where $x^{*}(y)$ and $x^{+}(y)$ are the two roots of the kernel quadratic

$$\lambda_1 x^2 - \big[\lambda_1 + \lambda_2(1 - y) + \mu + \theta_1\big]x + (\mu + \theta_1) = 0, \tag{10.6}$$

with $x^{*}(y)$ the root inside the unit disc, and $\pi(0, 0)$ is given by Corollary 7. As $\theta_1 \to 0^{+}$ the formula recovers Model-A (Theorem 2) exactly.

The theorem invokes the following Corollary:

Corollary 7. *Under Equation (10.1), the empty-state probabilities of Model-C_2^{H} are*

$$\pi_0 = \frac{(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1}{\mu(\mu + \theta_1) + \lambda_1 \theta_1}, \qquad \pi(0, 0) = \frac{\lambda\big[(\mu + \theta_1)(\mu - \lambda) + \lambda_1 \theta_1\big]}{\mu\big[\mu(\mu + \theta_1) + \lambda_1 \theta_1\big]}, \tag{10.7}$$

with $\lambda = \lambda_1 + \lambda_2$. At $\theta_1 \to 0^{+}$ these reduce to $\pi_0 \to 1 - \rho$ and $\pi(0, 0) \to \rho(1 - \rho)$, recovering Model-$A$.

Proof of Corollary 7. Two evaluations of the fundamental equation (10.4) suffice. First set $y = 1$: the term $\mu x(1-y)\pi(0,0)$ vanishes, the coefficient of $P(x,1)$ factors as $-(x-1)\big(\lambda_1 x - (\mu+\theta_1)\big)$, and the RHS becomes $(x-1)(\mu+\theta_1)P_y(1)$. Cancelling $(x-1)$ we obtain

$$P(x,1) \;=\; \frac{(\mu+\theta_1)\,P_y(1)}{\mu+\theta_1-\lambda_1 x} \;=\; \frac{P_y(1)}{1-\rho_C\,x}, \qquad \rho_C := \frac{\lambda_1}{\mu+\theta_1}. \tag{10.8}$$

This geometric marginal shows that the priority queue clears at the effective rate $\mu+\theta_1$. Evaluating at $x = 1$ and using $P(1,1) = 1-\pi_0$ gives

$$1-\pi_0 \;=\; \frac{P_y(1)}{1-\rho_C} \quad\Longrightarrow\quad P_y(1) \;=\; (1-\pi_0)(1-\rho_C). \tag{10.9}$$

Second, set $x = 1$ in Equation (10.4). The coefficient of $P(1,y)$ collapses to $\lambda_2 y(1-y)$ and the right-hand side to $\mu(1-y)\big[P_y(y)-\pi(0,0)\big]$, yielding the following relation

$$\lambda_2\, y\, P(1,y) \;=\; \mu\big[P_y(y)-\pi(0,0)\big]. \tag{10.10}$$

Letting $y \to 1$ and using Equation (10.9) and the cut identity of Theorem 1 ($\pi(0,0) = \lambda\pi_0/\mu$, Equation (10.3)), we have

$$\lambda_2(1-\pi_0) \;=\; \mu\big[P_y(1)-\pi(0,0)\big] \;=\; \mu(1-\pi_0)(1-\rho_C)-\lambda\pi_0.$$

Expanding and collecting the π_0 terms,

$$\big[\mu(1-\rho_C)+\lambda-\lambda_2\big]\pi_0 = \mu(1-\rho_C)-\lambda_2, \qquad \text{i.e.} \qquad \pi_0 = \frac{\mu(1-\rho_C)-\lambda_2}{\mu(1-\rho_C)+\lambda_1},$$

where $\lambda-\lambda_2 = \lambda_1$. Substituting $\mu(1-\rho_C) = \mu(\mu+\theta_1-\lambda_1)/(\mu+\theta_1)$ and multiplying numerator and denominator by $\mu+\theta_1$, the numerator becomes $\mu(\mu+\theta_1-\lambda_1)-\lambda_2(\mu+\theta_1) = (\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1$ and the denominator $\mu(\mu+\theta_1-\lambda_1)+\lambda_1(\mu+\theta_1) = \mu(\mu+\theta_1)+\lambda_1\theta_1$, yielding the first identity in Equation (10.7); the second follows directly from $\pi(0,0) = \lambda\pi_0/\mu$. The denominator $\mu(\mu+\theta_1)+\lambda_1\theta_1$ is strictly positive, so $\pi_0 > 0$ is equivalent to the stability condition (10.1). In the renewal–reward form $\pi_0 = 1/\big(1+\lambda\,\mathbb{E}[B]\big)$ of Theorem 1, this is a whole-system mean busy period $\mathbb{E}[B] = (\mu+\theta_1)/\big[(\mu+\theta_1)(\mu-\lambda)+\lambda_1\theta_1\big]$, which reduces to the $M/M/1$ value $(\mu-\lambda)^{-1}$ as $\theta_1 \to 0^+$. $\square$

10.3.1 Analytical approach for Model-C_2^{H}

Proof of Theorem 6. We again apply the Kernel Method to Equation (10.4). Dividing the coefficient of $P(x,y)$ by $y \neq 0$ and rearranging gives the kernel quadratic (10.6),

$$\lambda_1 x^2 - \big[\lambda_1+\lambda_2(1-y)+\mu+\theta_1\big]x + (\mu+\theta_1) = 0,$$

Its roots are

$$x^{\pm}(y) = \frac{A(y) \pm \sqrt{A(y)^2-4\lambda_1(\mu+\theta_1)}}{2\lambda_1}, \qquad A(y) := \lambda_1+\lambda_2(1-y)+\mu+\theta_1, \tag{10.11}$$

with product $x^*(y)\,x^+(y) = (\mu+\theta_1)/\lambda_1$ by Vieta's formulas. Evaluating the quadratic $f(x) := \lambda_1 x^2 - A(y)x + (\mu+\theta_1)$ at $x = 1$,

$$f(1) = \lambda_1 - A(y) + (\mu+\theta_1) = -\lambda_2(1-y) < 0 \qquad \text{for } y \in [0,1),$$

So $x = 1$ lies between the two roots and $x^*(y) < 1 < x^+(y)$ for $y \in [0,1)$; at $y = 1$, $x^*(1) = 1$ and $x^+(1) = (\mu+\theta_1)/\lambda_1 > 1$ (the latter by $\rho_C < 1$). Hence $x^*(y)$ is the unique root inside the unit disc. Differentiating Equation (10.11) at $y = 1$,

$$\frac{d}{dy}x^*(1) = \frac{\lambda_2}{\mu+\theta_1-\lambda_1}. \tag{10.12}$$

Setting $x = x^*(y)$ in Equation (10.4) forces the RHS to vanish,

$$\big[\mu(x^*(y)-y)+\theta_1 y(x^*(y)-1)\big]P_y(y) = \mu x^*(y)(1-y)\,\pi(0,0),$$

and, regrouping the bracket as $\mu(x^*-y)+\theta_1 y(x^*-1) = (\mu+\theta_1 y)x^* - y(\mu+\theta_1)$, the boundary function is

$$P_y(y) = \frac{\mu\, x^*(y)\,(1-y)\,\pi(0,0)}{(\mu+\theta_1 y)\,x^*(y)-y(\mu+\theta_1)}. \tag{10.13}$$

Write $\Delta(y) := (\mu + \theta_1 y)x^*(y) - y(\mu + \theta_1)$ for the denominator of Equation (10.13); at $y = 0$, $\Delta(0) = \mu\, x^*(0)$ (with $x^*(0) > 0$), so $P_y(0) = \pi(0,0)$, exactly as for Model-B_2^{H} at Equation (9.10).

Substituting Equation (10.13) into Equation (10.4) and writing the kernel as $-\lambda_1 y\,(x - x^*(y))(x - x^+(y))$, the right-hand side becomes

$$\big[\mu(x-y)+\theta_1 y(x-1)\big]P_y(y)-\mu x(1-y)\,\pi(0,0) = \frac{\mu(1-y)\,\pi(0,0)}{\Delta(y)}\Big\{\big[\mu(x-y)+\theta_1 y(x-1)\big]x^*(y)-x\,\Delta(y)\Big\}.$$

Substituting $\Delta(y)$, it simplifies to $(\mu + \theta_1)\, y\, (x - x^*(y))$. Dividing by the factored kernel cancels $(x - x^*(y))$, yielding

$$P(x,y) = \frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)\,y\,(x-x^*(y))/\Delta(y)}{-\lambda_1 y\,(x-x^*(y))(x-x^+(y))} = \frac{\mu(\mu+\theta_1)(1-y)\,\pi(0,0)}{\lambda_1\,(x^+(y)-x)\,\Delta(y)},$$

which is Equation (10.5) and concludes the proof. □

11 Results

Every closed-form expression derived above is first validated against an exact *continuous-time Markov chain* (CTMC), then used to compare the solved models and quantify the effect of the two waiting-room mechanisms.

11.1 Validation against the CTMC

The method employed is a truncated CTMC: it works by solving the linear system $\pi Q = 0$ directly, with no further approximation than the finite truncation at $N_{\max} = 40$. Each result is therefore verified against the exact limiting distribution at base parameters $\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1.0$, with the active mechanism parameter γ_1 or θ_1 set to 0.5 where applicable.

Table 4 collects the verdict for every result. All results are admitted within a machine tolerance level: lemma residuals and closed-form PGF errors fall between 10^{-7} and 10^{-15}, and the corollary errors lie on the diagonal to 10^{-4} across 60 random configurations each.

Table 4: Validation summary. All mathematical results are accepted; the PPGF approximation is conditionally accepted (accurate only for $\rho_1/\rho \geq 0.6$ and $\rho \leq 0.6$). Max errors reflect CTMC truncation, not formula error.

ID	**Result**	**Model(s)**	**Check**	**Max error**	**Verdict**
L1	Lemma 1	A, B, B$_2$, C$_2$	PGF eq. residual	$< 5 \times 10^{-9}$	Accept
L2	Lemma 2	A, B$_2$, C$_2$	PPGF dynamics residual	$< 2 \times 10^{-15}$	Accept
T-A	Theorem 2	A	$P(x,y)$ on 5×5 grid	$< 4 \times 10^{-8}$	Accept
C-A	Corollary 1	A, B, B$_2$	π_0, $\pi(0,0)$, 60 configurations	$< 10^{-4}$	Accept
T-B2	Theorem 3	B$_2$	$P(x,y)$ on 5×5 grid, $x < y$	$< 3 \times 10^{-7}$	Accept
T-C2	Theorem 4	C$_2$	$P(x,y)$ on 5×5 grid	$< 5 \times 10^{-12}$	Accept
C-C2	Corollary 5	C$_2$	π_0, $\pi(0,0)$, 60 configurations	$< 10^{-4}$	Accept
T-S	Approximation 1	A	ε_∞ over (ρ_1, ρ_2)	0–100%	Conditional
E-B	Theorem 5	B_2^{H}	$P(x,y)$ on 5×5 grid	$< 3 \times 10^{-7}$	Accept
E-C	Theorem 6	C_2^{H}	$P(x,y)$ on 5×5 grid	$< 3 \times 10^{-11}$	Accept

Only the approximation for Model-A in state space $\widetilde{S}$ resolves to a non-satisfactory precision level, only being acceptable in a priority-dominated setup. This was expected, as the approximation only neglected the problematic non-homogeneous term in Lemma 2, coming from the boundary-diagonal term.

Rows C-A and C-C2 combined validate Theorem 1: the identity $\pi(0,0) = \rho\,\pi_0$ and the busy-period law $\pi_0 = 1/(1+\lambda\,\mathbb{E}[B])$ hold to $< 10^{-4}$ across both the no-abandonment regime (Corollary 1, Models A, B, B_2) and the class-1 abandonment regime (Corollary 5, Model-C_2), confirming the single scalar $\mathbb{E}[B]$ as the only model-dependent input.

11.2 Structural comparison of the solved models

With every closed form validated, we compare the five solved models: Model-A (baseline), Model-B_2 (class-1 jockeying), Model-C_2 (class-1 abandonments) and the two head-of-line variants Model-

B_2^{H} and Model-C_2^{H}. The first three are more tractable simplifications of Model-X, while the head-of-line models are further simplifications of Model-B_2 and Model-C_2, respectively. Each produces a closed-form or integral-form expression for $P(x, y)$. Table 5 collects the key structural facts across all five solved models.

Table 5: Structural comparison of the five solved models. Replacing a length-proportional mechanism rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a head-of-line rate ($\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$) collapses the governing ODE back to the algebraic Model-A form, so $P(x, y)$ becomes rational again.

	Model-A	**Model-B_2**	**Model-B_2^{H}**	**Model-C_2**	**Model-C_2^{H}**
Extra mechanism (rate)	none	jockey $1 \to 2$, $\gamma_1 n_1$	jockey $1 \to 2$, $\gamma_1 \mathbf{1}_{\{n_1 \geq 1\}}$	abandon, $\theta_1 n_1$	abandon, $\theta_1 \mathbf{1}_{\{n_1 \geq 1\}}$
Fundamental equation	algebraic	1st-order ODE	algebraic	1st-order ODE	algebraic
Pinning of $P_y(y)$	kernel root $x^*(y)$	analyticity at $x = y$	kernel root $x^*(y)$	boundedness at $x = 1$	kernel root $x^*(y)$
$P(x, y)$ form	rational	integral $+ {}_1F_1$	rational	integral $+ {}_1F_1$	rational
Conserves total N?	yes	yes	yes	no	no
$\pi(0, 0)$	$\rho(1 - \rho)$	$\rho(1 - \rho)$	$\rho(1 - \rho)$	(8.3)	(10.7)
π_0	$1 - \rho$	$1 - \rho$	$1 - \rho$	$> 1 - \rho$, (8.4)	(10.7)
Stability condition	$\rho < 1$	$\rho < 1$	$\rho < 1$	$\lambda_2 \mathbb{E}[B_C] < 1$	(10.1)

Model-A: the rational baseline

Model-A is a classical, well-known model, so it acts as a baseline model to which all the rest should reduce as their active mechanism parameters approach zero. All performance measures have a closed-form representation, and it also shows some structural identities that persist across other models, such as the conservation of the total number of customers for other models without abandonments —for those with abandonments, it shows how the performance measures change.

Effect of jockeying: Model-A $\to$ Model-B_2

Adding one-way jockeying ($\gamma_1 > 0$, $\gamma_2 = 0$) couples the two queues: a class-1 customer waiting in the queue for class-2 may defect to the queue for class-2 at rate γ_1. The consequences are asymmetric.

What is preserved. Jockeying merely reallocates customers, so the total count of customers in the queues is preserved and remains to be that of the standard $M/M/1$ queueing model. Consequently $P(z, z)$, π_0, $\pi(0, 0)$, and $\mathbb{E}[N]$ are identical to Model-A for all $\gamma_1 > 0$.

What no longer holds. Jockeying breaks the Poisson arrival stream. In the case of Model-B_2, the arrivals of class-2, as they can either arrive by a normal Poisson process or by jockeying from the queue of class-1, which adds a state-dependent rate. The Pollaczek-Khinchine route is therefore not available.

What is new. The PGF satisfies a first-order linear ODE in x, for each fixed y. The solution is found by means of an integrating factor with a singularity at the point $x = y$; therefore the singularity must vanish since P is a PGF and, hence, analytic. This narrows it down to a ratio of Kummer functions.

Effect of abandonment: Model-A $\to$ Model-C_2

Adding class-1 abandonment ($\theta_1 > 0$, $\theta_2 = 0$, $\gamma_i = 0$) allows each waiting class-1 customer to leave the system at a per-customer rate θ_1.

What is preserved. Class-2 customers still see Poisson arrivals, as abandonment in class-1 does not alter the stream of arrivals to the queue for class-2. Therefore, the PASTA property still appries, and so does the Pollaczek-Khinchine probabilistic approach to derive $P_y(y)$.

What no longer holds. Since abandonment is a true departure from the system, the total number of customers is no longer preserved, requiring a more mathematically involved analysis compared to Corollary 2 to determine the stability conditions, the idle system probability π_0 and the busy-server-empty-queues probability $\pi(0, 0)$.

What is new. The PGF satisfies a first-order linear ODE in x, for fixed y, as in Model-B. Here the singularity is found at the boundary $x = 1$, with a positive exponent so that the integrating factor μ_I vanishes there and $\int_0^1 q\, \mu_I\, dt = 0$ is imposed by boundedness of the PGF.

Mechanism contrast: Models B_2 and $\mathbf{C}_2$

Both models reduce the fundamental equation to a first-order linear ODE in x and both close in confluent hypergeometric functions; but the mechanism that determines the boundary function $P_y(y)$ is structurally different.

	Model-C_2 ($\theta_1 > 0$)	**Model-B_2** ($\gamma_1 > 0$)
Convection coefficient	$\theta_1\, x(x-1)$	$\gamma_1\, x(x-y)$
Operative singular point	boundary $x = 1$	interior $x = y$
Exponent at singularity	$\beta(y) = \lambda_2(1-y)/\theta_1 > 0$ $(\mu_I \to 0)$	$\beta(y) = (1-y)(\lambda y-\mu)/(\gamma_1 y) < 0$ $(\mu_I \to \infty)$
Pinning principle	*boundedness*: $\int_0^1 q\, \mu_I\, dt = 0$	*analyticity*: branch coefficient $= 0$
$\pi(0,0)$	$M/G/1$+renewal (conservation broken)	$\rho(1-\rho)$ exactly (conservation holds)
Probabilistic backbone	class-2 is $M/G/1$; P-K with B_C	class-1 is $M/M/1{+}M$ (reneging $= \gamma_1$); no P-K

In words: for Model-C_2 the singularity sits on the boundary $x = 1$ with a positive exponent, so μ_I vanishes and $P_y(y)$ is determined by the PGF being bounded; and the arrival stream for class-2 remains Poisson, so the Pollaczek-Khinchine and $M/G/1$ renewal argument yield $P_y(y)$ and $\pi(0,0)$. In contrast, for Model-B_2 the singularity is found at the interior point $x = y$ with a negative exponent, so μ_I blows up and only analycity determines $P_y(y)$. Jockeying leaves no room for the Pollaczek-Khinchine identity; the $M/M/1{+}M$ reading of the class-1 marginal is the origin of the ${}_1F_1$ factors.

Common Erlang-A structure

A common object appears in both Model-B_2 and Model-c_2:

$$ {}_1F_1\left(1;\ \frac{\mu}{r}+1;\ \frac{\lambda_1}{r}\right), \tag{11.1} $$

with $r = \gamma_1$ (Model-B_2) or $r = \theta_1$ (Model-C_2). This comes from the Erlang-A substructure: in C_2 it equals $\mu\,\mathbb{E}[B_C]$, while in B_2 it controls the Kummer ratio in $P_y(y)$ through the exponent $\alpha(y) = \mu/(\gamma_1 y)$. In both cases, the function evaluates a negative argument $(-\lambda_1/r)$ in C_2 and $(-\lambda_1 y/\gamma_1)$ in B_2, and the ratio ${}_1F_1(\alpha+1;\cdot;\cdot)/{}_1F_1(\alpha;\cdot;\cdot)$ acts as a boundary correction to the algebraic formula found in Model-A.

Model-B_2 reduces to Model-A only when $\gamma_1 \to 0^+$, in which the exponents $\alpha, \beta \to \infty$ and the Kummer ratio collapses to the expression $x^*(y)(1-y)/(x^*(y)-y)$ found in Model-A. Similarly, Model-C_2 degenerates to Model-A uniformly as $\theta_1 \to 0^+$, and all formulas also recover their counterparts from Model-A

Head-of-line simplifications: Models B_2^{H} and C_2^{H}

The head-of-line variants of §§9–10 restrict each mechanism to the single customer at the head of the queue for class 1, replacing the length-proportional rate ($\gamma_1 n_1$ or $\theta_1 n_1$) by a constant one ($\gamma_1\, \mathbf{1}_{\{n_1 \geq 1\}}$ or $\theta_1\, \mathbf{1}_{\{n_1 \geq 1\}}$). As a consequence, the derivative terms in our fundamental equations disappear, leaving behind an algebraic expression. For Model-B_2^{H} the result is structurally identical to Model-A, the only change being $\mu \to \mu + \gamma_1 y$ in the kernel, and conservation is retained ($\pi_0 = 1-\rho$, $\pi(0,0) = \rho(1-\rho)$). For Model-C_2^{H} the broken conservation of the number of total customers introduces the factor $(\mu + \theta_1)$ and the stability boundary shifts to Equation (10.1). Each head-of-line model is hence the $k = 1$ step of a ladder that connects between Model-A and its full-rate parent (B_2 or C_2) as the head-of-line restriction is increased to more potential customers.

11.3 Quantitative comparison across load

Keeping $\mu = 1$ and the arrival split fixed at $\lambda_1{:}\lambda_2 = 3{:}4$, we sweep the offered load ρ across $[0.10, 0.95]$ and compute the exact stationary performance measures of the five solved models with

Table 6: Comparison of Models A, B$_2$ and B_2^{H} ($\gamma_1 = 0.5$), and C$_2$ and C_2^{H} ($\theta_1 = 0.5$) at three traffic intensities with $\lambda_1{:}\lambda_2 = 3{:}4$, $\mu = 1$. The primed models use the head-of-line rate $\mathbf{1}_{\{n_1 \geq 1\}}$; the unprimed use the full rate n_1.

ρ	Model	$E[N_1]$	$E[N_2]$	$E[N]$	$E[W_1]$	$E[W_2]$	π_0	$\pi(0,0)$
0.50	A	0.1364	0.3636	0.5000	0.6364	1.2727	0.5000	0.2500
	B$_2$	0.0767	0.4233	0.5000	0.3578	1.4817	0.5000	0.2500
	B_2^{H}	0.0833	0.4167	0.5000	0.3889	1.4583	0.5000	0.2500
	C$_2$	0.0712	0.2521	0.3233	0.3323	0.8823	0.5356	0.2678
	C_2^{H}	0.0778	0.2593	0.3370	0.3630	0.9074	0.5333	0.2667
0.70	A	0.3000	1.3333	1.6333	1.0000	3.3333	0.3000	0.2100
	B$_2$	0.1545	1.4788	1.6333	0.5151	3.6970	0.3000	0.2100
	B_2^{H}	0.1750	1.4583	1.6333	0.5833	3.6458	0.3000	0.2100
	C$_2$	0.1392	0.6967	0.8358	0.4639	1.7417	0.3696	0.2587
	C_2^{H}	0.1591	0.7424	0.9015	0.5303	1.8561	0.3636	0.2545
0.90	A	0.5651	7.5346	8.0997	1.4651	14.6506	0.1000	0.0900
	B$_2$	0.2626	7.8371	8.0997	0.6808	15.2388	0.1000	0.0900
	B_2^{H}	0.3115	7.7882	8.0997	0.8077	15.1437	0.1000	0.0900
	C$_2$	0.2292	1.9245	2.1537	0.5941	3.7421	0.2146	0.1931
	C_2^{H}	0.2760	2.2084	2.4844	0.7157	4.2941	0.2025	0.1823

the truncated CTMC. Figure 8 plots $\mathbb{E}[N]$,$\mathbb{E}[N_1]$, $\mathbb{E}[W_1] = \mathbb{E}[N_1]/\lambda_1$ (Little's Law), and π_0; Table 6 reads the entire set of descriptors at three reference loads.

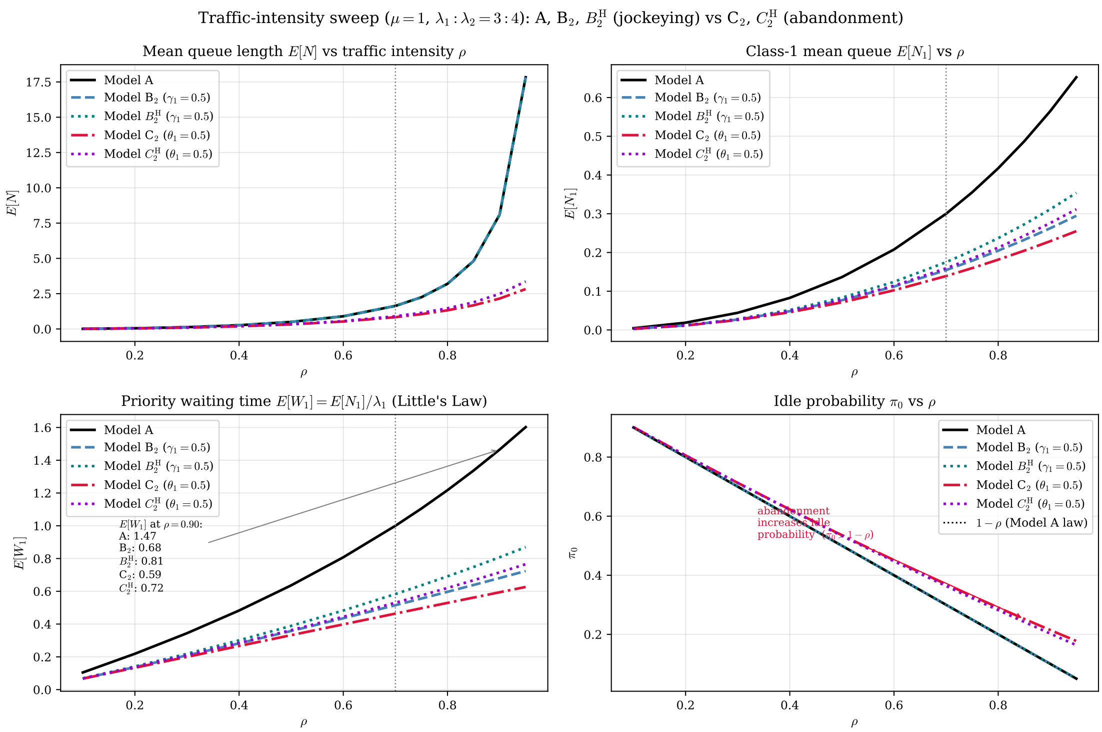

Figure 8: Performance versus traffic intensity ρ for the five solved models —the baseline A, the jockeying pair B_2, B_2^{H} ($\gamma_1 = 0.5$), and the abandonment pair C_2, C_2^{H} ($\theta_1 = 0.5$)— at $\mu = 1$ and $\lambda_1{:}\lambda_2 = 3{:}4$; the vertical line marks the canonical $\rho = 0.70$. *Top-left:* total queue $\mathbb{E}[N]$ —A, B_2 and B_2^{H} coincide because jockeying conserves N (Corollaries 2, 6), while C_2 and C_2^{H} lie strictly below because abandonment removes customers (C_2^{H} by less). *Top-right:* priority queue $\mathbb{E}[N_1]$ —the head-of-line variants drain it less than their full-rate cousins. *Bottom-left:* priority waiting time $\mathbb{E}[W_1]$. *Bottom-right:* idle probability π_0, with the $1-\rho$ law (Models A, B_2, B_2^{H}) shown dotted; C_2 and C_2^{H} sit above it for every ρ, as predicted by Corollaries 5, 7.

The two mechanisms differ fundamentally in the way they operate. Jockeying is an internal rearrangement of customers, whereas abandonment is a true departure: class-1 customers renege at rate $\theta_1 n_1$. This distinction has measurable consequences. Abandonment reduces the overall queue length and raises the idle probability, whereas jockeying merely redistributes the load.

This is why B_2 and A are virtually indistinguishable: jockeying alone cannot lighten the system. Their curves for $\mathbb{E}[N]$ and π_0 overlap across all traffic intensities. Model-C_2, by contrast, stands apart. The priority class sheds load before it reaches the server, keeping both the queue shorter and idle time higher.

The numbers back the claim. At $\rho = 0.70$ (Table 6), Model-C_2 achieves $\pi_0 = 0.3696$ compared to 0.3000 for Model-A and Model-B_2, with $\mathbb{E}[N] = 0.8358$ against 1.6333. At higher loads, the gap widens: $\rho = 0.90$ sees the baseline class-2 queue increase to $\mathbb{E}[N_2] = 7.5349$, while Model-C_2 keeps it at 1.9245. The two head-of-line models set at a middle ground, interpolating between A and their full-rate parents across every metric: Model-B_2^{H} does not alter the queue-conservation property of Model-B_2 but shifts the class-1 share, whereas Model-C_2^{H} sheds less load than Model-C_2 ($\pi_0 = 0.3636$, $\mathbb{E}[N] = 0.9015$ at $\rho = 0.70$).

Two more things to be noted. First, $\mathbb{E}[W_1]$ is smallest for Model-C_2 at every load, but $\mathbb{E}[W_i] = \mathbb{E}[N_i]/\lambda_i$ counts waiting time in queue-i, which is shortened by every departure mode. So smaller $\mathbb{E}[W_1]$ does not certify faster service. Second, jockeying actively *penalises* class 2: at $\rho = 0.70$, Model-B_2 raises $\mathbb{E}[W_2]$ to 3.6970 from the baseline 3.3333 (+10.9%).

11.4 Convergence to the baseline

Each of the four simplified models must recover the baseline Model-A as their parameters for the active mechanisms approach zero: Model-B_2 and Model-B_2^{H} as $\gamma_1 \to 0^+$, Model-C_2 and Model-C_2^{H} as $\theta_1 \to 0^+$. Some of these limits have already been sketched in thos work's derivations, and we recover $\pi_0 \to 1-\rho$ and $\pi(0,0) \to \rho(1-\rho)$ (Corollary 5). Here we confirm both limits numerically, on a single synoptic axis, and quantify the rate of approach.

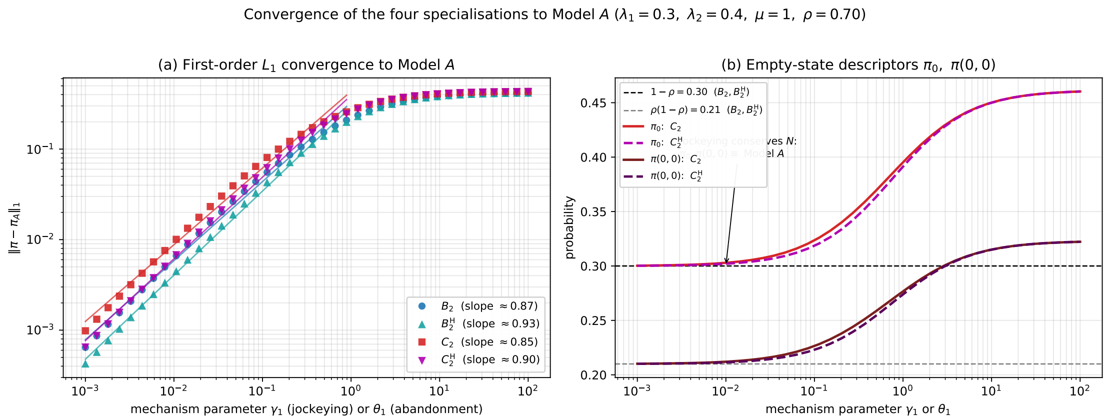

Figure 9: Convergence of the four specialisations to Model-A on a single axis ($\lambda_1 = 0.3$, $\lambda_2 = 0.4$, $\mu = 1$, $\rho = 0.70$). *(a)* Log–log plot of the L_1 distance $\|\pi - \pi_A\|_1$ between the full joint distributions against the distinguishing mechanism parameter (γ_1 for the jockeying pair B_2, B_2^{H}; θ_1 for the abandonment pair C_2, C_2^{H}); all four collapse onto the baseline at first order, with the fitted log–log slopes shown in the legend. *(b)* The empty-state descriptors π_0 and $\pi(0,0)$: the conserving models B_2 and B_2^{H} remain pinned to the Model-A values $1-\rho$ and $\rho(1-\rho)$ (dashed) for every γ_1, so they converge by interior redistribution alone, whereas C_2 and C_2^{H} drift above both lines and relax back onto them only as $\theta_1 \to 0^+$.

To measure how closely each simplified model approaches Model-A, we use the L_1 distance $\|\pi - \pi_A\|_1$ between their full joint distributions. As each model's distinguishing parameter decreases to zero (jockeying rate γ for Model-B_2, abandonment rate θ_1 for Model-C_2, and so on), all four variants converge back to the baseline. We capture this convergence by plotting the L_1-distance against the parameter on a log–log scale and measuring the slope.

Theory expects a slope of exactly 1. Our results, shown in Figure 9(a), fall between 0.85 and 0.93 —acceptableto validate the theoretical prediction. The small gap below unity seems to be motivated by having truncated the CTMC. When we discretize the state space for numerical computation and then fit a straight line to noisy data on a log–log plot, small numerical errors compound and pull the measured slope down slightly. The underlying convergence rate is indeed linear.

The two mechanisms are shown to converge to the baseline along fundamentally different paths, visible in Figure 9(b) by tracking the idle-state probabilities π_0 and $\pi(0,0)$.

Under jockeying: these probabilities remain frozen at their Model-A values. By Corollaries 4 and 6, we have $\pi_0 = 1-\rho$ and $\pi(0,0) = \rho(1-\rho)$ identically for every $\gamma_1 > 0$, due to conservation of the total number of customers. Therefore, both Model-B_2 and Model-B_2^{H} lie exactly on the Model-A baseline throughout the entire parameter sweep. Their convergence to Model-A is invisible at the level of idle probabilities; instead, it happens entirely in the interior, where mass gradually redistributes between the class-1 and class-2 coordinates.

Under abandonment: the idle-state probabilities do change as abandonment is a true departure. They only recover Model-A as $\theta_1 \to 0^+$. For Model-C_2 and Model-C_2^{H}, convergence is thus driven by both mechanisms: the idle state drifts, and the interior mass redistributes.

This conservation under jockeying versus drift under abandonment is the same conservation dichotomy that distinguishes the two mechanisms throughout the entire comparison (§11.2). Here, it simply appears at the convergence endpoint.

12 Conclusion and Future Work

This thesis analysed a two-class non-preemptive priority $M/M/1$ queue under two waiting-room mechanisms —jockeying and abandonment. It obtained the exact stationary bivariate PGF $P(x,y)$ in closed or integral form for five models: the baseline Model-A, one-way $(1 \to 2)$ jockeying Model-B_2, class-1 abandonment Model-C_2, and the head-of-line variants Model-B_2^{H} and Model-C_2^{H}.

One common property is shared by all solved models: the algebraic form of the fundamental equation that the PGF satisfies (Lemma 1) determins the tractability of the model with the methods employed in this work. As a result, we have less closed-form solutions than intended at the beginning of this thesos: having derivative terms in y complicates the problem, as one would need more advanced mathematical tools to determine the boundary function $P_y(y)$.

12.1 The obstruction

Remark 1 reads the fundamental equation as a competition between an algebraic coefficient and two derivative terms, and the analysis highlights a clean dichotomy, summarised model by model in the comparison of §11.2 (Table 5). Where the equation is solvable, the unknown boundary $P_y(y)$ function is determined by exactly one of four mechanisms. The first is evaluation at a kernel root lying strictly inside the unit disc, the purely algebraic route of the baseline Model-A (Section 5). The second is interior analyticity: when one-way jockeying survives, the factor $(x-y)$ places the operative singularity at the diagonal point $x = y$ inside the disc, and the boundary function is fixed by requiring analysity there (Model-B_2, Section 7). The third is boundary boundedness: when class-1 abandonment survives, the factor $(x-1)$ moves the singularity on the unit-circle boundary $x = 1$, where mere boundedness of the PGF suffices to determine the unknown (Model-C_2, Section 8). The fourth is head-of-line algebraic collapse: restricting a mechanism to the head-of-line customer replaces the length-proportional rate with a constant one, eliminates the derivative term in the equation, and recovers the purely algebraic form of Model-A, solvable by the same kernel argument (Models B_2^{H} and C_2^{H}, Sections 9–10).

The unsolvable cases—Model-B and Model-X expose structural limits, as the derivative terms in y make the task of determining $P_y(y)$ significantly more challenging than it already was. Model-B requires solving a Volterra-type integral equation whose forcing term depends on the boundary function itself; Model-X further compounds this difficulty with non-linear characteristic curves, although we have not determined the exact difficulty or feasibility of the problem. The hierarchy is therefore not a loose collection of cases, but a complete enumeration: four mechanisms that enable exact solutions and two obstructions that prevent them.

12.2 The mechanism dichotomy

The two mechanisms act in opposite registers. Jockeying *redistributes* delay at fixed total work: relocating a waiting customer conserves the total count, so $\mathbb{E}[N]$, π_0 and $\pi(0,0)$ are identical to the baseline for every γ_1. The mechanism's entire effect is to move delay between classes —the priority premium $\mathbb{E}[W_2]/\mathbb{E}[W_1]$ inflating to 22.38 at $\rho = 0.90$ against the Model-A value 10.00.

Abandonment instead *reduces* the load: the idle probability rises, every queue shortens, and total waiting times fall.

12.3 Operational reading and the deterioration-direction caveat

This subsections closes the healthcare-related loop opened in the Introduction.

These results close the healthcare loop opened in the Introduction. The original clinical motivation was the $2 \to 1$ direction, rather than the jockeying in the opposite direction: a non-priority patient deteriorating and moving into the priority class. That mechanism —an upgrade driven by clinical urgency— is precisely what Model-B captures via $\gamma_2 > 0$. It is also, as the analysis shows, the direction that leads to a Volterra-type integral equation (Section 6). The mechanism the clinical setting most wants is thus the one the analysis shows as irreducibly hard.

However, the solved models show some valuable results. Model-B_2 implements the opposite direction, $1 \to 2$: a priority patient is reclassified downward, whether by clinical recovery, administrative review, or triage revision. This is less common, but not unreal. More immediately useful is Model-C_2, which captures class-1 abandonment —the rate θ_1 at which a waiting high-priority patient exits the list through death, or transfer to another facility. Abandonment of priority patients is a realistic feature of any stretched healthcare system, and having exact closed-form expressions for π_0, $P(x, y)$, and the idle probability makes these results directly applicable for capacity planning and optimisation. The same, can be said about the head-of-line models, where it is not completely unreasonable that the first in line can be reallocated somewhere else.

12.4 Limitations

Several assumptions are intrinsic to the $M/M/1$ framework itself. All results rest on exponential interarrival, service, jockeying, and abandonment times —an idealisation that is most consequential for service times. Allowing class-dependent rates $\mu_1 \neq \mu_2$ would invalidate the analysis at its foundation, since the state-space construction and the definition of the limiting probabilities rely both on the server being indifferent to the class in service. Relaxing exponentiality entirely to general service times further compounds the difficulty; the intractability that already forces de Walal's approximate $M/G/1$ treatment illustrates how quickly the problem escapes closed-form reach.

The partial-PGF result on the alternative state space $\widetilde{S}$ is an approximation, not a theorem (Approximation 1), and should be read as a documented negative result. The rationale was as follows. Cohen [Coh56] applied a partial generating function on the standard state space (n_1, n_2) with respect to the class-2 variable, leaving the tractable class-1 subsystem untouched and transforming only in the harder direction. In the presence of jockeying, neither class has simple dynamics individually, but the total queue length $N = N_1 + N_2$ does. The alternative state space (n_2, n) was therefore introduced to exploit this total-count structure. The attempt does not succeed: rather than simplifying the problem, the change of coordinates makes the balance equations harder even for Model-A, as Lemma 2 demonstrates.

12.5 Future work

Several extensions follow directly from the machinery already built:

1. *Head-of-line versions of the remaining models.* The head-of-line models could be extended with bidirectional active mechanisms —namely Model-B^{H} for jockeying and Model-C^{H} for abandonment— rather than limiting them to strictly one direction, as the derivative terms in the fundamental equation would also disappear, and even a head-of-line model cattying both active mechanisms at the same time, namely Model-X^{H}. This is near-term low-hanging fruit that does not seem to require new mathematical tools.

2. *The k-customer head-of-line ladder.* The head-of-line variants let only the single customer at the head of the queue for class 1 jockey or abandon; the natural next step lets the first *two* waiting class-1 customers do so, then the first three, and so on (and likewise $\theta_1 \min(n_1, k)$). Writing $\min(n_1, k) = \sum_{j=1}^{k} \mathbf{1}_{\{n_1 \geq j\}}$, the bounded rate keeps the fundamental equation *algebraic* at the POTENTIAL price of k unknown boundary functions. This route has not been explored in this work, so it is unclear what it could yield. One would expect that such models would converge to Model-B_2 and Model-C_2 as $k \to \infty$.

3. *Numerical inversion of the Model-B Volterra equation.* Unlike Model-X, Model-B yields a first-kind Volterra equation with an *explicit* kernel and a *known* forcing term (Section 6). A person with more advanced mathematical resources could either try to find a solution analytically or inverting it via numerical approximation. Either way, this was considered beyond the scope of this thesis.

4. *Multi-server extension.* Carrying the mechanisms to $M/M/c$ would connect this exact single-server line to the approximate multi-server scheduling results of [HCD22], trading the exactness the single server buys for operational realism.

5. *Derive mean performance metrics.*The closed-form PGFs obtained for Model-B_2, Model-C_2, Model-B_2^{H}, and Model-C_2^{H} contain the full distributional information, but extracting mean queue lengths and waiting times requires differentiating at $(1,1)$, which in Model-B_2 and Model-C_2 involves boundary functions given as integrals. A systematic differentiation—applying L'Hôpital where necessary at the singularity $x = 1$—would yield closed-form expressions for $\mathbb{E}[N_1]$, $\mathbb{E}[N_2]$, and, via Little's law, $\mathbb{E}[W_1]$ and $\mathbb{E}[W_2]$, replacing the numerical estimates reported in Section 11 with exact ones.

6. *Calibration.* Fitting γ_1, θ_1 and the loads to real elderly-care waiting-list data [Zen99] would test whether the regimes identified here accurately describe operational reality.

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